
Local Langlands for $\mathrm{GL}_2(\mathbf{F})$

— *EYNTKA* Series —

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Contents

Contents	3
Introduction	7
1 Introduction to Smooth Representations	9
1.1 Locally Profinite Groups	9
1.2 Smooth Representations of Locally Profinite Groups	16
1.3 Measures and Duality	31
1.4 Hecke Algebra	40
2 Smooth Representations of GL_2: General Structure and Principal Series	41
2.1 Important Background: Representation of $GL_2(\mathbb{F}_p)$	41
2.2 Linear Groups over Local Fields	51
2.3 Representation of the Mirabolic Group	56
2.4 Jacquet Modules and Induced Representations	62
3 Cuspidal Representations	71
3.1 Cuspidal Representations and Coefficients	71
3.2 Intertwining and Compact Induction	75
3.3 Chain Orders and Fundamental Strata	79
3.4 Classification of Fundamental Strata	82
3.5 The Classification of Cuspidal Representations	83
3.5.1 Exhaustion	83

3.5.2	Cuspidal Types	84
3.6	Representations with Iwahori-Fixed Vectors	85
4	Quadratic Tame Extensions Paramaterizing Tame Cuspidals	87
4.1	Admissible Pairs	87
4.2	Construction of Representations	89
4.2.1	Level Zero Construction	89
4.2.2	Positive Level Construction	90
4.3	The Parametrization Theorem	91
4.4	Tame Intertwining Properties	92
4.5	A Certain Group Extension	93
5	Characterizing Representations using Functional Equations	97
5.1	Functional Equation for $GL(1)$	97
5.1.1	Local Constants	104
5.2	Functional Equation for $GL(2)$	109
5.3	Epsilon Factors for Cuspidal	118
5.4	Epsilon Factors for non-Cuspidal	118
5.5	Converse Theorems	119
6	Representation of Galois Theory: Weil Groups	121
6.1	Weil Group and Smooth Representations	121
6.2	Weil Group and Local Class Field Theory	131
6.3	L-functions and Epsilon-Factors	134
6.4	Deligne Representations	141
6.4.1	Definition and Algebraic Structure	142
6.4.2	Relation with ℓ -adic Representations	143
6.4.3	L-functions and Epsilon-factors	144
7	The Langlands Correspondence	147
7.1	Statement of the Correspondence	147
7.2	The Reducible Correspondence	149
7.3	The Tame Correspondence ($p \neq 2$)	149
7.4	The ℓ -adic Correspondence	150
7.5	Summary	151
A	The Projectivity Theorem	153

0.0.1 Inspiration Very heavily inspired by [1]

Introduction

The Langlands Program sets out to find a way to fully understand the absolute galois group $\text{Gal}(\overline{F}/F)$ for any field. This pursuits has spawned a web of connections linking Number Theory (Galois representations), Harmonic Analysis (automorphic forms), and Geometry (cohomology of algebraic varieties). At its heart lies a dictionary, the infamous (global) *Langlands Correspondence*, which translates arithmetic data into analytic data, and vice versa.

While the global theory is vast and often technically overwhelming, the *Local Langlands Correspondence* provides the essential vocabulary for this dictionary. It asserts that for a local field F (like $\mathbb{Q}_p$), the representation theory of the groups $\text{GL}_n(F)$ for $n \in \mathbb{N}_{>0}$ is mirrored by the representation theory of the absolute Galois group of F .

The Goal The goal of this book is to provide a complete, self-contained, and accessible proof of the Local Langlands Correspondence for the case $n = 2$.

This book is heavily based on the seminal work *The Local Langlands Conjecture for $GL(2)$* by Colin J. Bushnell and Guy Henniart. These notes are written in the *EYNTKA* (Everything You Need To Know About) style. The aim is not just to verify the truth of the theorems, but to expose the narrative arc of the discovery and hopefully build-up the appropriate knowledge to generalize to the n -dimensional case, and to at least understand start to build up the knowledge to prepare the reader for the infamous connection between elliptic curves and modular forms.

The Roadmap

1. **The Analytic Side (Chapters 1–3):** We begin by establishing the theory of *smooth representations*. Unlike the representation theory of finite groups or Lie groups, we must deal with vector spaces that are infinite-dimensional but are totally disconnected. We classify the “easy” representations (the Principal Series) using parabolic induction, and then tackle the “hard” representations (the Supercuspidals). We show that the cuspidal representations can be constructed explicitly via *Compact Induction* from open subgroups.

2. **The Arithmetic Side (Chapter 4–5):** How do we verify that a representation on one side matches a representation on the other? We need invariants that exist in both worlds. We develop the theory of *Local L-functions* and *Local Constants* (ε -factors). We build up to prove the *Converse Theorem*, which states that these invariants characterize $GL_2(F)$ representations: if the L and ε factors match after enough twisting, the representations must be identical.
3. **The Algebraic Side (Chapter 6):** We refine the Galois group into the *Weil Group* $\mathcal{W}_F$ and introduce *Deligne Representations*. This modification is necessary to account for the complex geometry of the “Special” (Steinberg) representations on the analytic side.
4. **The Correspondence (Chapter 7):** Finally, we bring the two sides together. We match the Principal Series to reducible Galois representations. For the cuspidal representations, we use the machinery of *Admissible Pairs* and *Fundamental Strata* to construct a precise bijection. We verify that this bijection preserves the arithmetic invariants, thereby proving the Local Langlands Conjecture.

0.0.2 Limitations of Book At the moment the Langlands correspondence for $p = 2$ is *not* covered as that would require another few chapters all on it’s own!! In other words, it is *much* harder. I also would love to cover the Jacquet-Langlands correspondence as done in [1], but that also will be covered another time. I also had no time to mention how one would start using this information to bridge the gap to the global case.

Prerequisites

The reader is assumed to be comfortable with:

- The basic theory of local fields (valuations, integers, p -adic numbers), as covered in [6].
- Galois theory, as covered in [3]
- Classical representation theory of finite groups (characters, induction, Schur’s Lemma).
- Familiarity topological groups, as covered in either [5], [3], or [6].
- Familiarity with the basic structure of profinite groups is useful, but will be covered.

Introduction to Smooth Representations

The Langlands program is, at its core, a bridge between number theory and harmonic analysis. On the number theoretic side, we have Galois groups $\text{Gal}(\overline{F}/F)$, which encode the arithmetic structure of fields. On the analytic side, we have groups of matrices like $\text{GL}_n(F)$. To build this bridge, we must first understand the ground on which the analytic side stands. The goal of this chapter is to establish the foundational language for the “analytic” side of the correspondence: the theory of *smooth representations* of locally profinite groups. Unlike the representation theory of finite groups or Lie groups, the topology of p -adic fields requires a unique approach. We cannot simply look at continuous representations on Hilbert spaces; instead, we must look at representations that respect the totally disconnected topology of $\mathbb{Q}_p$.

We begin in Section 1.1 by defining locally profinite groups and exploring the topology of F and $\text{GL}_n(F)$. In Section 1.2, we define smooth representations—actions that are locally constant—and show that they satisfy an analogue of Schur’s Lemma. In Section 1.3, we develop the necessary integration theory (Haar measure) to analyze these groups. Finally, in Section 1.4, we introduce the Hecke algebra $\mathcal{H}(G)$, an associative algebra that allows us to translate problems of smooth group representations into the language of modules over a ring, mirroring the relationship between finite groups and the group ring $\mathbb{C}[G]$.

1.1 Locally Profinite Groups

All finite groups will by assumption have the discrete topology. For a deeper review of topological groups, see [8, chapter 8].

The field $\mathbb{Q}_p$ is a *locally profinite groups* with its topology endowed by the metric $|\cdot|_p$. This is a

substantially different topology to $\mathbb{C}$, and hence a moment must be taken to understand better its structure.

Definition 1.1.1: P-adic Topology

Let $\mathbb{Q}_p$ be a p-adic field for a (finite) prime p and let $\mathbb{Z}_p$ be the corresponding ring of integers. Define the neighbourhood basis:

$$\mathbb{Z}_p \supseteq p\mathbb{Z}_p \supseteq p^2\mathbb{Z}_p \supseteq \dots$$

Giving $\mathbb{Z}_p$ a p-adic topology which can be easily extended to $\mathbb{Q}_p$. As each $p^k\mathbb{Z}_p$ is a closed ball, this matches the metric topology induced by $|\cdot|_p$:

$$B_r(a) = \{x \in \mathbb{Q}_p : |x - a|_p < r\}$$

which extends to $\mathbb{Q}_p$ by defining:

$$\left| \frac{m}{n} \right|_p = \frac{|m|_p}{|n|_p}$$

Let us establish some basic properties of this field.

Lemma 1.1.2: P-adic Field Totally Disconnected

The field $\mathbb{Q}_p$ with the topology induced by the metric $|\cdot|_p$ is totally disconnected. More generally any space X whose topology is induced by an ultra-metric is totally disconnected.

Proof:

Let $a, b \in \mathbb{Q}_p$. As $a \neq b$, there must exist a d such that:

$$0 < |a - b|_p = p^{-k} = d$$

Let:

$$B_d(a) = \{x \in \mathbb{Q}_p : |x - a|_p < d\}$$

By construction, $a \in B_d(a)$ and $b \notin B_d(a)$.

Let us show that

$$B_d(a)^c = \{x \in \mathbb{Q}_p : |x - a|_p \geq d\}$$

is also open. Take $x \in B_d(a)^c$ so that $D = |a - x|_p \geq d$ and consider $B_d(x)$ and take $y \in B_d(x)$ so that $\delta = |x - y|_p \leq d$. We want to show that $y \in B_d(a)^c$ so $|y - a|_p \geq d$. Then by the isosceles property of the ultra-metric, for any three points a, x, y two of $d(a, x)$, $d(a, y)$ or $d(x, y)$ must be to the larger one. As $|a - x|_p \geq d$, $|y - a|_p \geq d$, and so $y \in B_d(a)^c$, showing that $B_d(x) \subseteq B_d(a)^c$ showing that it is *also* open.

Thus, $\mathbb{Q}_p$ is totally disconnected, as we sought to show.

Lemma 1.1.3: P-adic Integers Open Subgroup

The subgroup $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ is an open subgroup with the topology induced by the valuation. Hence, it is also closed

This differs strongly from $\mathbb{C}$: if $H \subseteq \mathbb{C}$ is an open subgroup then as noted earlier it must also be closed, hence connected, and hence $H = \mathbb{C}$. As is $\mathbb{Q}_p$ is totally disconnected, it is possible to have clopen subsets that are not the entire set.

Proof:

First, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\}$. Take $a \in \mathbb{Z}_p$ (so that $|a| \leq 1$) and consider:

$$B_1(a) = \{x \in \mathbb{Q}_p : |x - a|_p < 1\}$$

I claim $x \in \mathbb{Z}_p$. Indeed:

$$|x|_p = |x + (-a + a)|_p \leq \max(|x - a|_p, |a|_p) = \max(t, 1) = 1$$

where $t < 1$ as $|x - a|_p < 1$. But then $x \in \mathbb{Z}_p$, and hence $B_1(a) \subseteq \mathbb{Z}_p$.

Thus, as $\mathbb{Q}_p$ is a topological group, $\mathbb{Z}_p$ is also closed.

There shall be instances where we work with other similar groups constructed from p -adic fields, for example finite extension of $\mathbb{Q}_p$, the ring of integers of $L/\mathbb{Q}_p$, $\overline{\mathbb{Q}_p}$, $\mathbb{C}_p$, $M_n(\mathbb{Q}_p)$, $M_n(\mathcal{O}_{L/\mathbb{Q}_p})$, $M_n(\mathfrak{m}_{\mathfrak{p}})$, various matrix subgroups of this groups (GL_n , SL_n , PGL_n , Sp_n , O_n , SU_n , etc.) and so forth. Thus, it is worth going over a general theory that encapsulates the topology of these spaces. Let us first recall some basic properties of topological groups:

- The topology of G is translation invariant: If $U \subseteq G$ is open, then Ux and xU is open for all $x \in G$
- If $H \leq G$, G/H has the quotient topology, and if $H \trianglelefteq G$ then G/H is a topological group.
- If K_1, K_2 are compact subset of G , so is $K_1 K_2$
- subgroups in relation to open/closed:
 - if $H \leq G$ is open, it must be closed ($\bigcup_{g \in G \setminus H} gH$ is the union of open sets and the complement of H). More strictly, if H is open if and only if it contains an open neighbourhood around the identity (if $e \in U \subseteq H$ with open U , then $hU \subseteq H$ and $H = \bigcup_{h \in H} hU$). If G is connected, it has no non-trivial proper open subgroups. We shall frequently be working with totally disconnected groups.
 - If $H \leq G$ is an open subgroup, then the quotient space G/H and $H \backslash G$ each have the discrete topology.
 - A closed subgroup $H \leq G$ need not be open (ex. $\mu_n \subseteq \mathbb{C}$). However if $H \leq G$ is closed and finite index, it is open (the finite union of closed is closed). If G is first-countable, H is closed if it contains all its limit points (more generally, it contains all the limits of nets).
 - A subgroup need not be open or closed (ex. $\mathbb{Q} \subseteq \mathbb{R}$).

- if H is a subgroup of G , so is $\overline{H}$ (the closure under limits/nets).
- If $G_0 \subseteq G$ is the connected component of the identity, then G_0 is closed, G_0 is the intersection of all normal subgroups of G , and G/G_0 is the largest totally disconnected quotient.
- symmetric neighbourhood:
 - For every open neighborhood of e , say for example U , there is a symmetric neighborhood (if $A \subseteq G$ such that $A = A^{-1}$, we'll say A is *symmetric*) V of e such that $V \subseteq U$
 - For every neighborhood U of e , there is a neighborhood V of e with $VV \subseteq U$
- Density: the groups we shall work with will often have dense subsets/subgroups. Sometimes the sequence definition of density will not be equivalent to the definition of density, hence a reminder for those who have internalized the sequence definition to the point that they forgot the general topological definition: A set $S \subseteq X$ is *dense* if it intersects every nonempty open subset of X : $U \cap S \neq \emptyset$ for all open $U \subseteq X$.

Lemma 1.1.4: Characterizing Hausdorffness

Let G be a topological group, $e \in G$ the identity, and $H = \bigcap_{\substack{U \subseteq G \text{ open} \\ e \in U}} U$. Then:

1. H is a subgroup
2. H is the closure of e , i.e. $\overline{e} = H$
3. G/H is Hausdorff
4. G is Hausdorff if and only if $H = \{e\}$

Proof:

exercise

Henceforth, we shall always assume Hausdorffness unless specified otherwise.

Definition 1.1.5: Profinite Group

Take a diagram finite groups with the discrete topology. Then a *profinite group* is a topological group that is the colimit of the system. Given any topological group G , we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

where N ranges over all finite index open normal subgroups ordered by containment.

Any group G can be given the profinite topology by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then $\varphi : G \rightarrow \widehat{G}$ is a continuous morphism and G is dense in $\widehat{G}$.

Example 1.1.6: Profinite Groups

We have the following “trivial” examples:

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The following is the “usual” example and is better suited for the intuition of profinite groups. Let $G = \mathbb{Z}$. Then

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$

where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p -adic completion of $\mathbb{Z}$. The main idea is that for any element $n \in \mathbb{Z}$, in chain of normal subgroups (say $p^k\mathbb{Z} \subseteq p^{k+1}\mathbb{Z} \subseteq \dots$) the element mod $p^k\mathbb{Z}$ eventually stabilize. For example, take $p = 7$ and $n = 129'654 = 2 \cdot 3^3 \cdot 7^4$. Then the value is zero for the until $\mathbb{Z}/7^5\mathbb{Z}$ where it becomes $12'005 \bmod 7^n$ for $n \in \{5, 6\}$ and then $129'654$ forever onward. $\mathbb{Z}_p$ allows for never-stabilizing chains (mirroring how stable sequences of rational numbers converge to rational numbers).

For Langlands, the important non-abelian example is the profinite completion of galois Galois groups, namely $\text{Gal}_K(\overline{K})$. See [3] or a quick overview or [2] for more details.

More generally, profinite groups can be seen as the object that captures all the finite behaviour “up to infinity” (or philosophically, all the “potential infinity”) of a system. In the $\mathbb{Z}_p$ example, it appeared when taking a solution mod each p^n and for galois it comes from amalgamating all symmetries of roots of some infinite family of (finite) polynomial. Another place this shall show up is in the definition of the étale fundamental group, see [ref:HERE](#).

The above examples show that the map $\varphi : G \rightarrow \widehat{G}$ could be injective, surjective, both.

Lemma 1.1.7: Profinite Completion Injective

Let G be a topological group and $\widehat{G}$ it's profinite completion. Then $G \rightarrow \widehat{G}$ is injective if the intersection all finite-index open normal subgroups of G is trivial

Proof:
exercise.

The completion comes equipped with a natural universal property:

Theorem 1.1.8: Universal Property of Profinite Completion

let $\varphi : G \rightarrow H$ be continuous homomorphism with H a profinite group. Then there is a unique continuous homomorphism such that the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & \widehat{G} \\ & \searrow \varphi & \downarrow \exists! \\ & & H \end{array}$$

Proof:

exercise (hint: recall G is dense in $\widehat{G}$).

A striking result is that all profinite groups can be characterized by a simple topological property:

Theorem 1.1.9: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

Proof:

Let G be profinite. Then it is a closed subgroup of a direct product of finite groups, hence compact and totally disconnected.

Conversely, suppose G is totally disconnected and compact. We must show it comes from an inverse system of finite groups. **will finish later.**

Corollary 1.1.10: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion, or even stronger:

$$G \cong \varprojlim G/U$$

where U ranges over all *open* normal subgroups ordered by contain need.

If G is isomorphic to its profinite completion *as a group* (i.e. is strongly complete) if and only if every finite index subgroup is open

Proof:

exercise.

Proposition 1.1.11: Dense Subsets of Profinite Group

Let $S \subseteq G$ be a subset of a profinite group. Then S is *dense* if and only if the natural maps $S \rightarrow G/U$ is surjective for all open $U \subseteq G$.

Proof:

Definition 1.1.12: Locally Profinite Group

A topological group G is *locally profinite* if it is Hausdorff and every open neighbourhood of the identity contains a compact open subgroup of G .

Notice how there is no mention of an inverse system. The reader should show that the definition implies G is totally disconnected, and hence if it is also compact then we recover a profinite group. Every discrete group is finite, closed subgroups of a locally profinite group are locally profinite, and so are quotients by closed normal subgroups. A locally profinite subgroup is locally compact. It is an exercise to show that any locally compact totally disconnected group is locally profinite. The canonical example for these notes of a locally profinite group is the non-Archimedean local field $(F, +)$ as proven in [6]. Similarly, $(F^\times, \times)$ is a profinite group by passing through e^x , or:

$$U_F^n = 1 + \mathfrak{p}^n$$

Being profinite is closed under products, hence F^n is profinite, and $(M_n(F), +)$ is locally profinite under addition. The group $\mathrm{GL}_n(F) \subseteq M_n(F)$ under multiplication is an open topological subgroup. It is locally profinite as $\mathrm{GL}_n(\mathcal{O}_F)$ is profinite (it is compact as it is homeomorphic to a closed subset of $\mathcal{O}_F^{n^2}$ which is a product of compact groups, and it is open as $\mathcal{O}_F = \{x \in F : |x|_p \leq 1\}$ is an open set. Note that this was all done in coordinates, more generally this applies to $\mathrm{End}_F(V)$ and $\mathrm{Aut}_F(V)$).

Let us now consider character's on locally profinite groups:

Theorem 1.1.13: Continuous Character

let $\varphi : G \rightarrow \mathbb{C}^\times$ be a group homomorphism where G is locally profinite. Then the following are equivalent:

1. φ is continuous
2. the kernel of φ is open

If φ satisfies one of these conditions and G is the union of its compact open subgroups (such as $\mathbb{Q}_p$), then $\mathrm{im} \varphi \subseteq S^1 \subseteq \mathbb{C}^\times$.

Proof:

If the kernel is open, then if $U \subseteq \mathbb{C}^\times$ and $K = \ker \varphi$,

$$\varphi^{-1}(U) = \bigcup_{g \in \varphi^{-1}(U)} gK$$

showing φ is continuous. Conversely, Let φ be continuous and U an open neighbourhood of 1. Then $\varphi^{-1}(U)$ is open, and thus by local-profiniteness contains a compact open subgroup $K \subseteq G$. As this applies to any U , we can pick U sufficiently small so that it contains no non-trivial subgroup of $\mathbb{C}^\times$. As the image of group is a group and $K \subseteq \varphi^{-1}(U)$, $K \subseteq \ker \varphi$. As K is open, $\ker \varphi$ contains an open subgroup and hence by topological group theory $\ker \varphi$ is open.

Next, S^1 is the unique maximal compact subgroup of $\mathbb{C}^\times$. If K' is a compact subgroup of G , $\varphi(K') \subseteq S^1$, giving the 2nd result as we sought to show.

This condition has interesting algebraic consequences, for example take the map $\mathbb{Z} \rightarrow S^1$ by sending 1 to $e^{2\pi i\theta}$ where θ is irrational. As $\mathbb{Z}$ is dense in $\mathbb{Z}_p$, this map extends to a map $\mathbb{Z}_p \rightarrow S^1$. It's kernel is $\{0\}$, i.e. it is not open, and so this map is not continuous with the profinite topology on $\mathbb{Z}_p$.

Corollary 1.1.14: Continuous Maps To General Linear Group

Let $\varphi : G \rightarrow \mathrm{GL}_n(\mathbb{C})$ be a group homomorphism where G is locally profinite. Then φ is continuous if and only if the kernel of φ is open.

If G is the union of compact open subgroup, the maps into $U(n)$ I'll prove this later.

Proof:

Note that $\mathrm{GL}_n(\mathbb{C})$ is a lie group and that there exists an open neighbourhood of I_n that contains no non-trivial subgroups, the argument follows.

Definition 1.1.15: Continuous Character

A *character* for a locally profinite group G is a continuous homomorphism

$$\varphi : G \rightarrow \mathbb{C}^\times$$

We shall often write 1_G or just 1 for the trivial character (i.e. the constant character).

If $\varphi(G) \subseteq S^1$, φ is called a *unitary character*.

summary of next section On page 12 in [1], he defines $\widehat{F}$ to be the set of all characters of F . As F is a local field, it is the union of its compact open subgroup $\mathfrak{p}^n$ for $n \in \mathbb{N}$, hence all characters are unitary. Then if $\varphi \neq 1$ is a character, there exists $d \in \mathbb{Z}$ such that $\mathfrak{p}^d \subseteq \ker(\varphi)$; the least such number is defined as the level of φ . Fixing a d , the set of characters of level $\leq d$ is a subgroup. This defines a duality which extends to the multiplicative set, in other words this is indeed the right notion of character!

1.2 Smooth Representations of Locally Profinite Groups

All representations shall be complex unless specified otherwise. When working with $\mathbb{Q}_p$ representations and more generally representations of locally profinite groups, it is important to take into account the topology, or else we will get representations that give no valuable information:

Example 1.2.1: “Useless” Representation

Take the algebraic closure of $\mathbb{Q}_p$, $\overline{\mathbb{Q}_p}$. By field theory, it embeds into $\mathbb{C}$ (using the axiom of choice). Thus there is an injective ring homomorphism $\mathbb{Q}_p \rightarrow \mathbb{C}$ and hence a (valid) representation:

$$\pi : \mathrm{GL}_n(\mathbb{Q}_p) \rightarrow \mathrm{GL}_n(\mathbb{C})$$

But this has disregarded all topological (and hence arithmetic) information of $\mathbb{Q}_p$.

The first step to fix this would be to only consider continuous representations as is done in the representation of topological groups. Often when G is not compact, we get a vector space V that is infinite dimensional with which is either a Banach or Hilbert space and $\text{GL}(V)$ is the set of all invertible bounded linear operators; this thread is further explored in [7].

When restricting to locally profinite groups, there is a useful stronger notion called *smooth*. It must be that this notion of smoothness is different from smoothness for manifolds as we will be looking at maps $\mathbb{Q}_p \rightarrow \mathbb{C}$ which don't share the local topology ($\mathbb{Q}_p$ is also totally disconnected, so there is no way to put a reasonable atlas on it). Instead, the intuition comes from how smoothness (or differentiability) means that zooming in close will get the same linear, or "constant" if we change coordinates in our perspective. In terms of locally profinite groups, as they are the union of open compact subgroups, a representation being constant on these groups abstractly shares this same property. This loose analogy turns out to be very practical, hence:

Definition 1.2.2: Smooth Map For Locally Profinite Groups

Let G be a locally profinite group. Then a map $f : G \rightarrow \mathbb{C}^n$ is *smooth* if it is locally constant, namely for any $x \in G$ there exists an open neighbourhood U containing x such that $f(x) = f(y)$ for all $x, y \in U$. The collection of all smooth functions $f : G \rightarrow \mathbb{C}$ is denoted $C^\infty(G)$.

If the function also has compact support, we denote the space $C_c^\infty(G)$ and call the space the *smooth compact support space* or *Schwartz-Bruhat Space*.

1.2.3 Precision in Vocabulary Via Harmonic Analysis There is a more precise comparison when considering the Fourier transform with distributions, so we'll assume [8] for the following.

1. On $\mathbb{R}$: For the Fourier transform, the space of "ideal" functions is the Schwartz space, consisting of functions that are both smooth (C^∞) and rapidly decaying. The Fourier transform is a perfect map on this space: the transform of a smooth, rapidly decaying function is another smooth, rapidly decaying function.
2. On $\mathbb{Q}_p$: There is an analogous theory of Fourier analysis we shall cover in chapter 5. The "ideal" space of functions (called the Schwartz-Bruhat space) consists of functions that are locally constant and have compact support, namely since such functions under the Fourier transform map back to $C_c^\infty(G)$. Note that in the real case the image of real smooth function with compact support is the Paley-Wiener space which are holomorphic function with a certain decay property, something we need not worry about in p -adic analysis! Thus, as Fourier relates smoothness to decay, in p -adic analysis being a compact support function "is" smooth from the Fourier perspective.

Proposition 1.2.4: Equivalence Notion of Smoothness

Let G be a locally profinite group and $f : G \rightarrow \mathbb{C}^n$ a map. Then f is smooth if and only if for any subset $U \subseteq \mathbb{C}^n$ (which is *not* necessarily open), $f^{-1}(U) \subseteq G$ is open.

Thus, if f is smooth, it is continuous with $\mathbb{C}^n$ having the discrete topology.

Proof:
exercise.

The key in the notion of smoothness being useful is that on $\mathbb{Q}_p$ there are non-trivial locally constant functions:

Example 1.2.5: Smooth Functions On Locally Profinite Groups

1. The p -adic valuation $|\cdot|_p : \mathbb{Q}_p^\times \rightarrow \mathbb{R}$ is smooth. To see this, take any $x \in \mathbb{Q}_p$ so that $|x|_p = p^{-n}$ for some $n \in \mathbb{Z}$ giving the range. Then the pre-image of p^{-n} is $p^n \mathbb{Z}_p^\times$ which is open.

Note that this is not smooth on $\mathbb{Q}_p$ because the pre-image of 0 is 0, i.e. there is no open neighbourhood of 0 on which the absolute value is constant. This can even be stretched to the absolute value on $\mathbb{R}$, when restricting to $\mathbb{R}^\times$ it is smooth, but naturally it is not smooth at 0.

2. The map $\mathrm{GL}_n(\mathbb{Q}_p) \rightarrow \mathbb{R}$ given by $g \mapsto |\det(g)|_p$ is smooth.

Often, a smooth function should be thought of as being well-defined modulo cosets of an open subgroup.

By identifying $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$, we get the notion of a smooth representation:

Definition 1.2.6: Smooth Representation

Let G be a locally profinite group. Then a *complex representation* (or just *representation*) of G is a pair (π, V) where V is a complex vector space and $\pi : G \rightarrow \mathrm{Aut}_{\mathbb{C}}(V)$ a group homomorphism.

A representation (π, V) is called *smooth* if for every $x \in V$, there is a compact open subgroup $K_x \subseteq G$ such that for all $k \in K_x$:

$$\pi(k) \cdot x = x$$

Equivalently, if V^K denotes the space of $\pi(K)$ -fixed vector in V ,

$$V = \bigcup_K V^K$$

where K ranges over all compact open subgroups of G .

A smooth representation is called *admissible* if V^K is finite dimensional for all compact open $K \subseteq G$.

Proposition 1.2.7: Smooth Representation Alternative

A representation $\pi : G \rightarrow \text{Aut}_{\mathbb{C}}(V)$ is smooth if

$$\text{Stab}_G(v) = \{g \in G : \pi(g) \cdot v = v\}$$

is an open subgroup of G for all $v \in V$

Proof:
exercise.

If (π, V) is a smooth representation of G , then any G -stable subspace of V is another smooth representation, and similarly if $U \subseteq V$ is a G -subspace, the induced representation of G on V/U is smooth. A representation (π, V) is *irreducible* if $V \neq 0$ and V has no nontrivial G -stable subspaces. A G -homomorphism between two representations (π_1, V_1) , (π_2, V_2) is a map that commutes with the group action induce by π_i .

Example 1.2.8: Smooth Representations

1. If $(\mathbb{C}, \chi)$ is a one-dimensional representation, by theorem 1.1.13 it is smooth if and only if it is continuous. Hence in one dimensions these two concepts are the same. Indeed, if $\chi(g) \cdot v = v$, as $\chi(g)$ is a complex number and v is nonzero we have:

$$\chi(g) = 1$$

Hence, χ is smooth if and only if the kernel is open (and as G is locally profinite it contains a compact subgroup), which by theorem 1.1.13 is true if and only if χ is continuous.

By corollary 1.1.14, any finite dimensional continuous representation is smooth.

2. Let $G = \mathbb{Q}_p^*$; let us look for maps $\psi : \mathbb{Q}_p^* \rightarrow \mathbb{C}^*$ which as $\mathbb{Q}_p$ is the union of compact open sets by theorem 1.1.13 we know every maps into S^1 . As $\mathbb{Q}_p^* \cong \mathbb{Z} \times \mathbb{Z}_p^*$ (since $x = p^n u$ for $n \in \mathbb{Z}$ and $u \in \mathbb{Z}_p^*$; namely $n = \nu_p(x)$), and ψ is a group homomorphism, the map is determined by where it maps $(1, 0)$ and $\mathbb{Z}_p^*$. Let us look at the two parts seperatly:

$$\psi((c, u)) = \psi((c, 1)) \cdot \psi((0, u)) = \psi_1(c) \cdot \psi_2(u)$$

Thus, for any choice of $s \in \mathbb{C}$, let:

$$\psi_1(\nu_p(x)) = p^{-s\nu_p(x)}$$

Next, lets look at the units. Note that this includes $1 \in \mathbb{Z}_p^*$, and we know that the kernel must be open, and as $\varphi(1) = 1$, that means that an open neighbourhood around $1 \in \mathbb{Z}_p^*$ maps to $1 \in \mathbb{C}^*$. The open neighborhoods around 1 are:

$$U_k = 1 + p^k \mathbb{Z}_p = \{u \in \mathbb{Z}_p^* : u \equiv 1 \pmod{p^k}\}$$

Thus, for some $k \in \mathbb{N}$ $\psi_2(U_k) = 1$ By the isomorphism theorem the cosets will be:

$$\mathbb{Z}_p^* / U_k \cong (\mathbb{Z}/p^k \mathbb{Z})^*$$

But now ψ_2 reduces to $(\mathbb{Z}/p^k\mathbb{Z})^* \rightarrow \mathbb{C}^*$ which is just the familiar roots of unity maps. Note how this is “ramified”. For this reason, ψ_1 is often called the *unramified component* while ψ_2 is called the *ramified component*.

Combining these together, if we let $\psi_1(x) = |x|_p^s$ and $\psi_2(x) = \psi(x)$:

$$\chi(x) = |x|_p^s \psi(x)$$

These classify all continuous/smooth characters of $\mathbb{Q}_p^*$.

3. Let us next consider the next dimension up: $G = \mathrm{GL}_n(\mathbb{Q}_p)$ and $V = \mathbb{C}$, that is smooth characters on $\mathrm{GL}_n(\mathbb{Q}_p)$. Then As $\mathbb{C}^*$ is commutative, $\chi : \mathrm{GL}_n(\mathbb{Q}_p) \rightarrow \mathbb{C}^*$ must be trivial on the commutator. It is an exercise in linear algebra to show that $[G, G] = \mathrm{SL}_n(\mathbb{Q}_p)^a$. As the identity has determinant 1, $I \in \mathrm{SL}_n(\mathbb{Q}_p)$ and as χ is a group homomorphism it must map to 1. Thus $\chi(\mathrm{SL}_n(\mathbb{Q}_p)) = 1$.

Next, for any $A, B \in \mathrm{GL}_n(\mathbb{Q}_p)$ such that $\det(A) = \det(B)$ so that $AB^{-1} \in \mathrm{SL}_n(\mathbb{Q}_p)$, we must have $\chi(A) = \chi(B)$ as χ is a homomorphism. But then, χ is only dependent on the determinant of a matrix.

From this, to can construct a character $\chi : \mathrm{GL}_n(\mathbb{Q}_p) \rightarrow \mathbb{C}^*$, we choose a 1-dimensional character $\psi : \mathbb{Q}_p^* \rightarrow \mathbb{C}^*$ and compose it with the determinant map $A \mapsto \det(A)$ so that:

$$\chi(A) = \psi(\det(A))$$

It can be shown that all characters are of this form.

4. Let us consider $G = \mathbb{Q}_p$ and $V = \mathbb{C}$ so that we are looking at characters $\chi : \mathbb{Q}_p \rightarrow \mathbb{C}^*$. These are all formed by the *standard character*. For any $x \in \mathbb{Q}_p$ the p -adic representation is of the form

$$x = \sum_n^{\infty} a_i p^i \quad n \in \mathbb{Z}, \quad a_i \in \{0, 1, \dots, p-1\}$$

The p -adic fractional part is the negative indexed summands; let $\{x\}_p$ represent this. Define:

$$\psi_p : \mathbb{Q}_p \rightarrow \mathbb{C}^* \quad \psi_p(x) = e^{2\pi i \{x\}_p}$$

Notice the kernel of this map is $\mathbb{Z}_p$, and hence it is continuous/smooth.

5. We can form characters on $\mathfrak{m}_p$ by restricting characters on $\mathbb{Q}_p$. Show that $\hat{\mathfrak{m}}_p \cong \mathbb{Q}_p/\mathbb{Z}_p$.
6. For a continuous non-smooth example, let $G = \mathbb{Z}_p$ and $V = C(\mathbb{Z}_p, \mathbb{C})$ be the $\mathbb{C}$ -vector space of continuous characters with the uniform-convergence topology. Then take the left-regular representation (ρ, V) :

$$(\rho(g) \cdot f)(x) = f(x - g)$$

The reader can show this is continuous. Let us look at the stabilizer of the vector $g \mapsto |g|_p$. Note this map is certainly continuous as the norm-map is continuous, and thus $|\cdot|_p \in C(\mathbb{Z}_p, \mathbb{C})$. Let us consider the stabilizer of $|\cdot|_p$ that is all, $g \in \mathbb{Z}_p$ such that :

$$g \cdot |\cdot|_p = |\cdot - g|_p = |\cdot|_p$$

Let $g \in \mathrm{Stab}_{\mathbb{Z}_p}(|\cdot|_p)$ so that:

$$|g - x|_p = |x|_p \forall x \in \mathbb{Z}_p$$

If $x \neq g$, then it easy to see that equality *fails*, hence the only stabilizer is 0. But then $\mathrm{Stab}_{\mathbb{Z}_p}(|\cdot|_p) = \{0\}$ which is *not* open.

7. (principal series representation) Let:

$$B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Q}_p) \right\}$$

Let us define a character on B by choosing two characters $\chi_1, \chi_2 : \mathbb{Q}_p^* \rightarrow \mathbb{C}^*$ given by:

$$\varphi \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) = \chi_1(a)\chi_2(c)$$

Now, define the complex vector space V to be smooth complex function $f : \mathrm{GL}_2(\mathbb{Q}_p) \rightarrow \mathbb{C}$ such that:

$$f(bg) = \psi(b)f(g) \quad b \in B, g \in \mathrm{GL}_2(\mathbb{Q}_p)$$

Then define the action to be:

$$(\rho(h) \cdot f)(g) = f(gh)$$

Then this is an infinite dimensional smooth representation; for a choice of χ_1, χ_2 denote it $I(\chi_1, \chi_2)$. It is almost always irreducible, but there shall be there are non-reducible representations given the right choices of χ_1, χ_2 , see example 1.2.13.

By Schur's Lemma, the abelian groups have no higher dimensional irreducible representations. For higher finite dimensional smooth representations of the non-abelian groups, [ref:HERE](#) shall show that they are *all* a direct sum of 1 dimensional representations, even though these groups are not abelian! This shows that to capture some of their more non-abelian structure of these groups, it will be important to look at (well-behaved) infinite dimensional representations, hence the need to add the smooth condition.

Note that by Lie-Kolchin theorem, (complex) irreducible representations of solvable connected groups are always 1-dimensional. Looking $\mathrm{GL}_b(F)$ where F is nonarchimedean in the Zariski topology, it is irreducible and hence connected and hence if it *were* solvable this would apply. It is however *not* the case that it is solvable.

^athis is true as long as the field F is not the field with two elements

Proposition 1.2.9: Generalized Maschke's Theorem

Let G be compact (thus profinite) and (π, V) an irreducible representation of G . Then V must be finite-dimensional.

Proof:

If $0 \neq v \in V$, then $v \in V^K$ for an open subgroup $K \subseteq G$. As G is profinite, $[G : K]$ is finite and so:

$$\{\pi(g) \cdot v : g \in G/K\}$$

spans V . We can then take the smaller $K' = \bigcup_{g \in G/K} gKg^{-1}$ which is an open normal subgroup of G , still of finite index, and the action is invariant on K' . Thus, V can be seen as an irreducible representation of the finite discrete group G/K' , and hence is certainly finite dimensional.

As there are many hierarchies of types of representations in representation theory (semi-stable,

crystalline, etc), we shall purposefully specify the category that has been described thus far:

Definition 1.2.10: Smooth Representation Category

Let G be a locally profinite group. Then $\text{Rep}_{\text{smooth},K}(G)$ is the smooth representation category whose objects are smooth representations (V, π) and morphisms are G -equivariant maps. The representations shall always be smooth, and hence we write $\text{Rep}_K(G)$ instead of $\text{Rep}_{\text{smooth},K}(G)$. The field K is dropped if it is clear from context; by default in this book it shall be $\mathbb{C}$.

By a similar argument as the usual category for representations of finite groups, $\text{Rep}(G)$ is an abelian category. When working with finite groups, we saw that representations can be thought of as $\mathbb{C}[G]$ -modules. With the smoothness condition imposed, there are more elements in the $\mathbb{C}[G]$ -module V than smooth representations; we shall need to limit to *Hecke algebras* as shall be discussed in section 1.4.

Let us next upgrade the standard decomposition result from finite group representation theory:

Proposition 1.2.11: Semisimple Equivalence

Let G be a locally profinite group, and let (π, V) be a smooth representation of G . The following conditions are equivalent:

1. V is the sum of its irreducible G -subspaces;
2. V is the direct sum of a family of irreducible G -subspaces;
3. any G -subspace of V has a G -complement in V .

Proof:

(1) $\Rightarrow$ (2). Take the set of irreducible G -subspaces U_i of V

$$\{U_i : i \in I\}$$

such that $V = \sum_{i \in I} U_i$. Consider the set $\mathcal{I}$ of subsets J of I such that the sum $\sum_{i \in J} U_i$ is direct. The set $\mathcal{I}$ is non-empty; Indeed, for an induction argument suppose we have a totally ordered set $\{J_a : a \in A\}$ of elements of $\mathcal{I}$. Put $J = \bigcup_{a \in A} J_a$. If the sum $\sum_{j \in J} U_j$ is not direct, there is a finite subset S of J for which $\sum_{j \in S} U_j$ is not direct. Since S must be contained in some J_a , this is impossible. Therefore $J \in \mathcal{I}$. We can now apply Zorn's Lemma to get a maximal element J_0 of $\mathcal{I}$. For this set, we have

$$V = \bigoplus_{a \in J_0} U_i,$$

as required for (2).

(2) $\Rightarrow$ (3). In (3), let W be a G -subspace of V . By (2), we can assume that $V = \bigoplus_{i \in I} U_i$, for a family (U_i) of irreducible G -subspaces of V . We consider the set $\mathcal{J}$ of subsets J of I for which $W \cap \sum_{i \in J} U_i = 0$. Again, the set $\mathcal{J}$ is nonempty and inductively ordered by inclusion. If J is a maximal element of $\mathcal{J}$, the sum $X = W + \bigoplus_{j \in J} U_j$ is direct. If $X \neq V$, there is $i \in I$ with $U_i \not\subset X$, so the sum $X + U_i$ is direct, and $J \cup \{i\} \in \mathcal{J}$, contrary to hypothesis. Thus (2) $\Rightarrow$ (3).

(3) $\Rightarrow$ (1) Let V_0 be the sum of all irreducible G -subspaces of V and write $V = V_0 \oplus W$, for some

G -subspace W of V . Assume for a contradiction that $W \neq 0$. By its definition, the space W has no irreducible G -subspace. However, there is a non-zero G -subspace W_1 of W which is finitely generated over G . By Zorn's Lemma, W_1 has a maximal G -subspace W_0 , and then W_1/W_0 is irreducible. We have $V = V_0 \oplus W_0 \oplus U$, for some G -subspace U of V , and hence a G -projection $V \rightarrow U$. The image of W_1 in U is an irreducible G -subspace of U , hence an irreducible subspace of V which is not contained in V_0 . This is nonsense, so $V = V_0$ and (3) $\Rightarrow$ (1).

Definition 1.2.12: Semisimple Representation

A representation (π, V) is G -semisimple if it satisfies one of the equivalent conditions of proposition 1.2.11.

Unlike finite groups, locally profinite groups will have (many) non-semisimple complex representations:

Example 1.2.13: Reducible non-semisimple representation

The details will be left for the reader. Let $\chi_1(x) = |x|_p$ and $\chi_2(x) = |x|_p^{-1}$ and take the principal series representation $I(\chi_1, \chi_2)$. This representation has a unique 1-dimensional subrepresentation which is a character of $\mathrm{GL}_2(\mathbb{Q}_p)$; denote it χ . The quotient of $I(\chi_1, \chi_2)$ by this representation is called the *Steinberg representation* and is denoted St . This fits into a short exact sequence:

$$0 \rightarrow \chi \rightarrow I(\chi_1, \chi_2) \rightarrow \mathrm{St} \rightarrow 0$$

However, this sequence *does not split*. Thus, it is *reducible* but *indecomposable*, hence *not semisimple*.

The compact opens subgroups shall fair better:

Lemma 1.2.14: Compact Open Semistable Representation

Let G be a locally profinite group, and let K be a compact open subgroup of G . Let (π, V) be a smooth representation of G . The space V is the sum of its irreducible K -subspaces:

$$V = \sum_{\substack{W \subseteq V \\ \text{irred. } K\text{-subspace}}} W$$

We say that V is K -semisimple.

Proof:

Let $v \in V$. As in example 1.2.8, v is fixed by an open normal subgroup K' of K , and it generates a finite-dimensional K -space W on which K' acts trivially. Thus W is effectively a finite-dimensional representation of the finite group K/K' and so is the sum of its irreducible K -subspaces. Since $v \in V$ was chosen at random, the lemma follows.

Thus, the following is a meaningful definition:

Definition 1.2.15: Isotypic Component

Let G be a locally profinite group and $K \subseteq G$ a compact open subgroup. Let $\widehat{K}$ denote the set of equivalence classes of irreducible representations. Let $\rho \in \widehat{K}$ and (π, V) be a smooth representation of G . Then:

$$V^\rho = \sum_{\substack{W \subseteq V \\ K\text{-subspace} \\ \text{of class } \rho}} W$$

is called the ρ -isotypic component of V .

By definition, V^K is the isotypic subspace for the class of trivial representation 1 of K .

Proposition 1.2.16: K -isotypic Decomposition

Let (π, V) be a smooth representation of G and let K be a compact open subgroup of G .

1. The space V is the direct sum of its K -isotypic components:

$$V = \bigoplus_{\rho \in \widehat{K}} V^\rho.$$

2. Let (σ, W) be a smooth representation of G . For any G -homomorphism $f : V \rightarrow W$ and $\rho \in \widehat{K}$, we have

$$f(V^\rho) \subset W^\rho \text{ and } W^\rho \cap f(V) = f(V^\rho).$$

Proof:

By proposition 1.2.11:

$$V = \bigoplus_{i \in I} U_i,$$

for a family of irreducible K -subspaces U_i of V . We let $U(\rho)$ be the sum of those U_i of class ρ . We then have

$$V = \bigoplus_{\rho \in \widehat{K}} U(\rho).$$

If W is an irreducible K -subspace of V of class ρ , then $W \subset U(\rho)$; otherwise, there would be a non-zero K -homomorphism $W \rightarrow U_i$, for some U_i of some class $\tau \neq \rho$. We deduce that $V^\rho = U(\rho)$ and (1) follows.

In (2), the image of V^ρ is a sum of irreducible K -subspaces of W , all of class ρ and therefore contained in W^ρ . Moreover, $f(V)$ is the sum of the images $f(V^\tau)$, $\tau \in \widehat{K}$, and $f(V^\tau) \subset W^\tau$. Since the sum of the W^τ is direct, $f(V)$ is the direct sum of the $f(V^\tau)$ and the second assertion follows.

We frequently use part (2) of the Proposition in the following context:

Corollary 1.2.17: Exactness reduced to compact subgroup

Let $a : U \rightarrow V$, $b : V \rightarrow W$ be G -homomorphisms between smooth representations U, V, W of G . The sequence

$$U \xrightarrow{a} V \xrightarrow{b} W$$

is exact if and only if

$$U^K \xrightarrow{a} V^K \xrightarrow{b} W^K$$

is exact, for every compact open subgroup K of G .

Proof:

exercise.

If H is a subgroup of G , we define

$$V(H) = \text{the linear span of } \{v - \pi(h)v : v \in V, h \in H\}.$$

In particular, $V(H)$ is an H -subspace of V .

Corollary 1.2.18: Splitting Using Compact Subgroups

Let G be a locally profinite group, and let (π, V) be a smooth representation of G . Let K be a compact open subgroup of G . Then

$$V(K) = \bigoplus_{\substack{\rho \in \hat{K} \\ \rho \neq 1}} V^\rho, \quad V = V^K \oplus V(K),$$

and $V(K)$ is the unique K -complement of V^K in V .

Proof:

The sum $W = \bigoplus V^\rho$, with ρ not trivial, is a K -complement of V^K in V . There is therefore a K -surjection $V \rightarrow V^K$ with kernel W . Clearly, $V(K)$ is contained in the kernel of any K -homomorphism $V \rightarrow V^K$; we conclude that W contains $V(K)$. On the other hand, if U is an irreducible K -space of class $\rho \neq 1$, then $U(K) = U$, so $V^\rho \subset V(K)$.

The next important concept is that of induced representation and how we can construct representations using them. Recall that in the finite case a subgroup $H \subseteq G$ which has a representation (W, σ) will have a natural induced representation $(\mathbb{C}[G]_{\mathbb{C}[H]}W, \text{Ind}_H^G \sigma)$ where the space $\mathbb{C}[G]_{\mathbb{C}[H]}W$ can be represented as H -equivariant functions and Ind_H^G acts on the set of these functions. The following generalizes this:

Definition 1.2.19: Induced Representation

Let G be a locally profinite group, and let $H \subseteq G$ be a closed subgroup of G so that H is also locally profinite. Let (σ, W) be a smooth representation of H .

Define the space X of functions $f : G \rightarrow W$ which satisfy:

1. equivariant: $f(hg) = \sigma(h)f(g)$, $h \in H$, $g \in G$
2. smoothness: there is a compact open subgroup K of G (depending on f) such that $f(gx) = f(g)$, for $g \in G$, $x \in K$ (so f is smooth i.e. continuous with W having the discrete topology)

Then define a homomorphism $\Sigma : G \rightarrow \text{Aut}_{\mathbb{C}}(X)$ by

$$\Sigma(g)f : x \mapsto f(xg), \quad g, x \in G.$$

The pair (Σ, X) provides a smooth representation of G and is said to be *smoothly induced by* σ . It is usually denoted

$$(\Sigma, X) = \text{Ind}_H^G \sigma.$$

Proposition 1.2.20: Induced Representation is A Functor

The map $\sigma \mapsto \text{Ind}_H^G \sigma$ gives a functor $\text{Rep}(H) \rightarrow \text{Rep}(G)$.

Proof:

exercise

There is a canonical H -homomorphism

$$\alpha_\sigma : \text{Ind}_H^G \sigma \longrightarrow W, \quad f \mapsto f(1).$$

The pair $(\text{Ind}_H^G \sigma, \alpha_\sigma)$ has the following fundamental property:

Theorem 1.2.21: Frobenius Reciprocity

Let H be a closed subgroup of a locally profinite group G . For a smooth representation (σ, W) of H and a smooth representation (π, V) of G , the canonical map

$$\begin{aligned} \text{Hom}_G(\pi, \text{Ind}_H^G \sigma) &\longrightarrow \text{Hom}_H(\pi, \sigma) \\ \varphi &\mapsto \alpha_\sigma \circ \varphi, \end{aligned}$$

is an isomorphism that is functorial in both variables π, σ .

Proof:

Let $f : V \rightarrow W$ be an H -homomorphism. We define a G -homomorphism $f_* : V \rightarrow \text{Ind}_H^G \sigma$ by letting $f_*(v)$ be the function $g \mapsto f(\pi(g)v)$. The map $f \mapsto f_*$ is then the inverse of the above map.

The following shows how this is a very useful functor:

Proposition 1.2.22: Induced Representation is Additive and Exact

The functor $\text{Ind}_H^G : \text{Rep}(H) \rightarrow \text{Rep}(G)$ is additive and exact.

need to re-phrase this!

Proof:

For a smooth representation (σ, W) of H , temporarily let $I(\sigma)$ denote the space of functions $G \rightarrow W$ satisfying the first condition $f(hg) = \sigma(h)f(g)$ of the definition above. Thus I is a functor to the category of abstract representations of G ; it is clearly additive and exact, while $\text{Ind}_H^G(\sigma) = I(\sigma)^\infty$. Thus Ind_H^G is surely additive, and 2.3 Exercise (3) shows it to be left-exact.

To prove it is right-exact, let $(\sigma, W), (\tau, U)$ be smooth representations of H and let $f : W \rightarrow U$ be an H -surjection. Take $\varphi \in I(\tau)^\infty$, and choose a compact open subgroup K of G which fixes φ . The support of φ is a union of cosets HgK , and the value $\varphi(g) \in U$ must be fixed by $\tau(H \cap gKg^{-1})$. By 2.3 Corollary 1 (applied to the group H and the trivial representation of its compact open subgroup $H \cap gKg^{-1}$), there exists $w_g \in W$, fixed by $\sigma(H \cap gKg^{-1})$, such that $f(w_g) = \varphi(g)$. We define a function $\Phi : G \rightarrow W$ to have the same support as φ and $\Phi(hgk) = \sigma(h)w_g$, for each $g \in H \backslash \text{supp} \varphi / K$. The function Φ is fixed by K , and hence lies in $I(\sigma)^\infty$. Its image in $I(\tau)^\infty$ is φ , as required.

Example 1.2.23: Induced Representation

1. here
2. here

As the compact open subgroups are the key subgroups to consider for this book, there is the following useful variation of induced representation.

Definition 1.2.24: Compact Induction

Given (σ, W) for a representation of closed $H \subseteq G$ and similarly defined X as in definition 1.2.19, define:

$$X_c = \{f \in X : \text{compactly supported modulo } H\}$$

that is the image of $\text{supp}(f)$ in $H \backslash G$ is compact (equivalently, $\text{supp}(f) \subseteq HC$ for some compact set $C \subseteq G$). The induced functor is denoted $\text{c-Ind}_H^G \sigma$.

Proposition 1.2.25: Compact Induction is Well-defined

The space X_c along with $\text{c-Ind}_H^G \sigma$ is a smooth representation

Proof:
exercise.

Proposition 1.2.26: Compact Induction Repreises Additive and Exact

The functor $\text{c-Ind}_H^G : \text{Rep}(H) \rightarrow \text{Rep}(G)$ is additive and exact.

Proof:
exercise.

Example 1.2.27: Compact Induction

1. here
2. here

There is a natural G -embedding:

$$\text{c-Ind}_H^G(\sigma) \rightarrow \text{Ind}_H^G(\sigma) \quad (\text{i.e. a natural transformation } \text{c-Ind}_H^G \Rightarrow \text{Ind}_H^G)$$

This map is an isomorphism if and only if $H \setminus G$ is compact. [1] says that there are examples of such a map being an isomorphism even when $H \setminus G$ is not compact with the right choice of G, H, σ .

The map is mostly interesting when $H \subseteq G$ is open in which case there is a natural H -homomorphism:

$$\alpha_\sigma^c : W \rightarrow \text{c-Ind}(\sigma) \quad w \mapsto f_w$$

go over the Frobenius for compact; skipping for now because I have too little intuition

1.2.28 Hypothesis From Now On For any compact open $K \subseteq G$, G/K is countable.

If G/K is countable, so is G/K' for any other compact open; for $K \cap K'$ is compact open and a finite index of K thus

$$\frac{G}{K \cap K'} \rightarrow G/K$$

has finite fibers, thus $G/K \cap K'$ and G/K' are both countable. All groups that will be considered in this book are have this property, but a few further results bellow don't require them. The main point is so that:

Lemma 1.2.29: Irreducible Smooth Representation At Most Countable

Let (π, V) be a smooth irreducible representation of a locally profinite group G . Then $\dim_{\mathbb{C}}(V)$ is countable

Proof:

Let $0 \neq v \in V$ and choose $K \subseteq G$ such that $v \in V^K$. As V is irreducible,

$$\{\pi(g) \cdot v : g \in G/K\}$$

spans V , and this set is countable completing the proof.

This also allows for the following:

Theorem 1.2.30: Schur's Lemma For Smooth Representations

Let (π, V) be a smooth irreducible representation of a locally profinite group G . Then

$$\text{End}_{\mathbb{C}}(V) = \mathbb{C}$$

Proof:

Let $0 \neq \varphi \in \text{End}_{\mathbb{C}}(V)$. As the image and kernel of φ must be G -subspaces, by irreducibility it must be bijective (which is enough in this category for invertible). Thus, $\text{End}_{\mathbb{C}}(V)$ is at least a division $\mathbb{C}$ -algebra.

Next, $\text{End}_{\mathbb{C}}(V)$ must have countable dimension for if $0 \neq v \in V$, the G -translate of v spans V , and so $\varphi \in \text{End}_{\mathbb{C}}(V)$ is uniquely determined by the value of $\varphi(v)$. Now, if $\varphi \in \text{End}_{\mathbb{C}}(V)$ where $\varphi \notin \mathbb{C}$, φ would be transcendental over $\mathbb{C}$ and hence the subset

$$\{(\varphi - a)^{-1} : a \in \mathbb{C}\}$$

would be linearly independent, and hence $\text{End}_{\mathbb{C}}(V)$ would have uncountable $\mathbb{C}$ -dimension, a contradiction. Thus, $\text{End}_{\mathbb{C}}(V) \cong \mathbb{C}$, as we sought to show.

Note that this is an if and only if when working with finite groups. If G is compact, this becomes an if and only if. This is false in general ([1] cites section 9.10).

Corollary 1.2.31: Central Character

Let (π, V) be an irreducible smooth representation of G and $Z = Z(G)$ is the center of G . Then Z acts on V via the character

$$\omega_{\pi} : Z \rightarrow \mathbb{C}^* \quad \text{i.e.} \quad \pi(z) \cdot v = \omega_{\pi}(z) \cdot v$$

for all $v \in V$ and $z \in Z$

Proof:

As $z \in G$, the left-multiplication map is an endomorphism, hence $\pi(z) \in \text{End}_G(V) \cong \mathbb{C}$. Thus $\pi(z)$ is some scalar multiple of id_V , hence $\pi(z) = \omega_{\pi}(z)$. To show it's a character, note that if K is a compact open subgroup of G such that $V^K \neq 0$, Then ω_{π} is trivial on the compact open subgroup $K \cap Z$, showing it is indeed a smooth representation, and hence ω_{π} is a character of Z .

Definition 1.2.32: Central Character

The function $Z \rightarrow \mathbb{C}^*$ defined in corollary 1.2.31 is called a *central character*.

Definition 1.2.33: Smooth Representation Admitting Central Character

Let (π, V) be a smooth representation of G and χ a central character of $Z(G)$. Then we say that (π, V) admits χ as a central character if:

$$\pi(z) \cdot v = \chi(z) \cdot v \quad \forall v \in V, z \in Z$$

Proposition 1.2.34: Irreducible Smooth Representation of Abelian Groups

Let G be an abelian locally profinite group. Then an smooth irreducible representation is one-dimensional

Proof:

By corollary 1.2.31, any representation π will act like the character at every element $g \in G$ since $Z(G) = G$. Now, if W is any one-dimensional subspace, as π acts by scaling, W is a G -subspace. Hence, any irreducible subspace must be one-dimensional, completing the proof.

Proposition 1.2.35: KZ Space

Let (π, V) be a smooth representation of G , admitting χ as a central character. Let K be an open subgroup of G such that KZ/Z is compact. Then:

1. Let $v \in V$. The KZ-space spanned by v is of finite dimension, and is a sum of irreducible KZ-spaces.
2. As representation of KZ , the space V is semisimple.

Proof:

The vector v is fixed by a compact open subgroup K_0 of K . The set KZ/K_0Z is finite, so the space W spanned by $\pi(KZ)v$ has finite dimension. Then W is K_0Z -semisimple. Since v was chosen arbitrarily, we get 2 as we sought to show.

The final elementary concept to adders is the dual space and how a representation (π, V) relates to it:

Definition 1.2.36: Smooth Dual (Contragradient)

here

this lemma is given next, may move it somewhere better. It is useful for the following discussion about smooth duals.

Lemma 1.2.37: Semisimple With Finite-index and Induction

Let G be a locally profinite group, and let H be an open subgroup of G of finite index.

1. If (π, V) is a smooth representation of G , then V is G -semisimple if and only if it is H -semisimple.
2. Let (σ, W) be a semisimple smooth representation of H . The induced representation $\text{Ind}_H^G \sigma$ is G -semisimple.

Proof:

Suppose that V is H -semisimple, and let U be a G -subspace of V . By hypothesis, there is an H -subspace W of V such that $V = U \oplus W$. Let $f : V \rightarrow U$ be the H -projection with kernel W . Consider the map

$$f^G : v \mapsto (G : H)^{-1} \sum_{g \in G/H} \pi(g) f(\pi(g)^{-1} v), \quad v \in V.$$

The definition is independent of the choice of coset representatives and it follows that f^G is a G -projection $V \rightarrow U$. We then have $V = U \oplus \text{Ker } f^G$ and $\text{Ker } f^G$ is a G -subspace of V . Thus V is G -semisimple (cf. 2.2 Proposition).

Conversely, suppose that V is G -semisimple. Thus G is a direct sum of irreducible G -subspaces (2.2), and it is enough to treat the case where V is irreducible over G . As representation of H , the space V is finitely generated and so admits an irreducible H -quotient U . Suppose for the moment that H is a normal subgroup of G . By Frobenius Reciprocity (2.4.2), the H -map $V \rightarrow U$ gives a non-trivial, hence injective, G -map $V \rightarrow \text{Ind}_H^G U$. As representation of H , the induced representation $\text{Ind}_H^G U = \bigoplus_{g \in G/H} gU$ is a direct sum of G -conjugates of U (cf. 2.5 Lemma). These are all irreducible over H , so $\text{Ind } U$ is H -semisimple. Proposition 2.2 then implies that $V \subset \text{Ind } U$ is H -semisimple.

In general, we set $H_0 = \bigcap_{g \in G/H} gHg^{-1}$. This is an open normal subgroup of G of finite index. We have just shown that the G -space V is H_0 -semisimple; the first part of the proof shows it is H -semisimple.

This completes the proof of (1), and (2) follows readily from the same arguments.

Proposition 1.2.38: Smooth Dual

here

1.3 Measures and Duality

The space $\mathbb{C}[G]$ is $\mathbb{C}$ -algebra isomorphic to $(\text{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ where $*$ is convolution. This requires summing over each $g \in G$. In this book, G is a.e. infinite, and so we replace summation with an appropriate measure on G .

Let G be a locally profinite group, $C_c^\infty(G)$ the space of functions $f : G \rightarrow \mathbb{C}$ which are locally constant and of compact support (i.e. smooth and of compact support). Let $f \in C_c^\infty(G)$. Then compact support and smoothness (locally constant), here exists open subgroup $K_1, K_2 \subseteq G$ such that

$$f(k_1g) = f(g) = f(gk_2) \quad \forall g \in G, k_i \in K_i$$

By letting $K = K_1 \cap K_2$, f must be a finite linear combination of characteristic functions of double cosets KgK . Now, $G \curvearrowright C_c^\infty(G)$ by left translation and right translation:

$$\begin{cases} \lambda : g \mapsto (\lambda_g f : x \mapsto f(g^{-1}x)) \\ \rho : g \mapsto (\rho_g f : x \mapsto f(xg)) \end{cases} \quad x, g \in G, f \in C_c^\infty(G)$$

Note that the left action is the one investing since the action is on functions. To see this, apply the operator λ_{g_2} to the function f . This produces a new function:

$$h = \lambda_{g_2} f$$

The definition of this new function h is:

$$h(y) = (\lambda_{g_2} f)(y) = f(g_2 y)$$

Now we apply the operator λ_{g_1} to h ; call the new function k :

$$k = \lambda_{g_1} h = \lambda_{g_1} (\lambda_{g_2} f) \quad \text{so} \quad k(z) = (\lambda_{g_1} h)(z) = h(g_1 z)$$

therefore:

$$k(z) = f(g_2(g_1 z)) = f(g_2 g_1 z)$$

showing why the left input needs to be the one inverted. The reader should check that $(C_c^\infty(G), \lambda)$ and $(C_c^\infty(G), \rho)$ are both smooth G -representation.

Definition 1.3.1: Haar Integral

A *right Haar integral* on G is a non-zero linear functional:

$$I : C_c^\infty(G) \rightarrow \mathbb{C}$$

such that

1. $I(\rho_g f) = I(f)$ for all $g \in G, f \in C_c^\infty(G)$
2. For all $f \in C_c^\infty(G)$ where $f \geq 0$, $I(f) \geq 0$.

And similarly for the left Haar integral.

Proposition 1.3.2: Existence of Haar Integral

There exists right Haar integral $I : C_c^\infty(G) \rightarrow \mathbb{C}$. A linear functional $I' : C_c^\infty(G) \rightarrow \mathbb{C}$ is a right Haar integral if and only if $I' = cI$ for some $c > 0$.

The same then applies to the left Haar integral.

some parts of this proof reference results I did not write down yet; I'll put the label from the textbook for now.

Proof:

Let K be a compact open subgroup of G . Denote by ${}^K C_c^\infty(G)$ the space of functions in $C_c^\infty(G)$ fixed by $\lambda(K)$. Then viewing ${}^K C_c^\infty(G)$ as G -space via right translation, it is then identical to the representation of G compactly induced from the trivial representation 1_K of K :

$${}^K C_c^\infty(G) = c\text{-Ind}_K^G 1_K.$$

Now, viewing $\mathbb{C}$ as the trivial G -space (and as a consequence K -space), by compact Frobenius reciprocity:

$$\dim_{\mathbb{C}} \text{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = \dim_{\mathbb{C}} \text{Hom}_K(\mathbb{C}, \mathbb{C}) = 1.$$

Next, let's show there exists a non-zero element $I_K \in \text{Hom}_G({}^K C_c^\infty(G), \mathbb{C})$ such that $I_K(f) \geq 0$ whenever $f \geq 0$, and if h_K is the characteristic function of K , then $I_K(h_K) > 0$. Indeed, for $g \in G$, let f_g denote the characteristic function of Kg . The set of functions f_g , $g \in K \backslash G$, then forms a $\mathbb{C}$ -basis of the space ${}^K C_c^\infty(G)$ (2.5 Lemma). The functional $I_K : f_g \mapsto 1$ has the required properties, noting that $h_K = f_1$.

Now, choose a descending sequence $\{K_n\}_{n \geq 1}$ of compact open subgroups K_n of G such that $\bigcap_n K_n = 1$. Then:

$$C_c^\infty(G) = \bigcup_{n \geq 1} {}^{K_n} C_c^\infty(G).$$

For each $n \geq 1$, there is a unique right G -invariant functional I_n on ${}^{K_n} C_c^\infty(G)$ which maps the characteristic function of K_n to $(K_1 : K_n)^{-1}$. We have $I_{n+1}|_{{}^{K_n} C_c^\infty(G)} = I_n$, and so the family $\{I_n\}$ gives a functional on $C_c^\infty(G)$ of the required kind. The uniqueness statement is immediate, completing the proof.

In the above, think $K = \mathbb{Z}_p$ so that $f_g \rightarrow 1$ so $\mu(\mathbb{Z}_p) = 1$. The rest of the compact sets are given by the translation invariance and taking the index $[H : K]$ when $K \subseteq H$. From this we get $\mu(a + p^n \mathbb{Z}_p) = \mu(p^n \mathbb{Z}_p) = p^{-n}$. This extends to define a left Haar integral on $\mathbb{Q}_p$. If we take a finite extension $F/\mathbb{Q}_p$, simply replace p with $\mathfrak{p}$ in the above. If we take $\text{GL}_k(\mathbb{Q}_p)$, recall the chain of compact open subsets:

$$K_m = \{g \in K_0 \mid g \equiv I \pmod{p^m}\} = I + p^m M_k(\mathbb{Z}_p)$$

where $K_0 = \text{GL}_k(\mathbb{Z}_p)$ has measure 1, and you do the same process.

Corollary 1.3.3: Left-Right Haar Integral

For $f \in C_c^\infty(G)$, define $\check{f} \in C_c^\infty(G)$ by $\check{f}(g) = f(g^{-1})$, $g \in G$. The functional

$$I' : C_c^\infty(G) \longrightarrow \mathbb{C},$$

$$I'(f) = I(\check{f}),$$

is a left Haar integral on G . Moreover, any left Haar integral on G is of the form cI' , for some $c > 0$.

Proof:

immediate.

From this, we can define a measure called the *Haar measure*. Let I be a left Haar integral on G and $\emptyset \neq S \subseteq G$ a compact open subset of a locally profinite group.

$$\mu_G(S) = I(\Gamma_S)$$

where Γ_S is the characteristic function of S . Then $\mu_G(S) > 0$ and μ_G is left-translation invariant:

$$\mu_G(gS) = \mu_G(S) \quad \forall g \in G$$

The reader should check that this is indeed a measure (see [8] for a review of measure theory)

Definition 1.3.4: Haar Measure

The measure μ_G defined above is called the *Haar measure*.

The function I is often written as:

$$I(f) = \int_G f(g) d\mu_G(g) \quad f \in C_c^\infty(G)$$

with the usual abbreviation of:

$$\int_S f(x) d\mu_G(x) := \int_G \Gamma_S(x) f(x) d\mu_G(x)$$

Definition 1.3.5: Unimodular Group

Let G be a locally profinite group (more generally a topological group with a Haar measure defined on it). Then G is *unimodular* if and only if the left Haar integral and is a right Haar integral.

Example 1.3.6: Left and Right Haar Measures

1. put the standard ones and interesting ones here, including the ones alluded earlier.

This is well-defined when working with functions with compact support. However, there shall be times when we have non-compact induction $\text{Ind}_H^G \sigma$ where $H \backslash G$ is not compact (in this case it is often countable). To that end, we extend the domain of integration of the Haar measure. Consider $f : G \rightarrow \mathbb{C}$ that is invariant under left translation by a compact subgroup $K \subseteq G$, and let μ_G be a left Haar measure on G . Then if the series:

$$\sum_{g \in K \backslash G} \int_{Kg} |f(x)| d\mu_G(x)$$

converges, so does the non absolute values version of the series, and hence we can define:

$$\int_G f(x) d\mu_G(x) = \sum_{g \in K \backslash G} \int_{Kg} f(x) d\mu_G(x)$$

Notice we are taking right cosets because the integral is left-invariant, and hence would have been a sum of constants and hence necessarily diverge. Furthermore, as μ_G is translation invariant this series is independent of choice of open compact group as we can take $K = K_1 \cap K_2$ which has finite index with both K_1 and K_2 .

Next, we shall often construct characters using tensor products necessitating extending a Haar measure. Let G_1, G_2 be locally profinite groups, and set $G = G_1 \times G_2$. Then G is locally profinite. An element $\sum_{1 \leq i \leq r} f_i^1 \otimes f_i^2$ of the tensor product $C_c^\infty(G_1) \otimes C_c^\infty(G_2)$ gives a function on G by

$$\Phi(g_1, g_2) = \sum_i f_i^1(g_1) f_i^2(g_2).$$

Then $\Phi \in C_c^\infty(G)$. Importantly, this gives an isomorphism

$$C_c^\infty(G_1) \otimes C_c^\infty(G_2) \cong C_c^\infty(G)$$

Let μ_j be a left Haar measure on G_j , $j = 1, 2$. Then there is a unique left Haar measure μ_G on G such that

$$\int_G f_1 \otimes f_2(g) d\mu_G(g) = \int_{G_1} f_1(g_1) d\mu_1(g_1) \int_{G_2} f_2(g_2) d\mu_2(g_2),$$

for $f_i \in C_c^\infty(G_i)$. One writes $\mu_G = \mu_1 \otimes \mu_2$.

For a general $f \in C_c^\infty(G)$, the function

$$f_1(g_1) = \int_{G_2} f(g_1, g_2) d\mu_2(g_2)$$

lies in $C_c^\infty(G_1)$. We have

$$\int_G f(g) d\mu_G(g) = \int_{G_1} f_1(g_1) d\mu_1(g_1),$$

and symmetrically: if f is of the form $f_1 \otimes f_2$, this is obvious, and such functions span $C_c^\infty(G)$.

Next, we naturally work over irreducible representations and hence smooth functions of the form $f : G \rightarrow V$ that need to be integrated. Let V be a complex vector space, and consider the space $C_c^\infty(G; V)$ of locally constant, compactly supported functions $f : G \rightarrow V$. This space is isomorphic to $C_c^\infty(G) \otimes V$ in the obvious way: a tensor $\sum_i f_i \otimes v_i$ gives the function

$$g \mapsto \sum_i f_i(g) v_i.$$

If μ_G is a left Haar measure on G , there is a unique linear map $I_V : C_c^\infty(G; V) \rightarrow V$ such that

$$I_V(f \otimes v) = \int_G f(g) d\mu_G(g) \cdot v.$$

and so define:

$$I_V(\varphi) := \int_G \varphi(g) d\mu_G(g), \quad \varphi \in C_c^\infty(G; V).$$

This has the same invariance properties as the Haar integral on scalar-valued functions.

Next, we shall have cases where the left and right Haar measure don't agree (recall example 1.3.6). By the uniqueness of the left/right Haar measure up to a constant, we can cleverly equate the left/right Haar integration up to a constant. Let μ_G be a left Haar measure on G . For $g \in G$, consider the functional

$$\begin{aligned} I_g : C_c^\infty(G) &\rightarrow \mathbb{C}, \\ f &\mapsto \int_G f(xg) d\mu_G(x). \end{aligned}$$

Then I_g is a left Haar integral on G , so there is a unique $\delta_G(g) \in \mathbb{R}_+^\times$ such that

$$\delta_G(g) \int_G f(xg) d\mu_G(x) = \int_G f(x) d\mu_G(x),$$

for all $f \in C_c^\infty(G)$. The function δ_G is a homomorphism $G \rightarrow \mathbb{R}_+^\times$. It is trivial if G is abelian; more generally, δ_G is trivial if and only if G is unimodular. If f is the characteristic function of a compact open subgroup K of G , then δ_G is trivial on K and hence a character of G .

Definition 1.3.7: Haar Module of G

The function δ_G as define above is the *module* of G .

Note that if G is compact, then $\delta_G = 1$ and G is unimodular. In the general case, any character $G \rightarrow \mathbb{R}_+^*$ is trivial on compact subgroups, since $\mathbb{R}_+^*$ has only the trivial compact subgroup.

Using the module, the functional

$$f \mapsto \int_G \delta_G(x)^{-1} f(x) d\mu_G(x), \quad f \in C_c^\infty(G),$$

is a right Haar integral on G . The mnemonic

$$d\mu_G(xg) = \delta_G(g) d\mu_G(x)$$

would help on how the left/right measures “commute”.

Next, many important irreducible representations shall be G -subspaces of simpler induced representations, very often by taking a character on a closed subgroup and inducing it. Let H be a closed subgroup of G , with module δ_H . Let $\theta : H \rightarrow \mathbb{C}^\times$ be a character of H . Consider the space of functions $f : G \rightarrow \mathbb{C}$ which are G -smooth under right translation, compactly supported modulo H , and satisfy the “induction” homogeneity property:

$$f(hg) = \theta(h)f(g), \quad h \in H, g \in G.$$

Definition 1.3.8: Induction Space of Smooth Functions

Label this space

$$C_c^\infty(H \backslash G, \theta)$$

and view it as a smooth G -space via right translation ρ .

Note that $C_c^\infty(H \backslash G, \theta) = \text{c-Ind}_H^G \theta$.

Theorem 1.3.9: Compact Induction on Haar Measure

Let $\theta : H \rightarrow \mathbb{C}^\times$ be a character of H . The following are equivalent:

1. There exists a non-zero linear functional $I_\theta : C_c^\infty(H \backslash G, \theta) \rightarrow \mathbb{C}$ such that $I_\theta(\rho_g f) = I_\theta(f)$, for all $g \in G$.
2. $\theta \delta_H = \delta_G|_H$.

When these conditions hold, the functional I_θ is uniquely determined up to constant factor.

Proof:

(1) $\Rightarrow$ (2) Let μ_G, μ_H be left Haar measures on G, H respectively. Define a G -map

$$C_c^\infty(G) \rightarrow C_c^\infty(H \backslash G, \theta) \quad f \mapsto \tilde{f} \quad \text{where} \quad \tilde{f}(g) = \int_H \theta \delta_H(h)^{-1} f(hg) d\mu_H(h).$$

This map satisfies

$$\widetilde{\lambda_k f} = \delta_H \theta(k)^{-1} \tilde{f},$$

for $k \in H$ and $f \in C_c^\infty(G)$. If this map is surjective, we shall be able to factor I_ω through this map, allowing to link the two moduli.

To that end, If K is a compact open subgroup of G , the space $C_c^\infty(G)^K$ by a simple generalization of proposition 2.2.6 this space is spanned by the characteristic functions of cosets $gK, g \in G/K$. On the other hand, each coset HgK supports, at most, a one-dimensional space of functions in $C_c^\infty(H \backslash G, \theta)^K$. These subspaces span $C_c^\infty(H \backslash G, \theta)^K$, so the map $f \mapsto \tilde{f}$ is surjective on K -fixed functions, proving surjectivity.

Suppose that the space $C_c^\infty(H \backslash G, \theta)$ admits a functional I_θ of the required kind. The map $f \mapsto I_\theta(\tilde{f})$ is then a non-trivial G -homomorphism

$$(C_c^\infty(G), \rho) \rightarrow \mathbb{C}.$$

However, the space $\text{Hom}_G(C_c^\infty(G), \mathbb{C})$ has dimension 1 by Frobenius Reciprocity:

$$\dim_{\mathbb{C}} \text{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = \dim_{\mathbb{C}} \text{Hom}_K(\mathbb{C}, \mathbb{C}) = 1.$$

and is spanned by a right Haar integral. Therefore, the function I_θ exists if and only if the right Haar integral $C_c^\infty(G) \rightarrow \mathbb{C}$ factors through the quotient map $C_c^\infty(G) \rightarrow C_c^\infty(H \backslash G, \theta)$. When it does hold, I_θ is determined up to a constant factor.

Now, the kernel of the map $f \mapsto \tilde{f}$ contains all functions $\lambda_h f - \delta_H \theta(h)^{-1} f$, for $h \in H$ and $f \in C_c^\infty(G)$. Applying the right Haar integral on G to such functions, we get

$$\begin{aligned} \int_G (\lambda_h f(g) - \delta_H \theta(h^{-1}) f(g)) \delta_G(g)^{-1} d\mu_G(g) \\ = (\delta_G(h^{-1}) - \delta_H \theta(h^{-1})) \int_G f(g) \delta_G(g)^{-1} d\mu_G(g). \end{aligned}$$

This vanishes identically if and only if $\delta_G(h^{-1}) = \delta_H \theta(h^{-1})$, for all $h \in H$. Thus (1) $\Rightarrow$ (2).

- (2) $\Rightarrow$ (1) Take a function $f \in C_c^\infty(G)$ such that $\tilde{f} = 0$. The function f is fixed by a compact open subgroup K , and it is enough to treat the case where $\text{supp } f \subset HgK$, for some $g \in K$. Thus f is a finite linear combination of the characteristic function of cosets $h_i g K$, $h_i \in H$. The condition $\tilde{f} = 0$ then amounts to

$$\mu_H(H \cap gKg^{-1}) \sum_i \theta \delta_H(h_i)^{-1} f(h_i g) = 0,$$

since $\theta \delta_H$ is trivial on the compact subgroup $H \cap gKg^{-1}$ of H . On the other hand,

$$\int_G f(x) \delta_G(x)^{-1} d\mu_G(x) = \mu_G(K) \delta_G(g)^{-1} \sum_i \delta_G(h_i)^{-1} f(h_i g).$$

If (2) holds, this will vanish, giving a well-defined non-zero functional I_θ , as we sought to show.

When the conditions of the proposition hold, the character θ takes only positive real values. Let $f \in C_c^\infty(G)$ satisfy $f(g) \geq 0$, for all $g \in G$; we then have $\tilde{f}(g) \geq 0$ for all g . Consequently:

Corollary 1.3.10: Existence of Compact Induction on Haar Measure

Suppose that the conditions of theorem 1.3.9 hold. There is then a non-zero linear functional I_θ on $C_c^\infty(H \backslash G, \theta)$ such that:

1. $I_\theta(\rho_g f) = I_\theta(f)$, for $f \in C_c^\infty(H \backslash G, \theta)$, $g \in G$;
2. $I_\theta(f) \geq 0$, for $f \in C_c^\infty(H \backslash G, \theta)$, $f \geq 0$.

These conditions determine I_θ uniquely, up to a positive constant factor, and we write:

$$I_\theta(f) = \int_{H \backslash G} f(g) d\mu_{H \backslash G}(g), \quad f \in C_c^\infty(H \backslash G, \theta),$$

Definition 1.3.11: Semi-invariant Measure

The measure $\mu_{H \backslash G}$ is called a positive *semi-invariant measure* on $H \backslash G$.

Since $\theta = \delta_H^{-1} \delta_G|_H$ is uniquely determined, there is no real need to refer to it again, and so it shall be taken as implicit.

This allows for taking the dual of a compact induced smooth representation:

Theorem 1.3.12: Duality Theorem

Let $\dot{\mu}$ be a positive semi-invariant measure on $H \backslash G$. Let (σ, W) be a smooth representation of H . Then there is a natural isomorphism:

$$(\text{c-Ind}_H^G \sigma)^\vee \cong \text{Ind}_H^G \delta_{H \backslash G} \otimes \check{\sigma},$$

depending only on the choice of $\dot{\mu}$.

This proof is transcribed. I will put it in my own words asap!

Proof:

We view $\delta_{H \backslash G} \otimes \check{\sigma}$ as acting on the same space $\check{W}$ as $\check{\sigma}$. Let $(\check{w}, w) \mapsto \langle \check{w}, w \rangle$ be the evaluation pairing $\check{W} \times W \rightarrow \mathbb{C}$. Let $\varphi \in \text{c-Ind } \sigma$, $\Phi \in \text{Ind } \delta_{H \backslash G} \otimes \check{\sigma}$, and consider the function

$$f : g \mapsto \langle \Phi(g), \varphi(g) \rangle, \quad g \in G.$$

This lies in $C_c^\infty(H \backslash G, \delta_{H \backslash G})$, so we have a G -invariant pairing

$$\begin{aligned} & \text{Ind } \delta_{H \backslash G} \otimes \check{\sigma} \times \text{c-Ind } \sigma \rightarrow \mathbb{C}, \\ & (\Phi, \varphi) \mapsto \int_{H \backslash G} \langle \Phi(g), \varphi(g) \rangle d\dot{\mu}(g). \end{aligned}$$

This induces a G -homomorphism $\text{Ind } \delta_{H \backslash G} \otimes \check{\sigma} \rightarrow (\text{c-Ind } \sigma)^\vee$. It is clearly natural in σ ; we show it is an isomorphism.

To do this, we take a compact open subgroup K of G and describe the space $(\text{c-Ind } \sigma)^K$ of K -fixed functions in $\text{c-Ind } \sigma$. The support of such a function, f say, is a finite union of cosets HgK . The value $f(g) \in W$ is then fixed by the compact open subgroup $H \cap gKg^{-1}$ of H . We choose a set $\mathcal{G}$ of representatives for the coset space $H \backslash G/K$. For each $g \in \mathcal{G}$, we choose a basis $\mathcal{W}_g$ of the space $W^{H \cap gKg^{-1}}$. For each $g \in \mathcal{G}$ and each $w \in \mathcal{W}_g$, there is a unique function $f_{g,w} \in (\text{c-Ind } \sigma)^K$, supported on HgK , such that $f_{g,w}(g) = w$. The set

$$\{f_{g,w} : g \in \mathcal{G}, w \in \mathcal{W}_g\}$$

is then a basis of $(\text{c-Ind } \sigma)^K$. That is, the space $(\text{c-Ind } \sigma)^K$ consists of the finite linear combinations of functions $f_{g,w}$, $g \in \mathcal{G}$, $w \in \mathcal{W}_g$.

Let $\mathcal{W}_g^*$ denote the basis of $\check{W}_g = \text{Hom}_{\mathbb{C}}(W^{H \cap gKg^{-1}}, \mathbb{C})$ dual to $\mathcal{W}_g$. For $g \in \mathcal{G}$, $\check{w} \in \mathcal{W}_g^*$, we define $f_{g,\check{w}} \in (\text{Ind } \delta_{H \backslash G} \otimes \check{\sigma})^K$ in the same way as before. We then have that the space $(\text{Ind } \delta_{H \backslash G} \otimes \check{\sigma})^K$ consists of all functions f such that, for any $g \in \mathcal{G}$, the restriction $f|_{HgK}$ is a finite linear combination of functions $f_{g,\check{w}}$, $\check{w} \in \mathcal{W}_g^*$.

For $g_1, g_2 \in \mathcal{G}$, $w \in \mathcal{W}_{g_1}$, $\check{w} \in \mathcal{W}_{g_2}^*$, we have

$$\langle f_{g_2,\check{w}}, f_{g_1,w} \rangle = \begin{cases} \dot{\mu}(Hg_1K) \langle \check{w}, w \rangle & \text{if } Hg_1K = Hg_2K, \\ 0 & \text{otherwise,} \end{cases}$$

for some $\dot{\mu}(Hg_1K) > 0$. The pairing thus identifies $(\text{Ind } \delta_{H \backslash G} \otimes \check{\sigma})^K$ with the linear dual of $(\text{c-Ind } \sigma)^K$, and hence $\text{Ind } \delta_{H \backslash G} \otimes \check{\sigma}$ with $(\text{c-Ind } \sigma)^\vee$, as required.

1.4 Hecke Algebra

Linear representations of finite groups are in bijective correspondence to the modules over the group algebra $\mathbb{C}[G]$. A similar conditions holds for smooth representations of locally profinite groups given the suitable generalization of $\mathbb{C}[G]$ to Hecke algebras.

Though not necessary, to avoid technicalities that don't add insight, throughout this entire section the locally profinite group G will be assumed to be unimodular.

Fix a Haar measure μ on G . Then for any $f_1, f_2 \in C_c^\infty(G)$, define the convolution:

$$f_1 * f_2(g) = \int_G f_1(x) f_2(x^{-1}g) d\mu(x)$$

This gives a function $(x,) \mapsto f_1(x) f_2(x^{-1}g) \in C_c^\infty(G \times G)$ and so by [ref:HERE](#) $f_1 * f_2 \in C_c^\infty(G)$. This operation is associative: Here's the equation in an align* environment:

$$\begin{aligned} f_1 * (f_2 * f_3)(g) &= \int \int f_1(x) f_2(y) f_3(y^{-1}x^{-1}g) d\mu(y) d\mu(x) \\ &= \int \int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(y) d\mu(x) \\ &= \int \int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(x) d\mu(y) \\ &= (f_1 * f_2) * f_3(g) \end{aligned}$$

Definition 1.4.1: Hecke Algebra

Let G be a unimodular locally profinite group. Then:

$$\mathcal{H}(G) = (C_c^\infty(G), *)$$

is an associative $\mathbb{C}$ -algebra called the *Hecke algebra* of G .

This is an example of a *non-unital* algebra if G is not finite. If G is commutative, so is $\mathcal{H}(G)$. Note that if μ, ν are two choices of Haar measures giving rise to $\mathcal{H}_\mu(G)$ and $\mathcal{H}_\nu(G)$ on $C_c^\infty(G)$, then there exists a constant $c > 0$ such that $\nu = c\mu$ and a map $f \mapsto c^{-1}f$, and hence

$$\mathcal{H}_\mu(G) \cong \mathcal{H}_\nu(G)$$

[The actual correspondence is proven here, going to write out the details later.](#)

Goal: An $\mathcal{H}(G)$ -module is smooth if $\mathcal{H}(G) * M = M$. If M is a smooth $\mathcal{H}(G)$ -module, then there exists a homomorphism $\pi : G \rightarrow \text{Aut}_{\mathbb{C}}(M)$ such that (π, M) is a smooth representation. A vice-versa.

Smooth Representations of GL_2 : General Structure and Principal Series

Having established the general framework of smooth representations, we now narrow our focus to the group $G = GL_2(F)$ for a non-Archimedean field F . This group can be thought of as the group being able to represent the “2D” information of the (local) absolute galois group of F . The goal of this chapter is to classify the “continuous spectrum” of representations of G , known as the *Principal Series*. These are the representations that can be constructed by parabolic induction from characters of the diagonal torus. In doing so, we will also identify the “discrete spectrum” – the Cuspidal representations – by what they are not.

We begin by looking at the representation theory of $GL_2(\mathbb{F}_q)$ for finite fields, which serves as a crucial finite model for the p -adic case. We then analyze the geometry of $GL_2(F)$ through its various decompositions (Iwasawa, Cartan, Bruhat). A key tool in our analysis will be the *Mirabolic subgroup* M (Section 2.3), whose representation theory is simple enough to serve as a lever for understanding G . Finally, using the Jacquet module (Section 2.4), we provide a complete classification of the principal series, determining when they are irreducible and how they decompose when reducible (leading to the Steinberg representation).

2.1 Important Background: Representation of $GL_2(\mathbb{F}_p)$

When analyzing $GL_2(\mathbb{Q}_p)$ and related groups, we shall draw inspiration of this special case. We shall first single out the following subgroups:

Definition 2.1.1: Matrix Subgroups

Let F be (an arbitrary) field. Then define the groups:

1. (Borel subgroup) $B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
2. (unipotent subgroup) $N = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(F) \right\} \cong (F, +)$
3. (torus subgroup) $T = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
4. (center) $Z = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in \mathrm{GL}_2(F) \right\} \cong F^*$

Note the Borel subgroup is not normal and its normalizer is G .

Lemma 2.1.2: Bruhat Decomposition

Let $G = \mathrm{GL}_2(F)$. Then its *Bruhat decomposition* is:

$$G = B \cup BwB$$

where

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Hence, $\{1, w\}$ is a set of representative of the coset space $B \backslash G / B$.

Furthermore, $B = NT = TN$, hence $B \cong N \rtimes T$

Proof:

Let $M \in \mathrm{GL}_2(F)$ so that $\det(M) = ad - bc \neq 0$ if $c = 0$, $ad \neq 0$, so $a, d \neq 0$. Then $M \in B$.

If $c \neq 0$, choose two arbitrary $B_1, B_2 \in B$, calculate $B_1 w B_2$, use the fact that $\det(B_1), \det(B_2) \neq 0$ to get an equation for each entry of M . Note that the bottom-left entry *must* be nonzero and hence $B \cap BwB = \emptyset$, proving the decomposition into disjoint cosets.

For the $B = NT = TN$, notice that:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & b/a \\ 0 & 1 \end{pmatrix} \in TN$$

It can also be factored as:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} 1 & b/d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \in NT$$

Showing they are a semidirect products completing the proof.

By constructing, any representative of the coset BwB has a nonzero entry in the $(2, 1)$ -position. As $B = TN = NT$ and w normalizes T ($wTw = T$):

$$BwB = NwB = BwN$$

Furthermore, the map:

$$B \times N \rightarrow BwN \quad (b, n) \mapsto bwn$$

is *bijective*.

Next, For any $M \in GL_2(F)$, define $\text{ch}_M(t) = \det(tI - M) = t^2 - \text{tr}(M)t + \det(M)$, that is $\text{ch}_M(t)$ is the characteristic polynomial of M , i.e. a monic quadratic F -polynomial and $\text{ch}_M(0) = \det(M) \neq 0$.

Proposition 2.1.3: Character Polynomial For $GL_2(F)$

Let F be an arbitrary field, $M \in GL_2(F)$, and $f(t) = \text{ch}_M(t) = \det(tI - M)$. Then:

1. If $f(t)$ is irreducible, the F -algebra $F[M] \subseteq M_2(F)$ is a field with $GL_2(F)$ -centralizer $F[M]^*$. Given $f(t) = t^2 + at + b$, then M is $GL_2(F)$ -conjugate to:

$$\begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}$$

Furthermore, an element $N \in GL_2(F)$ is GL_2 -conjugate to M if and only if $\text{ch}_N(t) = f(t)$.

2. If $f(t)$ has distinct roots $a, b \in F^*$, then M is $GL_2(F)$ -conjugate to:

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

3. If $f(t)$ has a repeated root $a \in F^*$, Then M is $GL_2(F)$ -conjugate to either:

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$$

In the first case, M lies in the center of $GL_2(F)$, and in the 2nd the $GL_2(F)$ -centralizer of M is $F[M]^*$.

Proof:

We shall do one and leave the rest as an exercise.

1. As $f(t)$ is irreducible, by Cayley-Hamilton $F[M] \cong F[t]/(f(t))$ where the characteristic polynomial $f(t)$ is also the minimal polynomial. For the conjugacy result, as M is irreducible it has no eigenvalues and so $\{v, Mv\}$ forms a basis for any $v \in F^2$. Finding the matrix representation of the linear transformation corresponding to M shall give the desired result
2. exercise
3. exercise

Now let $F = \mathbb{F}_q$ be a field with $q = p^l$ elements for some $l \in \mathbb{N}$, and let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Let us find the irreducible complex representations of G . First off:

Lemma 2.1.4: Order and Irreducible Representations of G

Let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Then:

$$|G| = (q^2 - 1)(q^2 - q)$$

Furthermore, G has $q^2 - 1$ conjugacy classes.

Thus, there are $q^2 - 1$ irreducible representations.

Proof:

Let the first column be any nonzero vector, giving $q^2 - 1$ possibilities. The second column cannot be a scalar multiple of the first. Given a choice v_1 of the first column, the second cannot be kv_1 for all $k \in \mathbb{F}_q$. This gives $q^2 - q$ possibilities. These choices are linearly independent, thus:

$$|G| = (q^2 - 1)(q^2 - q)$$

For the conjugacy classes we repeatedly use proposition 2.1.3. By the first case, as $\mathrm{GL}_2(\mathbb{F}_q)$ has exactly $(q^2 - q)/2$ irreducible monic polynomials of degree 2, there are $(q^2 - q)/2$ conjugacy classes. The second case will give $(q - 1)(q - 2)/2$ cases, and the third $2(q - 1)$ cases. Summing these up gives the desired result.

Thus, there are $q^2 - 1$ irreducible representations. As G is finite, we may take advantage of character theory, hence we shall construct some characters. First, note that $N \cong (\mathbb{F}_q, +)$ via $x \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$.

Fix a nontrivial character of $(\mathbb{F}_q, +)$ χ (ex. $\chi(x) = e^{2\pi i(x)/p}$). Then $(a\chi)(x) = \chi(ax)$ ranges over all characters of $(\mathbb{F}_q, +)$. From this, we can define a natural action of conjugation by T (or B) on N . Namely the action of T on N produces an action on the characters:

$$(t\chi)(n) = \chi(t^{-1}nt)$$

when interpreting n as a matrix in N via the isomorphism $N \cong (F_q, +)$ given above. Certainly all non-trivial characters are in the orbit of T , hence the conjugacy classes split into the single trivial character and all other ones.

2.1.5 Non-trivial Characters of N Therefore, all non-linear character characters on N are similar up to conjugation.

The trivial one shall be important in a moment in finding our first irreducible representation.

We first define the following: let χ_1, χ_2 be characters of $\mathbb{F}_q^*$. Define the character:

$$\chi = \chi_1 \otimes \chi_2 : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mapsto \chi_1(a)\chi_2(b)$$

This is a character of T . We may regard this as a character on B that is trivial on N via the quotient:

$$B \rightarrow B/N \cong T$$

This process is given a name:

Definition 2.1.6: Inflation

Let $H \trianglelefteq G$ be a normal subgroup and (π, V) a representation of G/H . Then the *inflation* of π from H to G is a new $\tilde{\pi} : G \rightarrow V$ such that $\tilde{\pi}(gH) = \pi(\bar{g})$; that is, it is trivial on the cosets of G/H .

After inflating from T to B , we may form the induced representation of $GL_2(\mathbb{F}_q) = G$

$$\text{Ind}_B^G \chi$$

and consider its irreducible components:

Lemma 2.1.7: Irreducible Components of G

Let π be an irreducible representation of G . Then the following are equivalent:

- $(\Rightarrow)$ π is equivalent to a G -subspace of $\text{Ind}_B^G \chi$ for a character χ of T seen as a character on B trivial on N .
- $(\Rightarrow)$ π contains the trivial character of N .

Proof:

First, π contains the trivial character of N if and only if it contains an irreducible representation σ of B containing to the trivial character of N . Indeed, restrict G to B , $\text{Res}_B^G \pi$ and then restrict further to N

$$\text{Res}_N^B(\text{Res}_B^G \pi)$$

By Maschke's theorem breakdown $\text{Res}_B^G \pi$ into irreducibles. Call this irreducible representation of B σ . As Res is additive over direct sums, we see the above if and only if holds. Now as we just proved above this is equivalent to σ being an inflation of a character T . By Frobenius reciprocity:

$$\text{Hom}_G(\text{Ind}_B^G \chi, \pi) \cong \text{Hom}_B(\chi, \text{Res}_B^G \pi)$$

As π is irreducible we can apply the same ideas as Schur's Lemma to finish the proof.

So, if π contains a trivial character of N , it is irreducible! By the lemma, we analyze the induced representations $\text{Ind}_B^G \chi$. To start, define the character of T :

$$\chi^w : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mapsto \chi_2(a)\chi_1(b)$$

where w is the permutation in G swapping columns. By abusing notation, we can say this is a character if B that is trivial on N .

Proposition 2.1.8: Analyzing Inflation of Torus Character

Let χ, ξ be character of T viewed as characters of B trivial on N . Then:

1. The space $\text{Hom}_G(\text{Ind}_B^G(\chi), \text{Ind}_B^G(\xi))$ is trivial, unless $\chi = \xi$ and $\chi = \xi^w$.
2. The space

$$\text{Hom}_G(\text{Ind}_B^G(\chi), \text{Ind}_B^G(\chi)) \quad \text{and} \quad \text{Hom}_G(\text{Ind}_B^G(\chi), \text{Ind}_B^G(\chi^w))$$

have the same dimension: it is 2 if $\chi = \chi^w$ and 1 otherwise.

Proof:

We use the canonical isomorphism of Frobenius Reciprocity

$$\text{Hom}_G(\text{Ind}_B^G(\chi), \text{Ind}_B^G(\xi)) \cong \text{Hom}_B(\text{Res}_B^G(\text{Ind}_B^G(\chi)), \xi) \quad (2.1)$$

The restriction-induction formula of Mackey gives

$$\text{Res}_B^G \text{Ind}_B^G(\chi) \cong \sum_{y \in B \backslash G/B} \text{Ind}_{B \cap B_y}^B \left(\text{Res}_{B_y \cap B}^{B_y \cap B}(\chi^y) \right), \quad (2.2)$$

where χ^y denotes the character $b \mapsto \chi(yby^{-1})$ of the group $B^y = y^{-1}By$. By Bruhat decomposition we only have to consider the cases $y = 1$ and $y = w$. The term in equation (2.2) corresponding to $y = 1$ is just χ , and so contributes a factor $\text{Hom}_T(\chi, \xi)$ to equation (2.1). The term corresponding to w contributes

$$\text{Hom}_B(\text{Ind}_T^B(\chi^w), \xi) \cong \text{Hom}_T(\chi^w, \xi).$$

Altogether, equation (2.1) reduces to

$$\text{Hom}_B(\text{Ind}_B^G(\chi), \text{Ind}_B^G(\xi)) \cong \text{Hom}_T(\chi, \xi) \oplus \text{Hom}_T(\chi^w, \xi),$$

and the result follow from Schur's lemma, completing the proof.

Corollary 2.1.9: Irreducible Representations Given Torus Character

Let χ be a character of T viewed as a character of B trivial on N . Then:

1. The representation $\text{Ind}_B^G \chi$ is irreducible if and only if $\chi \neq \chi^w$
2. If $\chi = \chi^w$, then $\text{Ind}_B^G \chi$ has length 2 with distinct component factors

Proof:

Apply Schur's lemma to the above proposition.

Example 2.1.10: Induced Character of Trivial Character

Consider the trivial character 1_B of B . The trivial character 1_G occurs in $\text{Ind}_B^G 1_B$, so

$$\text{Ind}_B^G 1_B = 1_G \oplus \text{St}_G,$$

for a unique irreducible representation St_G , called the Steinberg representation. Clearly, $\dim \text{St}_G = q$ since $[G : B] = q + 1$ and $\dim 1_G = 1$.

These irreducible representations are given a name:

Definition 2.1.11: Principle Series Representations

The representation Ind_B^G of a character χ of T is called the *principal series*. The term principle series is often extended to include the irreducible components, namely all irreducible representations of G containing the trivial character of N .

Corollary 2.1.12: Counting Principal Series

Up to isomorphism, the group $GL_2(F)$ has precisely $\frac{1}{2}(q^2 + q) - 1$ irreducible representations which contain the trivial character of N .

Proof:

This is a counting argument: if $\chi = \chi^w$, there are $2(q - 1)$ such characters (2 for the length 2). If they are not equal there are $\frac{1}{2} \cdot [(q - 1)^2 - (q - 1)]$ (where we half since $\text{Ind}_B^G \chi_1 \otimes \chi_2 = \text{Ind}_B^G \chi_2 \otimes \chi_1$). Adding them gives the desired result.

Notice this number is less than $q^2 - 1$, meaning not all irreducible representations are of this form. In particular there are $\frac{1}{2}(q^2 - q)$ such irreducible representations. An irreducible representation need not be trivial on N as we shall be given a name:

Definition 2.1.13: Cuspidal Representation

An irreducible representation of $GL_2(F)$ not containing the trivial character of N is called a *cuspidal representation*.

A cuspidal representation σ must contain some non-trivial character of N . As it is non-trivial on N , any two non-trivial characters of N are B -conjugate (see remark 2.1.5), so σ contains all non-trivial characters of N . This adds complexity, in particular the cuspidal representations cannot be constructed directly by induction.

To to that end, let $L/\mathbb{F}_q$ be a quadratic extension so that it has q^2 elements and L^* has $q^2 - 1$ elements. Let $F = \mathbb{F}_q$ to simply notation. The characters θ of L^* correspond to the $(q^2 - 1)$ primitive roots of unity, of which there are $q^2 - 1 - 1 = q^2 - 2$. A character is called regular if it is *not* invariant under the Frobenius automorphism, that is $\theta(x^q) \neq \theta(x)$. A character $\theta_j(g^k) = \omega^{jk}$ is *not regular* if j divides $q + 1$, as can be quickly verified. There are $(q^2 - 1)/(q + 1) = q - 1$ non-regular characters, and hence $q(q - 1)$ regular characters. Choose a basis of L so that $L \cong_F F$ and $GL_2(F)$ is associated with $\text{Aut}_F(L)$. There is a natural action L^* on L , which corresponds to a subgroup of

$\mathrm{GL}_2(F) \cong \mathrm{Aut}_F(L)$. Identify:

$$L^* \cong E \subseteq \mathrm{GL}_2(F)$$

Importantly, note that the elements of G with irreducible characteristic polynomial is conjugate to an element of E . For, if $0 \neq \lambda \in L$, then the corresponding matrix in $\mathrm{GL}_2(F)$ must have λ as its eigenvalue, and as it must be irreducible the other root must be the galois conjugate λ^q , but this shows the correspondence.

Now, let θ be a regular character of E and ψ a non-trivial character of N . Define θ_ψ of ZN as:

$$\theta_\psi : \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} u \mapsto \theta(a)\psi(u)$$

Note that $\mathrm{Ind}_{ZN}^G \theta_\psi$ is independent of choice of ψ up to equivalence (see remark 2.1.5).

Theorem 2.1.14: Cuspidal Representation of Finite Matrix Group

Let θ be a regular character of E and ψ a non-trivial character of N .

1. The virtual representation

$$\pi_\theta = \mathrm{Ind}_{ZN}^G \theta_\psi - \mathrm{Ind}_E^G \theta$$

is an irreducible cuspidal representation of G of dimension $q - 1$.

2. Let θ_1, θ_2 be regular characters of E ; then $\pi_{\theta_1} \cong \pi_{\theta_2}$ if and only if $\theta_2 = \theta_1$ or $\theta_2 = \theta_1^q$.
3. Every irreducible cuspidal representation of G is of the form π_θ , for some regular character θ of E .

Proof:

1. As we are working with finite groups, representations are characterized by characters. The character of the virtual representation π_θ is given by the difference of the characters of the two induced representations:

$$\mathrm{tr} \pi_\theta(g) = \mathrm{tr}(\mathrm{Ind}_{ZN}^G \theta_\psi)(g) - \mathrm{tr}(\mathrm{Ind}_E^G \theta)(g)$$

We use the general formula for the character of an induced representation, namely if (σ, W) is a representation of a subgroup $H \leq G$, then the character of $\Pi = \mathrm{Ind}_H^G \sigma$ is:

$$\mathrm{tr}(\Pi(g)) = \frac{1}{|H|} \sum_{x \in G \text{ s.t. } x^{-1}gx \in H} \sigma(x^{-1}gx)$$

We compute this for each part and for each type of conjugacy class in $G = \mathrm{GL}_2(k)$. First take $\Pi_1 = \mathrm{Ind}_{ZN}^G \theta_\psi$ so $H = ZN = B$ with $|H| = (q - 1)q$. The character is $\sigma_1 = \theta_\psi$.

- Let $g = z \in Z$. Since z is central, $x^{-1}zx = z$ for all $x \in G$. The element z is in ZN , so the condition is always met.

$$\mathrm{tr}(\Pi_1(z)) = \frac{1}{(q - 1)q} \sum_{x \in G} \theta_\psi(z) = \frac{|G|}{(q - 1)q} \theta(z) = \frac{(q^2 - 1)(q^2 - q)}{(q - 1)q} \theta(z) = (q^2 - 1)\theta(z)$$

- Let $g = zu$ with $z \in Z, u \in N, u \neq 1$. An element with a repeated eigenvalue is only conjugate to other elements with repeated eigenvalues. The conjugates of g that lie in ZN are only those of the form $z'u'$. A more detailed analysis of the character sum shows that the sum over the unipotent parts averages to -1 when paired against a non-trivial character ψ . This yields:

$$\text{tr}(\Pi_1(zu)) = -\theta(z)$$

- Let $g = y \in E \setminus Z$. The characteristic polynomial of y is irreducible over k . Every element of ZN has its eigenvalues in k , so its characteristic polynomial is reducible. Thus, no conjugate of y can be in ZN . The sum in the character formula is empty.

$$\text{tr}(\Pi_1(y)) = 0$$

Next, consider $\Pi_2 = \text{Ind}_E^G \theta$. Here $H = E^\times$ is a subgroup of order $q^2 - 1$. The character is $\sigma_2 = \theta$.

- Let $g = z \in Z$. Since $z \in F^\times \subset E^\times$, the condition $x^{-1}zx \in E^\times$ is always met. Next, consider $\Pi_2 = \text{Ind}_E^G \theta$. Here $H = E^\times$, a subgroup of order $q^2 - 1$. The character is $\sigma_2 = \theta$.

$$\text{tr}(\Pi_2(z)) = \frac{1}{q^2 - 1} \sum_{x \in G} \theta(z) = \frac{|G|}{q^2 - 1} \theta(z) = (q^2 - q)\theta(z)$$

- Let $g = zu$ with $u \neq 1$. The element g has a repeated eigenvalue. No conjugate of g can have an irreducible characteristic polynomial, so no conjugate of g lies in $E^\times \setminus Z$. Conjugation into Z is impossible for $u \neq 1$. The sum is empty.

$$\text{tr}(\Pi_2(zu)) = 0$$

- Let $g = y \in E \setminus Z$. The centralizer of y in G is $E^\times$. The conjugacy class of y intersects $E^\times$ in exactly two elements: y and its Galois conjugate y^q . The formula for the character of an induced representation on an element $h \in H$ simplifies in this case to $\sum_i \theta(g_i h g_i^{-1})$ where g_i are representatives for $H \backslash G / H$. For GL_2 , this gives:

$$\text{tr}(\Pi_2(y)) = \theta(y) + \theta(y^q) = \theta(y) + \theta^q(y)$$

We now subtract the character values to get:

- For $g = z \in Z$:

$$\text{tr}\pi_\theta(z) = (q^2 - 1)\theta(z) - (q^2 - q)\theta(z) = (q - 1)\theta(z)$$

- For $g = zu, u \neq 1$:

$$\text{tr}\pi_\theta(zu) = -\theta(z) - 0 = -\theta(z)$$

- For $g = y \in E \setminus Z$:

$$\text{tr}\pi_\theta(y) = 0 - (\theta(y) + \theta^q(y)) = -(\theta(y) + \theta^q(y))$$

This completes the derivation of the character:

Conjugacy Class Representative g	Character Value $\mathrm{tr}\pi_\theta(g)$
$z \in Z$	$(q-1)\theta(z)$
zu where $z \in Z, u \in N, u \neq 1$	$-\theta(z)$
$y \in E \setminus Z$	$-(\theta(y) + \theta^q(y))$
g not conjugate to an element of $ZN \cup E$	0

Now, we can take the inner product of the character with itself to get:

$$\frac{1}{|G|} \sum_{g \in G} |\mathrm{tr}\pi_\theta(g)|^2 = 1$$

Now, recall that a representation V is irreducible if given the basis of characters and the inner product with itself, we get 1. In this case, we may have gotten a virtual character, that is the coefficient was actually negative. This is easily checkable not to be the case by computing that

$$\mathrm{tr}\pi_\theta(e) = q - 1 = \dim(V)$$

If it were virtual, this would be negative, showing it is indeed a character and furthermore π_τ is an irreducible representation. Next, as

$$\sum_{u \in N} |\mathrm{tr}\pi_\theta(u)| = 0$$

we see that $\pi_\theta|_N$ does not contain the trivial character of N , and hence π_θ is cuspidal.

- Two representations are isomorphic if they have the same character. It is easy to see from the character table above that $\pi_{\theta_1} \cong \pi_{\theta_2}$ if and only if $\theta_2 = \theta_1$ or $\theta_2 = \theta_1^q$.
- As E^* has $q^2 - q$ characters and the group F has $q - 1$ we get a total $q^2 - q - (q - 1) = q^2 - q$ number of regular characters given of this form, which combining with corollary 2.1.12 gives us the total of $q^2 - 1$ irreducible representations, and hence we found all irreducible representations by lemma 2.1.4, completing the proof.

The book presents this interesting relation between characters and determinants as an exercise that I want to put down because it's pretty cool:

Example 2.1.15: Character and Determinants

Let ψ be a non-trivial character of N , and consider the representation

$$F = \mathrm{Ind}_N^G \psi.$$

σ be an irreducible representation of G . Show that:

- If $\sigma = \varphi \circ \det$, for a character φ of $k^\times$, then σ does not occur in F .
- Otherwise, σ occurs in F with multiplicity one.

2.2 Linear Groups over Local Fields

Given the additional structure, we have some more interesting decompositions. We have B , N , T , and Z as before as these are defined for any field F . Note that $B/N \cong T$ and $B \cong T \ltimes N$ are homeomorphisms.

Proposition 2.2.1: Iwasawa Decomposition

Let B be the Borel subgroup and $K = GL_2(\mathfrak{o})$. Then:

$$G = BK$$

Proof:

Take $g \in G$; if the $(2,1)$ -entry of g is zero, then $g \in B$. Otherwise, post-multiplying by the permutation matrix $w \in K$ if necessary, we can assume $v_F(g_{21}) \geq v_F(g_{22})$. We can then post-multiply by a lower triangular matrix in K to achieve $g_{21} = 0$.

Corollary 2.2.2: Compact Quotient

The quotient space

$$B \backslash G$$

is compact

Proof:

$B \backslash G$ is the continuous image of the compact subgroup $K = GL_2(\mathfrak{o})$.

Proposition 2.2.3: Cartan Decomposition

Let ϖ be a prime element of F . Then the matrices:

$$\begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix} \quad a, b \in \mathbb{Z}, \quad a \leq b$$

form a set of representatives for the coset space $K \backslash G / K$.

Proof:

Permuting rows and columns, using the permutation matrix in K , we can arrange for the largest entry of g (in absolute value) to be in the $(1,1)$ -place. Multiplying by elementary matrices from K , we can then arrange for g to be diagonal, and unit factors can be absorbed into K . This gives

$$G = \bigcup_{a \leq b} K \begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix} K.$$

We have to prove this union is disjoint. That is, we have to recover the integers a, b from the coset KgK , where

$$g = \begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix}.$$

First, we have $a + b = \nu_F(\det h)$, for any $h \in KgK$. Next, the group index $(K : K \cap hKh^{-1})$ depends only on the coset KhK , and

$$(K : K \cap gKg^{-1}) = 1$$

if $b = a$, or

$$(K : K \cap gKg^{-1}) = (q + 1)q^{b-a-1}$$

if $b > a$, as we sought to show.

Corollary 2.2.4: Countable Quotient

If K is a compact open subgroup of G , the set G/K is countable.

Proof:

As observed under box 1.2.28, it is enough to show that G/K is countable for one choice of K : we take $K = GL_2(\mathfrak{o})$. The space $K \backslash G/K$ is certainly countable, and each double coset KgK contains only finitely many cosets $g'K$.

We next define the Iwahori decomposition. First, the standard Iwahori subgroup of $GL_2(F)$ is the compact open subgroup $I \leq G$:

$$I = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, d \in U_F, b \in \mathfrak{o}, c \in \mathfrak{p} \right\}$$

Let $N' = N^w$ denote the subgroup of lower triangular unipotent matrices of G . Then:

Proposition 2.2.5: Iwahori Decomposition

If I is the Iwahori subgroup, then:

$$I = (I \cap N')(I \cap T)(I \cap N)$$

More precisely, the product map

$$I \cap N' \times I \cap T \times I \cap N \longrightarrow I$$

is bijective, and a homeomorphism, for any ordering of the factors on the left hand side.

Proof:

exercise.

This is useful for the following. Let $K = GL_2(\mathfrak{o})$; under the canonical surjection $K \rightarrow GL_2(k)$, the image of I is the standard Borel subgroup of $GL_2(k)$. The Bruhat decomposition for $GL_2(k)$ implies

$$K = I \cup IwI.$$

Combining this with the Iwasawa decomposition for G , we obtain the more symmetric double-coset decomposition

$$G = BI \cup BwI = B(I \cap N') \cup Bw(I \cap N)$$

The cosets BwI, BI are both open in G .

Furthermore, by letting $L = \mathfrak{o} \oplus \mathfrak{o}, L' = \mathfrak{o} \oplus \pi$, the Iwahori subgroup I is then the common G -stabilizer of the two lattices L, L' .

Haar Measures

A reminder that F is a nonarchemidian field.

Proposition 2.2.6: Smooth Functions Over F

The vector-space $C_c^\infty(F)$ is spanned by indicator functions (characteristic functions) of sets $a + \mathfrak{p}^n$ for $a \in F$ and $n \in \mathbb{Z}$

Proof:

Certainly the indicator function of $a + \mathfrak{p}^m$ lies in $C_c^\infty(F)$. Conversely, let $\varphi \in C_c^\infty(F)$. Since φ has compact support, there exists $n \in \mathbb{Z}$ such that $\text{supp } \varphi \subset \mathfrak{p}^n$. Also, φ is fixed under translation by a compact open subgroup of F , hence by $\mathfrak{p}^m$, for some $m \in \mathbb{Z}$. Thus φ is a linear combination of characteristic functions of sets $a + \mathfrak{p}^m, a \in \mathfrak{p}^n/\mathfrak{p}^m$.

Let us now define the Haar integral on our desired groups. Let μ be the Haar measure on F (automatically left and right as F is abelian), and let Φ_0 denote the indicator function (or characteristic function) on $\mathfrak{o}$. Then as $\mathfrak{o}$ is compact:

$$\int_F \Phi_0(x) d\mu(x) = c_0 \quad c \in \mathbb{R}_{>0}$$

If Φ_b is the indicator function on $a + \mathfrak{p}^b$ for $a \in F$ and $b \in \mathbb{Z}$, by translation invariance:

$$\int_F \Phi_b(x) d\mu(x) = c_0 q^{-b}$$

And as indicator functions span $C_c^\infty(F)$, we can use this to integrate any function. this integral has the following “homogeneity” property: if $\Phi \in C_c^\infty(F)$ and $y \in F^*$, then by the above equation we deduce:

$$\int_F \Phi(xy) d\mu(x) = \|y\|^{-1} \int_F \Phi(x) d\mu(x)$$

where $\|y\| = q^{\nu_F(y)}$. Thus, define the Haar measure μ^* on F^* given by:

$$\mu^*(x) = \frac{\mu(x)}{\|x\|}$$

Then if $\Phi \in C_c^\infty(F^*)$, the function $x \mapsto \|x\|^{-1} \Phi(x)$ lies in $C_c^\infty(F)$ (and vanishes at 0), so:

$$\int_{F^\times} \Phi(x) d\mu^\times(x) = \int_F \Phi(x) \|x\|^{-1} d\mu(x), \quad \text{for } \Phi \in C_c^\infty(F^\times).$$

and hence this certainly defines a Haar integral on F^* .

Now, viewing $M_2(F)$ under addition, we can define its Haar measure by interpreting it as the product of four copies of F , with the measure given by the tensor product of 4 copies of a Haar measure on F . Let $G = GL_2(F)$. Then:

Proposition 2.2.7: Haar measure on $GL_2(F)$

Let μ be a Haar measure on $M_2(F)$. For $\Phi \in C_c^\infty(GL_2(F))$, the function $x \mapsto \Phi(x) \|\det x\|^{-2}$ (vanishing on $M_2(F) \setminus GL_2(F)$) lies in $C_c^\infty(M_2(F))$. The functional

$$\Phi \mapsto \int_A \Phi(x) \|\det x\|^{-2} d\mu(x), \quad \Phi \in C_c^\infty(G),$$

is a left and right Haar integral on $GL_2(F)$, i.e. $GL_2(F)$ is unimodular.

Proof:

Let $A = M_2(F)$ and $G = GL_2(F)$. Take $g \in G$ and consider the functionals

$$\Phi \mapsto \begin{cases} \int_A \Phi(gx) d\mu(x), \\ \int_A \Phi(xg) d\mu(x), \end{cases} \quad \Phi \in C_c^\infty(A).$$

Each is a Haar integral on A and differs from the initial one by a positive constant (depending on g). To evaluate this constant, take Φ to be the indicator function of $\mathfrak{m} = M_2(\mathfrak{o})$. In the first instance, the function $x \mapsto \Phi(gx)$ is the characteristic function of the lattice $\mathfrak{m}' = g^{-1}\mathfrak{m}$. Thus

$$\int_A \Phi(gx) d\mu(x) = \mu(\mathfrak{m}') = \mu(\mathfrak{m}) \frac{[\mathfrak{m}' : \mathfrak{m} \cap \mathfrak{m}']}{[\mathfrak{m} : \mathfrak{m} \cap \mathfrak{m}']}$$

This quotient of indices depends only on the double coset KgK , $K = GL_2(\mathfrak{o})$. Taking g in diagonal form the Cartan decomposition (proposition 2.2.3), we get

$$\int_A \Phi(gx) d\mu(x) = \|\det g\|^{-2} \int_A \Phi(x) d\mu(x).$$

The second instance is treated in the same way to get

$$\int_A \Phi(xg) d\mu(x) = \|\det g\|^{-2} \int_A \Phi(x) d\mu(x).$$

The proposition then follows from simple manipulations. For example, if $\Phi \in C_c^\infty(G)$,

$$\begin{aligned} \int_A \Phi(xg) \|\det x\|^{-2} d\mu(x) &= \|\det g\|^{-2} \int_A \Phi(x) \|\det xg^{-1}\|^{-2} d\mu(x) \\ &= \int_A \Phi(x) \|\det x\|^{-2} d\mu(x), \end{aligned}$$

as we sought to show.

Having defined a Haar measure on $GL_2(F)$, let us turn to the relevant subgroups. Let N, T, B, Z be

as in definition 2.1.1. The subgroup N , Z , and T , are taken care of as $N \cong F$, $Z \cong F^*$, and $T \cong F^* \times F^*$, leaving B . As $B \cong T \ltimes N$, define a linear functional on $C_c^\infty(B) := C_c^\infty(T) \otimes C_c^\infty(N)$ by:

$$\Phi \mapsto \int_T \int_N \Phi(tn) d\mu_T(t) d\mu_N(n)$$

where μ_T , μ_N are the Haar measure of T and N respectively induced by the aforementioned isomorphisms. Certainly, this functional is left B -invariant, and hence a left-Haar integral on B . By the above functional, we may take $\mu_B := \mu_T \otimes \mu_N$. Importantly, this tensor product doesn't commute, taking into account the semidirect product defining B . Using μ_B , we may write:

$$\Phi \mapsto \int_B \Phi(b) d\mu_B(b)$$

As the tensor product does not commute, μ_B is not unimodular, and hence there is a non-trivial module:

Proposition 2.2.8: Module of Bruhat Group

The module δ_B of the group B is given by

$$\delta_B : tn \mapsto \|t_2/t_1\|, \quad n \in N, t = \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} \in T. \quad (2.3)$$

Proof:

Setting

$$c = sm, \quad m \in N, \quad s = \begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix} \in T,$$

we get

$$\int_B \Phi(bc) d\mu_B(b) = \int_T \int_N \Phi(ts s^{-1} n s m) d\mu_T(t) d\mu_N(n).$$

We use the obvious isomorphism $N \rightarrow F$ to identify μ_N with a certain Haar measure μ_F on F . For $\varphi \in C_c^\infty(N)$, we then have

$$\begin{aligned} \int_N \varphi(s^{-1} n s) d\mu_N(n) &= \int_F \varphi \left(\begin{pmatrix} 1 & s_1^{-1} x s_2 \\ 0 & 1 \end{pmatrix} \right) d\mu_F(x) \\ &= \|s_1 s_2^{-1}\| \int_N \varphi(n) d\mu_N(n). \end{aligned}$$

By definition,

$$\int_B \Phi(bc) d\mu_B(b) = \delta_B(c)^{-1} \int_B \Phi(b) d\mu_B(b),$$

completing the proof.

This defines the desired semi-invariant measure:

Corollary 2.2.9: semi-invariant measure on $GL_2(\mathfrak{o})$

The space $C_c^\infty(B \backslash G, \delta_B^{-1})$ admits a positive semi-invariant measure $\dot{\mu}$, where δ_B is given by proposition 2.2.8. If $K = GL_2(\mathfrak{o})$, there is a Haar measure μ_K on K such that

$$\int_{B \backslash G} f(g) d\dot{\mu}(g) = \int_K f(k) d\mu_K(k),$$

for $f \in C_c^\infty(B \backslash G, \delta_B^{-1})$.

Proof:

The character δ_B is trivial on the compact group $K \cap B$. Restriction of functions is an isomorphism $C_c^\infty(B \backslash G, \delta_B^{-1}) \rightarrow C_c^\infty(K \cap B \backslash K, 1)$, where 1 denotes the trivial character of $K \cap B$. The semi-invariant measure $\dot{\mu}$ thus restricts to a semi-invariant measure on $C_c^\infty(K \cap B \backslash K, 1)$, but so does any Haar measure on K .

Note that μ_K is effectively just the restriction of a Haar measure μ_G on G . Using this, we see there is a left Haar measure μ_B on B such that

$$\int_G \varphi(g) d\mu_G(g) = \int_K \int_B \varphi(bk) d\mu_B(b) d\mu_G(k), \quad \varphi \in C_c^\infty(G). \quad (2.4)$$

showing that the measure and integral “commute” with the decomposition $G = BK$.

2.3 Representation of the Mirabolic Group

A lot of the representation of $GL_2(F)$ can be simplified by first finding the representation of the following subgroup:

Definition 2.3.1: Mirabolic Subgroup

The group $M \subseteq GL_2(F) = G$ given by:

$$M = \left\{ \begin{pmatrix} a & x \\ 0 & 1 \end{pmatrix} : a \in F^*, x \in F \right\}$$

is called the *mirabolic subgroup* of G . Letting $S = T \cap M \cong F^*$, then $M \cong S \ltimes N$.

My professor once said that mirabolic should be remembered as “miracle-parabolic” group, an apt mnemonic.

Let us start by analyzing the smooth representations of N from which we shall do induction. Let (π, V) be a smooth representation and ϑ a character of N . Let $V(\vartheta)$ be the subspace generated by all $\pi(n)v - \vartheta(n)v$. This is an N subspace, and $V_\vartheta = V/V(\vartheta)$ is well-defined and, by construction, as the maximal N -quotient where the action of N acts via the character ϑ . If ϑ is trivial, then $V(\vartheta) = V(N)$ and so $V_\vartheta = V_N$.

From this, we can define the following lemma that can be thought of as the generalization of the test in Fourier analysis where $\int f e^{2i\pi k} dx = 0$:

Lemma 2.3.2: Generalized Harmonic Decomposition

Let μ_N be a Haar measure on N and ϑ a character of N .

1. Let (π, V) be a smooth representation of N and $v \in V$. The vector v lies in $V(\vartheta)$ if and only if there is a compact open subgroup N_0 of N such that

$$\int_{N_0} \vartheta(n)^{-1} \pi(n) v d\mu_N(n) = 0 \quad (2.5)$$

2. The process $(\pi, V) \mapsto V_\vartheta$ is an exact functor from $\text{Rep}(N)$ to the category of complex vector spaces.

Think of the integral as the generalized projection operator, $\int_{N_0} \vartheta(n)^{-1} \pi(n) v d\mu_N(n) = 0$. Multiplying by $\vartheta(n)^{-1}$ replaces the $e^{-2i\pi k}$.

Proof:

1. First assume that ϑ is trivial on the trivial character. The group $N \cong F$ is the union of an ascending sequence of compact open subgroups. So, if

$$v = \sum_{i=1}^r v_i - \pi(n_i) v_i \in V(N),$$

there is a compact open subgroup N_0 of N containing all n_i . Then the relation given by equation 2.5 holds for this choice of N_0 .

Conversely, let $v \in V$ and suppose the integral relation holds. Then there is an open normal subgroup N_1 of N_0 such that $v \in V^{N_1}$. The space V^{N_1} carries a representation of the finite group N_0/N_1 . Therefore, in the obvious notation, $V^{N_1} = V^{N_1}(N_0/N_1) \oplus V^{N_0}$ and the map

$$w \mapsto \mu_N(N_0)^{-1} \int_{N_0} \pi(n) w d\mu_N(n), \quad w \in V^{N_1},$$

is the N_0 -projection $V^{N_1} \rightarrow V^{N_0}$. This has kernel $V^{N_1}(N_0/N_1) \subset V(N)$. But then this proves (1) for the trivial character of N .

Now let ϑ be an arbitrary character of N , and consider the representation (π', V') of N , where $V' = V$ and $\pi'(n) = \vartheta(n)^{-1} \pi(n)$. Then $V(\vartheta) = V'(N)$ and so (1) follows in general.

2. right-exactness is clear as we are taking the quotient. For the other way, if $f : U \rightarrow V$ is injective we want that the induced $\bar{f}$ is injective. If $\bar{f}(v) = 0$, then for injectivity it must be in $V(\vartheta)$. As it is equal to $\bar{0}$ we get:

$$\int_{N_0} \vartheta(n)^{-1} \pi(n) \cdot (f(v)) d\mu_N(n) = 0$$

As f is an N -homomorphism, it commutes with the action given by π

$$\int_{N_0} \vartheta(n)^{-1} f(\pi(n) \cdot v) d\mu_N(n) = 0$$

and by linearity f comes out, and by injectivity of f we can drop f , and finally by part 1 we get that $v \in V(\vartheta)$ giving injectivity of $\bar{f}$, completing the proof.

Some important remarks. If (π, V) is a smooth representation of N (or of M), then $V(N)$ is an N - (or M -) subspace of V . The exact sequence

$$0 \rightarrow V(N) \rightarrow V \rightarrow V_N \rightarrow 0$$

gives an exact sequence

$$0 \rightarrow V(N)_N \rightarrow V_N \rightarrow V_N \rightarrow 0$$

in which the map $V_N \rightarrow V_N$ is the identity. Therefore

$$V(N)_N = 0 \quad \text{and} \quad V(N)(N) = V(N)$$

If $\vartheta \neq 1$, as N acts trivially on $V/V(N)$, we have $(V/V(N))_\vartheta = 0$ and so the inclusion $V(N) \rightarrow V$ induces an isomorphism

$$V(N)_\vartheta \cong V_\vartheta.$$

Proposition 2.3.3: Character Invariant Subspace Never Whole Space

Let (π, V) be a smooth representation of N and let $0 \neq v \in V$. Then there exists a character ϑ such that $v \notin V(\vartheta)$.

Proof:

We write

$$N_j = \begin{pmatrix} 1 & \mathfrak{p}^j \\ 0 & 1 \end{pmatrix}, \quad j \in \mathbb{Z}.$$

Take $v \in V$, $v \neq 0$. We choose $j_0 \in \mathbb{Z}$ such that N_{j_0} fixes v . For $j \leq j_0$, let V_j denote the N_j -space generated by v . This is the direct sum of isotypic components $V_j^{\eta_j}$, as η ranges over the characters of N_j trivial on N_{j_0} . For each $j \leq j_0$, there exists η_j such that $V_j^{\eta_j} \neq 0$; by definition, we have

$$\int_{N_j} \eta_j(n)^{-1} \pi(n) v \, dn \neq 0.$$

The N_{j-1} -space generated by $V_j^{\eta_j}$ is contained in V_{j-1} , so we may choose η_{j-1} such that $\eta_{j-1}|_{N_j} = \eta_j$. This satisfies the necessary properties to take the limit of these characters, and as N is a limit of the N_j we get that there exists a character ϑ of N such that, for all $j \leq j_0$, we have

$$\int_{N_j} \vartheta(n)^{-1} \pi(n) v \, dn \neq 0.$$

Therefore $v \notin V(\vartheta)$, as required.

Corollary 2.3.4: Trivial If Trivial On Characters

Let (π, V) be a smooth representation of N . If $V_\vartheta = 0$ for all characters ϑ of N , then $V = 0$

Proof:

immediate

Now, let (π, V) be a smooth representation of M . Then we can deduce:

Corollary 2.3.5: Trivial Representation of Mirabolic Group

Let (π, V) be a smooth representation of M . Suppose $V_N = 0$ and for some non-trivial character ϑ of N , $V_{\vartheta} = 0$. Then $V = 0$.

Proof:

Now let (π, V) be a smooth representation of M . The space $V(N)$ is then an M -subspace of V and V_N carries a natural representation of $M/N = S$: On the other hand, $\pi(S)$ permutes transitively the subspaces $V(\vartheta)$, $\vartheta \neq 1$: for $s \in S$, $\pi(s) \cdot V(\vartheta) = V(\vartheta')$, where $\vartheta'(n) = \vartheta(s^{-1}ns)$. The corollary follows immediately.

Now, fix a non-trivial character ϑ of N . We start by looking at $\text{Ind}_N^M \vartheta$ and $\text{c-Ind}_N^M \vartheta$. As $N \trianglelefteq M$ and all non-trivial characters ϑ, ϑ' of N are conjugate, the induced and compacted induced representations are M -isomorphic.

Theorem 2.3.6: $\text{c-Ind}_N^M \vartheta$ is Irreducible

Let ϑ be a non-trivial character of N and set $\mathcal{W} = \text{Ind}_N^M \vartheta$, $\mathcal{W}^c = \text{c-Ind}_N^M \vartheta$. Let $\alpha : \mathcal{W} \rightarrow \mathbb{C}$ denote the canonical map $f \mapsto f(1)$.

1. We have $\mathcal{W}(N) = \mathcal{W}^c(N) = \mathcal{W}^c$ and $(\mathcal{W}/\mathcal{W}^c)(N) = 0$.
2. The map α induces isomorphisms $\mathcal{W}_{\vartheta} \cong \mathbb{C}$ and $\mathcal{W}_{\vartheta}^c \cong \mathbb{C}$.

As a consequence, $\text{c-Ind}_N^M \vartheta$ is irreducible over M .

Proof:

1. Let $f \in \mathcal{W}$ and $n \in N$. For $a \in S$, we have $f(an) = \vartheta(ana^{-1})f(a)$. As there is an integer j such that N_j fixes f by proposition 2.3.3, $f(a) = 0$ if $\|\det a\|$ is sufficiently large. On the other hand, $f(an) = f(a)$ if $\|\det a\|$ is sufficiently small, so $nf - f$ vanishes at a if $\|\det a\|$ is sufficiently small. This implies that $nf - f \in \mathcal{W}^c$. Thus $\mathcal{W}(N) \subset \mathcal{W}^c$ and N acts trivially on $\mathcal{W}/\mathcal{W}^c$.

Next, to prove that $\mathcal{W}^c(N) = \mathcal{W}^c$, let ψ be the character of F defined by

$$\psi(x) = \vartheta \left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \right).$$

Take $a \in F^\times$, $j \in \mathbb{Z}_{\geq 1}$. Let $f_{a,j} \in \mathcal{W}^c$ be the function such that

$$f_{a,j} : \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} au & 0 \\ 0 & 1 \end{pmatrix} \mapsto \psi(x), \quad u \in U_F^j,$$

and which vanishes elsewhere. By similar arguments as done before, the functions $f_{a,j}$ span $\mathcal{W}^c$ over $\mathbb{C}$. We have

$$\begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} f_{a,j} \left(\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix} \right) = \psi(bz) f_{a,j} \left(\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix} \right).$$

We deduce that $nf_{a,j}$ has the same support as $f_{a,j}$, $n \in N$. We can certainly find $x \in F$ such that the function $u \mapsto \psi(au x)$, $u \in U_F^j$, is constant, equal to c say, with $c \neq 1$. If $n = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, then $nf_{a,j} - f_{a,j} = (c - 1)f_{a,j}$, whence $f_{a,j} \in \mathcal{W}^c(N)$ and so $\mathcal{W}^c(N) = \mathcal{W}^c$. This implies $\mathcal{W}(N) = \mathcal{W}^c$.

2. The map α induces a surjection $\mathcal{W}_\vartheta \rightarrow \mathbb{C}$. On the other hand, since N acts trivially on $\mathcal{W}/\mathcal{W}^c$, the inclusion $\mathcal{W}^c \rightarrow \mathcal{W}$ induces an isomorphism $\mathcal{W}_\vartheta^c \cong \mathcal{W}_\vartheta$. To prove (2), therefore, it is enough to show that any $f \in \mathcal{W}^c$ with $f(1) = 0$ belongs to $\mathcal{W}^c(\vartheta)$. A function f , vanishing at 1, is a finite linear combination of functions $f_{a,j}$ with $a \notin U_F^j$, so it is enough to treat such functions. However, as $a \notin U_F^j$, there exists $x \in F$ such that the function $u \mapsto \psi(au x) - \psi(x)$, $u \in U_F^j$, is a non-zero constant. Taking $n = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, the same calculation as before shows that $nf_{a,j} - \vartheta(n)f_{a,j}$ is a non-zero constant multiple of $f_{a,j}$, whence $f_{a,j} \in \mathcal{W}^c(\vartheta)$ and $\mathcal{W}^c(\vartheta) = \mathcal{W}^c$.

Finally, for the irreducibility, let V be an M -subspace of $\mathcal{W}^c$. As $\mathcal{W}_N^c = 0$, the spaces $V_N, (\mathcal{W}^c/V)_N$ are both zero. The sequence

$$0 \rightarrow V_\vartheta \rightarrow \mathcal{W}_\vartheta^c \rightarrow (\mathcal{W}^c/V)_\vartheta \rightarrow 0$$

is exact. As $\dim \mathcal{W}_\vartheta^c = 1$, we conclude that $\dim V_\vartheta$ is 0 or 1. In the first case, $V = 0$ by corollary 2.3.4. In the second, $(\mathcal{W}^c/V)_\vartheta = 0$ and so $\mathcal{W}^c = V$, showing it is irreducible and completing the proof.

The following from the textbook [1] summarizes the key points:

1. A function $f \in W$ is determined by its restriction to $S \cong F^\times$. The restriction $f|_{F^\times}$ is a smooth function on $F^\times$.
2. A smooth function φ on $F^\times$ is of the form $\varphi = f|_{F^\times}$, for some $f \in W$, if and only if there exists $c > 0$ such that $\varphi(x) = 0$ for all x satisfying $\|x\| > c$.
3. A function $f \in W$ lies in W^c if and only if $f|_{F^\times} \in C_c^\infty(F^\times)$.

2.3.7 Lack of Irreducibility of Dual The above shows the representation $\text{Ind}_N^M \vartheta$ is never irreducible, for any non-trivial character ϑ of N . The Duality Theorem (theorem 1.3.12) implies

$$(\text{c-Ind}_N^M \vartheta)^\vee \cong \text{Ind}_N^M \check{\vartheta}.$$

Thus $\text{c-Ind}_N^M \vartheta$ provides an example of an irreducible smooth representation with reducible contragredient.

We can now finish classifying *all* irreducible representations of the mirabolic group M . The following theorem is the key technical tool to show determine we found them all. As before let ϑ be a non-trivial character of N . Let (π, V) be a smooth representation of M . Frobenius Reciprocity gives a canonical isomorphism

$$\mathrm{Hom}_N(V, V_\vartheta) \cong \mathrm{Hom}_M(V, \mathrm{Ind}_N^M V_\vartheta).$$

Let $q : V \rightarrow V_\vartheta$ denote the quotient map and let $q_\star$ be the map $V \rightarrow \mathrm{Ind}_N^M V_\vartheta$ corresponding to q under this isomorphism: for $v \in V$,

$$q_\star(v) = (m \mapsto q(\pi(m)v))$$

Theorem 2.3.8: Invariant Space equivalent to $\mathrm{c}\text{-Ind}_N^M V_\vartheta$

Let (π, V) be a smooth representation of M . The M -homomorphism $q_\star : V \rightarrow \mathrm{Ind}_N^M V_\vartheta$ induces an isomorphism $V(N) \cong \mathrm{c}\text{-Ind}_N^M V_\vartheta$.

Proof:

The N -space V_ϑ is a direct sum of copies of ϑ . Therefore $\mathrm{Ind}_N^M V_\vartheta$ is a direct sum of copies of $\mathrm{Ind}_N^M \vartheta$. By theorem 2.3.6

$$(\mathrm{Ind}_N^M V_\vartheta)(N) = \mathrm{c}\text{-Ind}_N^M V_\vartheta = (\mathrm{c}\text{-Ind}_N^M V_\vartheta)(N).$$

The M -homomorphism $q_\star : V \rightarrow \mathrm{Ind}_N^M V_\vartheta$ surely maps $V(N)$ to $(\mathrm{Ind}_N^M V_\vartheta)(N) = \mathrm{c}\text{-Ind}_N^M V_\vartheta$.

Let $W = \mathrm{Ker} q_\star \cap V(N)$ and $C = \mathrm{c}\text{-Ind}_N^M V_\vartheta / q_\star(V(N))$. By lemma 2.3.2 the natural map $W_N \rightarrow V(N)_N$ is injective, so $W_N = 0$. Likewise, $(\mathrm{c}\text{-Ind}_N^M V_\vartheta)_N$ is zero, so $C_N = 0$.

Now, the map $q_\star : V \rightarrow \mathrm{c}\text{-Ind}_N^M V_\vartheta$ induces a map

$$q_{\star, \vartheta} : V_\vartheta = V(N)_\vartheta \rightarrow (\mathrm{c}\text{-Ind}_N^M V_\vartheta)_\vartheta.$$

By theorem 2.3.6(2), the canonical N -map $\mathrm{Ind}_N V_\vartheta \rightarrow V_\vartheta$ induces an isomorphism $\alpha_\vartheta : (\mathrm{c}\text{-Ind}_N V_\vartheta)_\vartheta \rightarrow V_\vartheta$. The composite map $\alpha_\vartheta \circ q_{\star, \vartheta} : V_\vartheta \rightarrow V_\vartheta$ is the identity. However, the kernel of this map is W_ϑ and its cokernel is isomorphic to C_ϑ . As the spaces $W_N, W_\vartheta, C_N, C_\vartheta$ are all zero, by corollary 2.3.5 both W and C are trivial and $q_\star : V(N) \rightarrow \mathrm{c}\text{-Ind}_N^M V_\vartheta$ is an isomorphism, completing the proof.

Theorem 2.3.9: Classification of Irreducible Representation of Mirabolic Group

Let (π, V) be an irreducible smooth representation of M . Either:

1. $\dim V = 1$ and π is the inflation of a character of $M/N \cong F^\times$
2. $\dim V$ is infinite and $\pi \cong \mathrm{c}\text{-Ind}_N^M \vartheta$, for any character $\vartheta \neq 1$ of N .

In case (1), $\dim V_N = 1$ and $V_\vartheta = 0$ for $\vartheta \neq 1$. In case (2), $V_N = 0$ and $\dim V_\vartheta = 1$ for all $\vartheta \neq 1$.

Proof:

If $V(N) = 0$, then N acts trivially on V . The group M/N is abelian, so Schur's Lemma 2.6 implies $\dim V = 1$ and we are in case (1).

If $V(N) \neq 0$, then $V(N) = V$ and $V_N = 0$. Therefore $V_\vartheta \neq 0$ for all characters $\vartheta \neq 1$ of N , and so $\dim V$ is infinite. Theorem 2.3.8 implies that $V = V(N)$ is M -isomorphic to $\text{c-Ind}_N^M V_\vartheta$. The N -space V_ϑ is a direct sum of copies of ϑ , so V is a direct sum of copies of $\text{c-Ind}_N^M \vartheta$ and, since it is irreducible, $V \cong \text{c-Ind}_N^M \vartheta$.

2.4 Jacquet Modules and Induced Representations

We now classify the irreducible smooth representations of $GL_2(F) = G$. In this section, we deal completely with *principal series representations*, that is irreducible representations where $V_N \neq 0$: This is the equivalent of looking to see if π contains the trivial character of N . The terminology comes from functional analysis when taking the spectrum of a topological group with the continuous spectrum associated to the principal (as in main part) series of characters and discrete spectrum's will be later associated to the cuspidal representations. It shall be the case that much of the important representation theory on the Galois side will be in relation to the cuspidal, but that cannot be properly studied without first grasping the whole picture of the representation of $GL_2(F)$.

Definition 2.4.1: Principal Series Representation

Let (π, V) be an irreducible representations of $GL_n(F)$ (notice $n \in \mathbb{N}_{>0}$). Then it is called a *principal series* if $V_N \neq 0$.

When $V_N = 0$ these shall be called *cuspidal representations* or *supercuspidal representations*, which could be thought of as the “singular” case (and a singularity in the form of a cusp does appear when looking at the geometric picture modular forms; see [ref:HERE](#)), and are dealt with in [ref:HERE](#).

Let (π, V) be a smooth representation of $GL_2(F)$ and $V(N)$ denote the subspace of V spanned by the vectors $v - \pi(x)v$, for $v \in V$ and $x \in N$. The space $V_N = V/V(N)$ inherits a representation π_N of $B/N = T$, which is smooth. This representation is given a name:

Definition 2.4.2: Jacquet Module

The representation (π_N, V_N) is called the *Jacquet module* of (π, V) at N . This gives the *Jacquet functor*

$$\begin{aligned} \text{Rep}(G) &\rightarrow \text{Rep}(T), \\ (\pi, V) &\mapsto (\pi_N, V_N), \end{aligned}$$

which is exact and additive.

Let (σ, W) be a smooth representation of T . View σ as a smooth representation of B which is trivial on N , and form the smooth induced representation $\text{Ind}_B^G \sigma$. We shall often abbreviate $\text{Ind}_B^G \sigma = \text{Ind } \sigma$, since B and G are the only groups involved for most of this section. This is often called *parabolic induction*.

If (π, V) is a smooth representation of G , Frobenius Reciprocity gives an isomorphism

$$\mathrm{Hom}_G(\pi, \mathrm{Ind} \sigma) \cong \mathrm{Hom}_B(\pi, \sigma).$$

However, σ is trivial on N so any B -homomorphism $\pi \rightarrow \sigma$ factors through the quotient map $\pi \rightarrow \pi_N$. Thus:

$$\mathrm{Hom}_G(\pi, \mathrm{Ind} \sigma) \cong \mathrm{Hom}_T(\pi_N, \sigma).$$

Proposition 2.4.3: Jacquet Module and Principal Series Representation

Let (π, V) be an irreducible smooth representation of G . The following are equivalent:

1. The Jacquet module V_N is non-zero.
2. The representation π is isomorphic to a G -subspace of a representation $\mathrm{Ind}_B^G \chi$, for some character χ of T .

Proof:

(1) $\Rightarrow$ (2) We have to show that the Jacquet module V_N admits an irreducible T - (or B -) quotient.

Choose $v \in V, v \neq 0$. Since V is irreducible over G , any element of V is a finite linear combination of translates $\pi(g)v$ of v , for various $g \in G$. Write $K = \mathrm{GL}_2(\mathfrak{o})$. The vector v is fixed by a subgroup K' of K of finite index; let $\{v_1, v_2, \dots, v_r\}$ be the distinct elements of the form $\pi(k)v, k \in K$. In particular, $r \leq (K : K')$. Since $G = BK$, the elements $v_1, \dots, v_r$ generate V over B , and their images generate V_N over T .

Thus V_N is finitely generated as a representation of T . We choose a minimal generating set $\{u_1, \dots, u_t\}, t \geq 1$, say. A standard Zorn's Lemma argument shows that V_N has a T -subspace U , containing $u_1, \dots, u_{t-1}$, and maximal for the property $u_t \notin U$. Then U is a maximal T -subspace of V_N and V_N/U is an irreducible representation of T , hence a character.

(2) $\Rightarrow$ (1) As $\mathrm{Hom}_G(\pi, \mathrm{Ind} \sigma) \cong \mathrm{Hom}_T(\pi_N, \sigma)$, we get

$$\mathrm{Hom}_T(\pi_N, \chi) \cong \mathrm{Hom}_G(\pi, \mathrm{Ind} \chi) \neq 0,$$

so $\pi_N \neq 0$.

Next, let μ_N be a Haar measure on N and let $t \in T$. The measure $S \mapsto \mu_N(t^{-1}St)$ is the Haar measure $\delta_B(t)\mu_N$:

$$\int_N f(txt^{-1}) d\mu_N(x) = \delta_B(t) \int_N f(x) d\mu_N(x), \quad f \in C_c^\infty(N).$$

As before, let w denote the permutation matrix

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

If σ is a smooth representation of T , we can form the representation $\sigma^w : t \mapsto \sigma(wtw^{-1})$, and view it as a representation of B which is trivial on N .

Let α_σ denote the canonical B -map $\text{Ind } \sigma \rightarrow \sigma$ given by $f \mapsto f(1)$. It induces a canonical T -map $(\text{Ind } \sigma)_N \rightarrow \sigma$, which we continue to denote α_σ .

Theorem 2.4.4: Restriction-induction Lemma

Let (σ, U) be a smooth representation of T and set $(\Sigma, X) = \text{Ind}_B^G \sigma$. There is an exact sequence of representations of T :

$$0 \rightarrow \sigma^w \otimes \delta_B^{-1} \rightarrow \Sigma_N \xrightarrow{\alpha_\sigma} \sigma \rightarrow 0.$$

First, a technical lemma:

Lemma 2.4.5: Technical Lemma

Recall that $G = B \cup BwN$. A function $f \in X$ lies in V if and only if $\text{supp } f \subset BwN$, that is if $f \in X$ then $f \in V$ if and only if there is a compact open subgroup N_0 of N (depending on f) such that $\text{supp } f \subset BwN_0$.

Proof:

A function $f \in X$ lies in V if and only if $f(1) = 0$. Since f is G -smooth, such a function f vanishes on a set BN'_0 , where N'_0 is some compact open subgroup of N' . The identity (for $x \neq 0$)

$$\begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} = \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -x^{-1} & 0 \\ 0 & x \end{pmatrix} w \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \in Bw \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix}$$

implies that $\text{supp } f \subset BwN_0$, for some compact open subgroup N_0 of N , as required.

Proof:

(of theorem 2.4.4)

By definition, X is the space of G -smooth functions $f : G \rightarrow U$ such that $f(bg) = \sigma(b)f(g)$, $b \in B, g \in G$. The canonical map $\alpha_\sigma : X \rightarrow U$ amounts to restriction of functions to B . Set $V = \text{Ker } \alpha_\sigma$. Thus V provides a smooth representation of B and there is an exact sequence

$$0 \rightarrow V_N \rightarrow X_N \rightarrow U \rightarrow 0.$$

We have to identify the T -representation V_N .

Let $f \in V$; in view of the preceding lemma, we can define a function $f_N : T \rightarrow U$ by

$$f_N(x) = \int_N f(xwn) dn = \sigma(x)f_N(1), \quad x \in T.$$

By 8.1 Lemma, the kernel of the map $f \mapsto f_N$ is $V(N)$, and so $f \mapsto f_N(1)$ gives a bijective map

$V_N \rightarrow U$. Taking $t \in T$ and $f \in V$, we have

$$\begin{aligned} (tf)_N(x) &= \int_N f(xwnt) \, dn \\ &= \delta_B(t^{-1}) \int_N f(xt^w wn) \, dn = \delta_B^{-1}(t)(t^w f_N)(x). \end{aligned}$$

Thus $f \mapsto f_N(1)$ is a B -homomorphism $V \rightarrow \sigma^w \otimes \delta_B^{-1}$ inducing a T -isomorphism $V_N \cong \sigma^w \otimes \delta_B^{-1}$.

Proposition 2.4.6: Non-cuspidal Irreducible Representation Are Admissible

Let (π, V) be an irreducible smooth representation of $GL_2(F)$ which is not cuspidal. The representation π is admissible.

Proof:

By definition, $V_N \neq 0$. By proposition 2.4.3, π is equivalent to a sub-representation of $\text{Ind}_B^G \chi$, for some character χ of T . It is enough, therefore, to prove that $\text{Ind} \chi$ is admissible.

We fix a compact open subgroup K of G ; shrinking it if necessary, we may assume that $K \subset K_0 = GL_2(\mathfrak{o})$. The space X^K of K -fixed points in $\text{Ind} \chi$ consists of the functions $f : G \rightarrow \mathbb{C}$ satisfying

$$f(bgk) = \chi(b)f(g), \quad b \in B, g \in G, k \in K. \quad (2.6)$$

We have $G = BK_0$, so the set $B \backslash G/K$ is finite, and each double coset BgK supports, at most, a one-dimensional space of functions satisfying equation 2.6. Thus X^K is finite-dimensional, as we sought to show.

Note that irreducible *cuspidal* representations of $GL_2(F)$ are likewise admissible, and hence all irreducible representations of $GL_2(F)$ are admissible (see [ref:HERE](#) for the cuspidal equivalent, corollary 10.2 in [1]). We shall now find a more granular classification of principal series. First recall from example 1.2.8 that smooth characters on $GL_n(F)$ factored via the determinant and a character of F^* . Define the following:

Definition 2.4.7: Twist On Representation

Let (π, V) is a smooth representation of G and φ a character of F^* . A *twist* $\varphi\pi$ is a smooth representation $(\varphi\pi, V)$ of G given by:

$$\varphi\pi(g) = \varphi(\det g)\pi(g), \quad g \in G$$

Similarly for characters of T : if $\chi = \chi_1 \otimes \chi_2$ is a character of T and φ is a character of $F^\times$, put $\varphi \cdot \chi = \varphi\chi_1 \otimes \varphi\chi_2$. If we inflate $\varphi \cdot \chi$ to a representation of B trivial on N , we get $\varphi \cdot \chi = (\varphi \circ \det|_B) \otimes \chi$. It follows immediately that

$$\text{Ind}_B^G(\varphi \cdot \chi) \cong \varphi \text{Ind}_B^G \chi$$

This allows us to make convenient adjustments to the character χ without changing the essential structure of the induced representation. With this definition, we make precise the structure of representations of the form $\text{Ind}_B^G \chi$. The main step is:

Theorem 2.4.8: Irreducibility Criterion

Let $\chi = \chi_1 \otimes \chi_2$ be a character of T , and set $(\Sigma, X) = \text{Ind}_B^G \chi$.

1. The representation (Σ, X) is reducible if and only if $\chi_1 \chi_2^{-1}$ is either the trivial character or the character $x \mapsto \|x\|^2$ of $F^\times$.
2. Suppose that (Σ, X) is reducible. Then:
 - (a) the G -composition length of X is 2;
 - (b) one composition factor of X has dimension 1, the other is of infinite dimension;
 - (c) X has a 1-dimensional G -subspace if and only if $\chi_1 \chi_2^{-1} = 1$;
 - (d) X has a 1-dimensional G -quotient if and only if $\chi_1 \chi_2^{-1}(x) = \|x\|^2$, $x \in F^\times$.

We will refine this to a classification of the irreducible principal series representations

Let:

$$V = \{f \in X : f(1) = 0\}.$$

This is a B -subspace of X and we have an exact sequence

$$0 \rightarrow V \rightarrow X \rightarrow C \rightarrow 0,$$

where the one-dimensional space $C = X/V$ carries the character χ of T . By the Restriction-Induction Lemma theorem 2.4.4, so $V_N \cong \delta_B^{-1} \chi^w$.

Proposition 2.4.9: Irreducible Kernel

Let W be the kernel $V(N)$ of the canonical map $V \rightarrow V_N$. The space W is irreducible over B .

First an important technical lemma:

Lemma 2.4.10: Representation of V

For $f \in V$, define a function $f_N \in C_c^\infty(N)$ by $f_N(n) = f(un)$, $n \in N$. The map

$$\begin{aligned} V &\longrightarrow C_c^\infty(N), \\ f &\longmapsto f_N \end{aligned}$$

is an N -isomorphism.

Proof:

As in theorem 2.4.4, the support of $f \in V$ is contained in $Bw\mathcal{N}$, for some compact open subgroup $\mathcal{N}$ of N . The assertions follow immediately (observing that the notation f_N here is not the same as that in theorem 2.4.4).

Proof:

(of proposition 2.4.9)

We shall prove that W is irreducible as a representation of the mirabolic group M : recall under lemma 2.3.2 it was shown that $W_N = 0$ and $W = W(N)$. For $\varphi \in C_c^\infty(N)$ and $a \in F^\times$, we define $a\varphi \in C_c^\infty(N)$ by

$$a\varphi \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \chi_2(a)\varphi \begin{pmatrix} 1 & a^{-1}x \\ 0 & 1 \end{pmatrix}.$$

This gives an action of $F^\times$ on $C_c^\infty(N)$ which we regard as an action of the group S of matrices $\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$. We combine this with the natural action of N to give $C_c^\infty(N)$ the structure of a smooth representation of M . With this structure, the map in lemma 2.4.10 is an M -isomorphism is an M -isomorphism.

Let ϑ be a non-trivial character of N . The map $f \mapsto \vartheta f$ is a linear automorphism of $V = C_c^\infty(N)$ carrying $V(N)$ to $V(\vartheta)$. Since $V/V(N)$ is one dimensional, so is $\dim V_\vartheta = 1$. However, since N acts trivially on V_N , the inclusion $W \rightarrow V$ induces an isomorphism $W_\vartheta \cong V_\vartheta$, hence $W_\vartheta \cong \vartheta$. Then apply theorem 2.3.9 to get $W = W(N) \cong \text{c-Ind}_N^M \vartheta$ which, by corollary 2.3.5 is irreducible.

As a direct consequence of the Proposition, we have:

Corollary 2.4.11: Length of Induced Representation by Character

As a representation of B or of M , $\text{Ind}_B^G \chi$ has composition length 3. Two of the composition factors have dimension one, and the third is of infinite dimension. In particular, the G -composition length of the representation $\text{Ind}_B^G \chi$ is at most 3.

Proposition 2.4.12: Detecting one-dimensional Subrepresentations

The following are equivalent:

- ($\Rightarrow$) (1) $\chi_1 = \chi_2$
- ($\Rightarrow$) (2) X has a one-dimensional N -subspace

When these conditions hold,

1. X has a unique one-dimensional N -subspace X_0 ;
2. X_0 is a G -subspace of X , and it is not contained in V .

Proof:

If (1) holds, by twisting we may as well take $\chi_1 = \chi_2 = 1$. The (non-zero) constant function then spans a one-dimensional G -subspace of X .

Conversely, let $f \in X$ span an N -stable subspace of dimension 1. Thus N acts on f by right translation as a character. The support of f is left-invariant under B , so $\text{supp } f$ is either G or BwN . The second case is impossible since if $f(1) = 0$, by the restriction induction lemma

(theorem 2.4.4) the support of f is confined to BwN_0 , for some compact open subgroup N_0 of N . Thus, $\text{supp } f = G$ and f vanishes nowhere so that $f(1) \neq 0$, and $f \notin V$. Then the canonical N -map $X \rightarrow \mathbb{C} = X/V$ identifies the N -space $\mathbb{C}f$ with the trivial N -space $\mathbb{C}$. It follows that N fixes f under right translation. Take $x \in F^\times$, and consider the identity

$$w \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -x^{-1} & 0 \\ 0 & x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}.$$

Taking x so that $\|x\|$ is sufficiently large, then f is fixed under right translation by $\begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}$. So, as f is fixed by N , we have

$$f(w) = \chi_1(-1)\chi_1^{-1}(x)\chi_2(x)f(1),$$

for all $x \in F^\times$ of sufficiently large absolute value. Thus $\chi_1 = \chi_2 = \varphi$, say, and $f(g) = \varphi(\det g)f(1)$. We have proved (1) $\Leftrightarrow$ (2), and that the one-dimensional N -subspace is uniquely determined.

We now prove theorem 2.4.8:

Proof:

(of theorem 2.4.8)

Assume X is reducible. By corollary 2.4.11 its G -length is 2 or 3, and it has either a finite-dimensional G -subspace or a finite-dimensional G -quotient

Assume the first alternative. Thus X has a one-dimensional N -subspace, and we are in the situation of proposition 2.4.12: X has a one-dimensional G -subspace L , and $\chi_1 = \chi_2 = \varphi$, say. Moreover, G acts on L as the character $\varphi \circ \det$ and $L \cap V = 0$. The quotient $Y = X/L$ is therefore B -isomorphic to V . If X has G -length 3, then Y has G -length 2. However, V has B -length 2 and a unique B -quotient, which is of dimension 1. This gives a G -quotient of Y on which G must act as a character $\varphi' \circ \det$. This would force $\varphi' \otimes \varphi'$ to appear as a factor in the Jacquet module $Y_N \cong \varphi \cdot \delta_B^{-1}$, which it cannot. Thus X has G -length 2 and we are in case (2)(c) of the Irreducibility Criterion (theorem 2.4.8).

In the other alternative, X has a finite-dimensional G -quotient. The representation $\check{X}$ therefore has a finite-dimensional G -subspace, and we are back with the first alternative. By the Duality Theorem (theorem 1.3.12), $\check{X} \cong \text{Ind}_B^G \delta_B^{-1} \check{\chi}$, so we are in the case (2)(d) of the Irreducibility Criterion (theorem 2.4.8).

Thus, in part (1) of the theorem, we have shown that X reducible implies χ has the stated form. The converse is given by proposition 2.4.12 and the dual case. We have also proved statements (a)–(d).

Having fully determined how to get irreducible representations from inductions/inflation of characters, to get a classification of the irreducible (non-cuspidal) representations of G , we need to investigate the homomorphisms between induced representations:

Proposition 2.4.13: Homomorphism between induced Representations

Let χ, ξ be characters of T . The space $\text{Hom}_G(\text{Ind}_B^G \chi, \text{Ind}_B^G \xi)$ has dimension 1 if $\xi = \chi$ or $\chi^w \delta_B^{-1}$, zero otherwise.

Proof:

We use Frobenius Reciprocity:

$$\text{Hom}_G(\text{Ind}_B^G \chi, \text{Ind}_B^G \xi) \cong \text{Hom}_T((\text{Ind } \chi)_N, \xi).$$

The Jacquet module $(\text{Ind } \chi)_N$ fits into the exact sequence

$$0 \rightarrow \chi^w \delta_B^{-1} \rightarrow (\text{Ind } \chi)_N \rightarrow \chi \rightarrow 0.$$

If we assume that $\chi \neq \chi^w \delta_B^{-1}$, then this sequence splits and the result follows immediately. The equation $\chi = \chi^w \delta_B^{-1}$ amounts to $\chi_1(x) = \|x\| \chi_2(x)$, $x \in F^\times$. In this case, $\text{Ind } \chi$ is irreducible and the result again follows.

By way of some examples, we examine in more detail the case where $\text{Ind}_B^G \chi$ is reducible. Thus there is a character φ of $F^\times$ such that $\chi = \varphi \cdot 1_T$ or $\chi = \varphi \cdot \delta_B^{-1}$. Twisting does not affect the situation materially, so we assume $\varphi = 1$. Consider first the case $\text{Ind}_B^G 1_T$. The irreducible G -quotient of $\text{Ind}_B^G 1_T$ is called the *Steinberg representation* of G , and is denoted St_G :

$$0 \rightarrow 1_G \rightarrow \text{Ind}_B^G 1_T \rightarrow \text{St}_G \rightarrow 0. \quad (2.7)$$

Its dimension is infinite and $(\text{St}_G)_N \cong \delta_B^{-1}$. Likewise, if φ is a character of $F^\times$, we have an exact sequence

$$0 \rightarrow \varphi_G \rightarrow \text{Ind}_B^G \varphi_T \rightarrow \varphi \cdot \text{St}_G \rightarrow 0,$$

where $\varphi_G = \varphi \circ \det$ and $\varphi_T = \varphi \otimes \varphi$. (Representations of the form $\varphi \cdot \text{St}_G$ are sometimes called *special*.)

Taking the smooth dual of (2.7), we get an exact sequence

$$0 \rightarrow \text{St}_G^\vee \rightarrow \text{Ind}_B^G \delta_B^{-1} \rightarrow 1_G \rightarrow 0. \quad (2.8)$$

The proposition implies

$$\text{St}_G \cong \text{St}_G^\vee. \quad (2.9)$$

2.4.14 Remark The proposition also implies that the space $\text{End}_G(\text{Ind } 1_T)$ has dimension 1, while $\text{Ind } 1_T$ is not irreducible. Thus the converse of Schur's Lemma fails in this context, as remarked in theorem 1.2.30.

Observe also the imperfect parallelism between the proposition above and the corresponding result (theorem 2.3.6) for the finite field case.

We introduce a new notation. If σ is a smooth representation of T , we define

$$\iota_B^G \sigma = \text{Ind}_B^G (\delta_B^{-1/2} \otimes \sigma). \quad (2.10)$$

This provides another exact functor $\text{Rep}(T) \rightarrow \text{Rep}(G)$, known as *normalized* or *unitary smooth induction*. It gives rise to more convenient combinatorics, for example:

$$(\iota_B^G \sigma)^\vee \cong \iota_B^G \check{\sigma}. \quad (2.11)$$

In this language, the Irreducibility Criterion (theorem 2.4.8) and proposition 2.4.13 say:

Lemma 2.4.15: Normalized Induction Properties

1. Let $\chi = \chi_1 \otimes \chi_2$ be a character of T . The representation $\iota_B^G \chi$ is reducible if and only if $\chi_1 \chi_2^{-1}$ is one of the characters $x \mapsto \|x\|^{\pm 1}$ of $F^\times$ or, equivalently, $\chi = \varphi \cdot \delta_B^{\pm 1/2}$ for some character φ of $F^\times$.
2. Let χ, ξ be characters of T . The space $\text{Hom}_G(\iota_B^G \chi, \iota_B^G \xi)$ is not zero if and only if $\xi = \chi$ or $\xi = \chi^w$.

Gathering up our earlier arguments and results, we get:

Theorem 2.4.16: Classification Theorem

The following is a complete list of the isomorphism classes of irreducible, non-cuspidal representations of G :

1. the irreducible induced representations $\iota_B^G \chi$, where $\chi \neq \varphi \cdot \delta_B^{\pm 1/2}$ for any character φ of $F^\times$;
2. the one-dimensional representations $\varphi \circ \det$, where φ ranges over the characters of $F^\times$;
3. the special representations $\varphi \cdot \text{St}_G$, where φ ranges over the characters of $F^\times$.

The classes in this list are all distinct except that, in (1), we have $\iota_B^G \chi \cong \iota_B^G \chi^w$.

Proof:

The examples in (2.7) show that every irreducible, non-cuspidal representation of G appears in this list. The relations between the irreducibly induced ones are given by proposition 2.4.13. The same result also implies that no special representation $\varphi \cdot \text{St}_G$ can be equivalent to $\varphi' \cdot \text{St}_G$, $\varphi' \neq \varphi$, or any irreducibly induced representation.

Cuspidal Representations

In the previous chapter, we classified the principal series representations. However, this left the *cuspidal* representations. These are the “atomic” particles of the theory; they cannot be constructed by inducing from parabolic subgroups, and they have no analogue in the representation theory of $\mathrm{GL}_n(\mathbb{C})$. They are the representations that carry the “actual” arithmetic information that is used in the Langlands Correspondence. The goal of this chapter is to define, analyze, and provide a general construction for cuspidal representations. Unlike the principal series, which are naturally constructed via the Bruhat decomposition, cuspidal representations are constructed using the compact open subgroups of G .

We start in Section 3.1 by giving a more workable definition of cuspidal representations via their matrix coefficients (compact support modulo the center). In Section 3.2, we introduce the machinery of *intertwining* and Hecke algebras for compact open subgroups. This leads to the main result of the chapter: a method to construct irreducible cuspidal representations by *compact induction* from representations of open subgroups that contain the center. We conclude by introducing the concept of *Fundamental Strata* (Section 3.3), which serves as the stratifying for these representations, classifying them by their level of ramification.

3.1 Cuspidal Representations and Coefficients

Our exploration of non-principal series representations interestingly enough starts by revisiting the definition of the cuspidal representation. So far, we have defined it negatively: a smooth representation (π, V) of $\mathrm{GL}_2(F)$ is cuspidal if the Jacquet module (π_N, V_N) is trivial, or equivalently, if it is not isomorphic to a composition factor of an induced representation $\mathrm{Ind}_B^G \chi$. This definition, being the “negative” of non-principal series hides the great importance these representations have in the Langlands program. Hence, we start by giving a more practical and algebraic definition.

The key insight from this new viewpoint is that irreducible cuspidal representations have an algebraic

property: they are *projective objects* in the appropriate subcategory of $\text{Rep}(G)$, as shall be elaborated on in appendix A.

Let (π, V) be a smooth representation of $G = \text{GL}_2(F)$. From vectors $v \in V$ and $\tilde{v} \in \check{V}$, we get a smooth function on G by

$$\gamma_{\tilde{v} \otimes v} : g \mapsto \langle \tilde{v}, \pi(g)v \rangle.$$

Let $C(\pi)$ be the vector space spanned by the functions $\gamma_{\tilde{v} \otimes v}$ for $\tilde{v} \in \check{V}, v \in V$. The functions $f \in C(\pi)$ are called the *(matrix) coefficients* of π . The space $\check{V} \otimes V$ carries a smooth representation of the group $G \times G$, while $G \times G$ acts on the function space $C(\pi)$ by translation: the first factor acts by left translation and the second by right translation. The map $\tilde{v} \otimes v \mapsto \gamma_{\tilde{v} \otimes v}$ is then a surjective $G \times G$ -homomorphism $\check{V} \otimes V \rightarrow C(\pi)$.

The space $C(\pi)$ is primarily, but not exclusively, of interest in the case where π is irreducible. When π is irreducible, the centre Z of G acts on V via the central character ω_π of π and

$$\gamma(zg) = \omega_\pi(z)\gamma(g), \quad z \in Z, g \in G, \gamma \in C(\pi).$$

The support of a coefficient is therefore invariant under translation by Z .

Definition 3.1.1: γ -Cuspidal Representation

Let (π, V) be an irreducible smooth representation of G ; one says that π is *γ -cuspidal* if every $\gamma \in C(\pi)$ is compactly supported modulo Z .

The term “ γ -cuspidal” is convenient, but temporary. As we shall see, it is equivalent to the standard definition. We first achieve some technical control.

Proposition 3.1.2: Admissibility and γ -Cuspidality

1. If (π, V) is an irreducible γ -cuspidal representation of G , then π is admissible.
2. Let (π, V) be an irreducible admissible representation of G , and suppose that some non-zero coefficient of π is compactly supported modulo Z ; then π is γ -cuspidal.

Proof:

1. Suppose for a contradiction that π is not admissible. We choose a compact open subgroup K such that V^K has infinite dimension. This dimension is countable (by lemma 1.2.29). The dimension of $\check{V}^K \cong \text{Hom}_{\mathbb{C}}(V^K, \mathbb{C})$ is therefore uncountable.

We fix a non-zero $v \in V^K$ and consider the map $\Gamma_v : \check{V}^K \rightarrow C(\pi)$ given by $\tilde{v} \mapsto \gamma_{\tilde{v} \otimes v}$. Since the translates gv ($g \in G$) span V , the map Γ_v is injective. Its image is a space of functions f on G satisfying

$$f(zkgk') = \omega_\pi(z)f(g), \quad g \in G, z \in Z, k, k' \in K,$$

and supported on a finite union of cosets $ZKgK$ (due to the γ -cuspidal property). The dimension of $\Gamma_v(\check{V}^K)$ is therefore at most countable, while Γ_v is injective and $\dim \check{V}^K$ is uncountable. This gives the desired contradiction.

2. The smooth dual $(\check{\pi}, \check{V})$ is irreducible and admissible (proposition 1.2.38). We view the space $\check{V} \otimes V$ as a smooth representation of $G \times G$, and hence as a smooth module over

$\mathcal{H}(G \times G) = \mathcal{H}(G) \otimes \mathcal{H}(G)$. If K is a compact open subgroup of G , we have

$$(\check{V} \otimes V)^{K \times K} = (e_K \otimes e_K) * (\check{V} \otimes V) = \check{V}^K \otimes V^K.$$

If K is sufficiently small, the spaces $\check{V}^K, V^K$ are finite-dimensional simple modules over $\mathcal{H}(G, K)$. The Jacobson Density Theorem implies that $\check{V}^K \otimes V^K$ is a simple module over $\mathcal{H}(G, K) \otimes \mathcal{H}(G, K) \cong \mathcal{H}(G \times G, K \times K)$. This holds for all sufficiently small K , so $\check{V} \otimes V$ is an irreducible admissible $G \times G$ -space.

The surjective $G \times G$ -homomorphism $\gamma : \check{V} \otimes V \rightarrow C(\pi)$ is therefore an isomorphism and $C(\pi)$ is irreducible over $G \times G$. If $\gamma \in C(\pi)$ is non-zero, any $\gamma' \in C(\pi)$ is a finite linear combination of functions $(g, h)\gamma$ for $(g, h) \in G \times G$. If γ is compactly supported modulo Z , then so is γ' .

We now justify the terminology by showing equivalence with the standard definition.

Theorem 3.1.3: Equivalence of Cuspidal Definitions

Let (π, V) be an irreducible smooth representation of G . Then π is cuspidal if and only if it is γ -cuspidal.

Proof:

We first assume that π is cuspidal ($V_N = 0$), and show it is γ -cuspidal. Let ϖ be a prime element of F , and put

$$t = \begin{pmatrix} \varpi & 0 \\ 0 & 1 \end{pmatrix}.$$

The set T^+ of powers t^n , $n \geq 0$, provides a family of representatives for $ZK \backslash G/K$, where $K = \mathrm{GL}_2(\mathfrak{o})$ (by Cartan Decomposition, proposition 2.2.3).

Lemma 3.1.4: Vanishing of Coefficients

Let $v \in V, \tilde{v} \in \check{V}$. There exists $m \geq 0$ such that $\gamma_{\tilde{v} \otimes v}(t^n) = 0$ for all $n \geq m$.

Proof:

(of Lemma) We choose a compact open subgroup N_1 of N which fixes $\tilde{v}$. Since $V_N = 0$, we have $v \in V(N)$ and by lemma 2.3.2 there is a compact open subgroup N_2 of N such that $\int_{N_2} \pi(x)v \, dx = 0$. We then have $\int_{N_0} \pi(x)v \, dx = 0$ for any compact open subgroup N_0 containing N_2 . However, there exists $m \geq 0$ such that $t^a N_2 t^{-a} \subset N_1$ for all $a \geq m$. For such a we have (for certain positive constants k_1, k_2):

$$\begin{aligned} \langle \tilde{v}, \pi(t^a)v \rangle &= k_1 \int_{N_1} \langle \check{\pi}(x^{-1})\tilde{v}, \pi(t^a)v \rangle \, dx \\ &= k_1 \int_{N_1} \langle \check{\pi}(t^{-a})\tilde{v}, \pi(t^{-a}xt^a)v \rangle \, dx \\ &= k_2 \int_{t^{-a}N_1t^a} \langle \check{\pi}(t^{-a})\tilde{v}, \pi(x)v \rangle \, dx = 0, \end{aligned}$$

since $t^{-a}N_1t^a \supset N_2$.

Returning to the proof of the theorem, fix a non-zero coefficient $f = \gamma_{\tilde{v} \otimes v}$. Write $K = \mathrm{GL}_2(\mathfrak{o})$ and let K' be an open normal subgroup of K fixing both $\tilde{v}$ and v . Let $k_1, \dots, k_r$ be coset representatives for K/K' . Thus, if $g \in G$, there exists $n \geq 0$ such that $ZKgK = ZKt^nK = \bigcup_{i,j} ZK'k_i^{-1}t^n k_j K'$. It follows that

$$\mathrm{supp} f \subset \bigcup_{1 \leq i,j \leq r} ZK'(\mathrm{supp} f_{ij} \cap T^+)K',$$

where f_{ij} denotes the coefficient function $x \mapsto f(k_i x k_j^{-1})$. This set is compact modulo Z by the lemma. Thus π is γ -cuspidal.

Converse: Let π be irreducible and γ -cuspidal. By proposition 3.1.2, π is admissible. By proposition 1.2.38, the dual $(\check{\pi}, \check{V})$ is irreducible and admissible. Let $K_n = 1 + \mathfrak{p}^n M_2(\mathfrak{o})$. Take $v \in V$ and choose $n \geq 1$ so that v is fixed by $\pi(K_n)$. For $\tilde{v} \in \check{V}^{K_n}$, the function $g \mapsto \langle \tilde{v}, \pi(g)v \rangle$ is compactly supported modulo Z ; we deduce that $\langle \tilde{v}, \pi(t^a)v \rangle = 0$ for all a sufficiently large. Since $\check{V}^{K_n}$ is finite dimensional, there is a constant c such that this holds for all $\tilde{v} \in \check{V}^{K_n}$ and all $a \geq c$. This implies $\pi(e_{K_n})\pi(t^a)v = 0$ for $a \geq c$. Writing $K_n = N_n T_n N'_n$ (Iwahori decomposition), we set $K_n^{(a)} = t^{-a} K_n t^a$. We have, for $a \geq c$:

$$0 = \pi(e_{K_n})\pi(t^a)v = \pi(t^a)\pi(e_{K_n^{(a)}})v = \pi(t^a) \sum_{x \in N_{n-a}/N_n} \pi(x)\pi(e_{K_n^{(a)} \cap K_n})v.$$

However, v is fixed by $K_n^{(a)} \cap K_n$, so this reduces to a sum over N which vanishes. We deduce that $v \in V(N)$. This applies to all $v \in V$, so π is cuspidal.

Combining the previous results, we obtain a crucial structural fact for $\mathrm{GL}_2(F)$.

Corollary 3.1.5: Admissibility of Irreducible Representations

Every irreducible smooth representation of $G = \mathrm{GL}_2(F)$ is admissible.

Proof:

If π is not cuspidal, it is admissible by proposition 2.4.6. If π is cuspidal, it is γ -cuspidal by theorem 3.1.3, and thus admissible by proposition 3.1.2.

3.2 Intertwining and Compact Induction

In this section, we describe a general method for constructing irreducible cuspidal representations of $G = \mathrm{GL}_2(F)$ using compact induction from open subgroups. While the formal framework applies to arbitrary locally profinite groups, its power lies in the specific application to the filtrations of compact subgroups in G .

3.2.1 General Context Throughout this section, G denotes a unimodular locally profinite group satisfying the countability hypothesis. Z denotes the centre of G . For a compact open subgroup K , $\widehat{K}$ denotes the set of isomorphism classes of irreducible smooth representations of K .

Definition 3.2.2: Intertwining

For $i = 1, 2$, let K_i be a compact open subgroup of G and let $\rho_i \in \widehat{K_i}$. Let $g \in G$. The element g *intertwines* ρ_1 with ρ_2 if

$$\mathrm{Hom}_{K_1^g \cap K_2}(\rho_1^g, \rho_2) \neq 0,$$

where $K_1^g = g^{-1}K_1g$ and ρ_1^g denotes the representation $x \mapsto \rho_1(gxg^{-1})$ of K_1^g .

This property depends only on the double coset K_1gK_2 . The representations ρ_1 and ρ_2 are said to *intertwine in G* if there exists some $g \in G$ which intertwines them. For a single pair (K, ρ) , the element g intertwines ρ if it intertwines ρ with itself.

Definition 3.2.3: Occurrence

Let K be a compact open subgroup of G , and let (π, V) be a smooth representation of G . The representation π is said to *contain* ρ , or ρ *occurs* in π , if

$$\mathrm{Hom}_K(\rho, \pi) \neq 0.$$

These definitions extend to the case where K is open and compact modulo Z , provided π admits a central character.

Proposition 3.2.4: Intertwining implies Common Occurrence

For $i = 1, 2$, let K_i be a compact open subgroup of G and let $\rho_i \in \widehat{K_i}$. Let (π, V) be an irreducible smooth representation of G which contains both ρ_1 and ρ_2 . Then there exists $g \in G$ which intertwines ρ_1 with ρ_2 .

Proof:

For each i , V decomposes into K_i -isotypic components (proposition 1.2.16). The hypothesis is equivalent to $V^{\rho_i} \neq 0$ for $i = 1, 2$. Let e_2 denote the K_2 -projection $V \rightarrow V^{\rho_2}$. Since π is irreducible and $V^{\rho_1} \neq 0$, the spaces $\pi(g^{-1})V^{\rho_1} = V^{\rho_1^g}$, $g \in G$, span V . Therefore, there exists $g \in G$ such that $e_2 \circ \pi(g^{-1})$ induces a non-zero map $V^{\rho_1} \rightarrow V^{\rho_2}$. This is the required element g .

There is a criterion for intertwining involving the Hecke algebra. Recall the idempotent $e_\rho \in \mathcal{H}(G)$ associated to $\rho \in \widehat{K}$ defined by $e_\rho(x) = \frac{\dim \rho}{\mu(K)} \text{tr} \rho(x^{-1})$ for $x \in K$ and 0 otherwise.

Proposition 3.2.5: Hecke Algebra Criterion

Let K be a compact open subgroup of G , let $g \in G$, and $\rho \in \widehat{K}$. The following are equivalent:

1. there exists $f \in e_\rho * \mathcal{H}(G) * e_\rho$ such that $f|_{KgK} \neq 0$;
2. g intertwines ρ .

Proof:

Consider the space $C^\infty(KgK)$ of G -smooth functions on the coset KgK . This carries a smooth representation of $K \times K$ by $(k_1, k_2)f : x \mapsto f(k_1^{-1}xk_2)$. Let H denote the group of pairs $(k, g^{-1}kg) \in K \times K$ for $k \in K \cap gKg^{-1}$. The map $f \mapsto f(g)$ is an H -homomorphism $C^\infty(KgK) \rightarrow \mathbb{C}$ (with H acting trivially). By Frobenius Reciprocity, this induces a $K \times K$ -isomorphism

$$C^\infty(KgK) \cong \text{Ind}_H^{K \times K}(1_H).$$

Condition (1) amounts to $e_\rho * C^\infty(KgK) * e_\rho \cong V^{\rho \otimes \check{\rho}} \neq 0$. Equivalently, $\text{Hom}_{K \times K}(\rho \otimes \check{\rho}, V) \cong \text{Hom}_H(\rho \otimes \check{\rho}, 1_H) \neq 0$. This last relation is equivalent to the representation $k \mapsto \rho(k) \otimes \check{\rho}(g^{-1}kg)$ of $K \cap gKg^{-1}$ having a fixed vector, which means $\text{Hom}_{K \cap gKg^{-1}}(\rho, \rho^{g^{-1}}) \neq 0$. This is precisely the condition that g intertwines ρ .

To make this condition operational, one associates a specific algebra of functions to the representation ρ . Let K be an open subgroup of G , containing and compact modulo the centre Z . Let (ρ, W) be an irreducible smooth representation of K . Let $\mathcal{H}(G, \rho)$ be the space of functions $f : G \rightarrow \text{End}_{\mathbb{C}}(W)$ which are compactly supported modulo Z and satisfy

$$f(k_1 g k_2) = \rho(k_1) f(g) \rho(k_2), \quad k_i \in K, g \in G.$$

The support of any $f \in \mathcal{H}(G, \rho)$ is a finite union of double cosets KgK . Under the convolution operation

$$\varphi_1 * \varphi_2(g) = \int_{G/Z} \varphi_1(x) \varphi_2(x^{-1}g) d\mu(x),$$

$\mathcal{H}(G, \rho)$ becomes an associative $\mathbb{C}$ -algebra with identity. It is called the ρ -spherical Hecke algebra (or intertwining algebra).

Lemma 3.2.6: Support of Hecke Functions

Let $g \in G$; there exists $\varphi \in \mathcal{H}(G, \rho)$ with support KgK if and only if g intertwines ρ .

Proof:

Let $f \in \text{End}_{\mathbb{C}}(W)$. For a fixed g , the assignment $kgk' \mapsto \rho(k)f\rho(k')$ is well-defined if and only if $f \in \text{Hom}_{K^g \cap K}(\rho^g, \rho)$. Since ρ is semisimple, the spaces $\text{Hom}(\rho^g, \rho)$ and $\text{Hom}(\rho, \rho^g)$ have the same dimension.

This algebra is intimately related to the endomorphisms of the compactly induced representation. Let $(\Sigma, X) = \text{c-Ind}_K^G \rho$. The space X consists of functions $f : G \rightarrow W$, compactly supported modulo Z , satisfying $f(kg) = \rho(k)f(g)$. For $\varphi \in \mathcal{H}(G, \rho)$ and $f \in X$, the convolution $\varphi * f$ lies in X . This defines a homomorphism of $\mathbb{C}$ -algebras:

$$\mathcal{H}(G, \rho) \longrightarrow \text{End}_G(\text{c-Ind}_K^G \rho).$$

Proposition 3.2.7: Hecke Algebra Isomorphism

The map $\mathcal{H}(G, \rho) \rightarrow \text{End}_G(\text{c-Ind}_K^G \rho)$ is an isomorphism of $\mathbb{C}$ -algebras.

Proof:

The relation $\text{End}_G(\text{c-Ind} \rho) \cong \text{Hom}_K(\rho, \text{c-Ind} \rho)$ from Frobenius Reciprocity (theorem 1.2.21) is used here. Let $\varphi_0 : w \mapsto \varphi_0^w$ be the canonical map $W \rightarrow X$ where φ_0^w is supported on K with value $\rho(k)w$. The isomorphism is composition with φ_0 . Let $\varphi \in \text{Hom}_K(W, X)$. Define $\Phi(g) \in \text{End}_{\mathbb{C}}(W)$ by $\Phi(g)w = \varphi(w)(g)$. One verifies that $\Phi \in \mathcal{H}(G, \rho)$ and this map provides the inverse.

These algebraic structures provide the necessary foundation for the central result of this section, which establishes a method for constructing irreducible cuspidal representations.

Theorem 3.2.8: Construction of Cuspidal Representations

Let K be an open subgroup of $G = \text{GL}_2(F)$, containing and compact modulo Z . Let (ρ, W) be an irreducible smooth representation of K . Suppose that an element $g \in G$ intertwines ρ if and only if $g \in K$. Then the compactly induced representation $\text{c-Ind}_K^G \rho$ is irreducible and cuspidal.

Proof:

Write $(\Sigma, X) = \text{c-Ind}_K^G \rho$. First, it is shown that Σ has a non-zero coefficient compactly supported modulo Z . By the Duality Theorem (theorem 1.3.12), $\check{X} \cong \text{Ind}_K^G \check{\rho}$. This contains $\text{c-Ind}_K^G \check{\rho}$ as a subspace. Canonical embeddings are used to identify W and $\check{W}$ with functions in X and $\check{X}$ supported on K . For $w \in W, \check{w} \in \check{W}$, the coefficient $\gamma_{\check{w} \otimes w}$ is non-zero and supported in K .

Consequently, it suffices to prove that X is irreducible. If it is, it is admissible (by corollary 3.1.5) and thus γ -cuspidal (by proposition 3.1.2), hence cuspidal. The centre Z acts on X via the central character ω_ρ . Thus X is the direct sum of its K -isotypic components. Any K -map $W \rightarrow X$ has image in X^ρ . Thus:

$$\text{Hom}_K(W, X^\rho) = \text{Hom}_K(W, X) \cong \text{End}_G(X) \cong \mathcal{H}(G, \rho).$$

The condition that g intertwines ρ iff $g \in K$ implies (by lemma 3.2.6) that functions in $\mathcal{H}(G, \rho)$ are supported on K . Thus $\dim \mathcal{H}(G, \rho) = \dim \text{End}_K(\rho) = 1$. Therefore, $\text{End}_G(X) \cong \mathbb{C}$. Since

$X^\rho \cong W$ is irreducible as a K -module and generates X as a G -module, standard arguments imply X is irreducible.

This construction principle can be immediately applied to classify the representations of level zero. Let $K = \mathrm{GL}_2(\mathfrak{o})$ and $K_1 = 1 + \mathfrak{p}M_2(\mathfrak{o})$. Let I_1 denote the preimage in K of the group of upper triangular unipotent matrices in $\mathrm{GL}_2(k)$.

Theorem 3.2.9: Classification of Level Zero Representations

Let (π, V) be an irreducible smooth representation of G , and suppose that π contains the trivial character of K_1 . Exactly one of the following holds:

1. π contains a representation λ of K , inflated from an irreducible cuspidal representation $\tilde{\lambda}$ of $\mathrm{GL}_2(k)$;
2. π contains the trivial character of I_1 .

In the first case, π is cuspidal, and there exists a representation Λ of ZK such that $\Lambda|_K \cong \lambda$ and

$$\pi \cong \mathrm{c}\text{-Ind}_{ZK}^G \Lambda.$$

Proof:

The group K stabilizes V^{K_1} , so it decomposes into representations of $K/K_1 \cong \mathrm{GL}_2(k)$. Let λ be one such component, inflated from $\tilde{\lambda}$. Either $\tilde{\lambda}$ is cuspidal (in the finite group sense) or it is not. If not, it contains the trivial character of $N(k)$, and hence λ contains the trivial character of I_1 .

It must be shown that the cases are mutually exclusive and that the structure in case (1) holds.

Lemma 3.2.10: Intertwining of Level Zero Cuspidals

For $i = 1, 2$, let $\tilde{\rho}_i$ be irreducible representations of $\mathrm{GL}_2(k)$ and ρ_i their inflations to K . Suppose $\tilde{\rho}_1$ is cuspidal.

1. The representations ρ_i intertwine in G if and only if $\tilde{\rho}_1 \cong \tilde{\rho}_2$.
2. An element $g \in G$ intertwines ρ_1 if and only if $g \in ZK$.

Proof:

Let g intertwine ρ_2 with ρ_1 . The element g can be taken to be diagonal of the form $\mathrm{diag}(\varpi^a, 1)$ for $a \geq 0$ (Cartan decomposition). If $a \geq 1$, $K_1^g \cap K$ contains the group N_0 of upper triangular unipotent matrices in K . ρ_2^g is trivial on this group. Since $\tilde{\rho}_1$ is cuspidal, it does not contain the trivial character of N_0 , so intertwining is impossible for $a \geq 1$. Thus $g \in ZK$.

By the lemma and proposition 3.2.4, the two cases cannot occur together. If $\tilde{\lambda}$ is cuspidal, π contains an extension Λ of λ to ZK . There is a non-trivial map $\mathrm{c}\text{-Ind}_{ZK}^G \Lambda \rightarrow \pi$. By the Lemma part (2) and theorem 3.2.8, $\mathrm{c}\text{-Ind} \Lambda$ is irreducible. Thus the map is an isomorphism.

3.3 Chain Orders and Fundamental Strata

This section constructs a family of compact open subgroups of $G = \mathrm{GL}_2(F)$, along with specific characters of these subgroups. This structure provides the “genetic code” for admissible representations.

Lattice Chains and Orders

Let $V = F \oplus F$ be the standard 2-dimensional vector space over F . $G = \mathrm{Aut}_F(V)$ and $A = \mathrm{End}_F(V) = M_2(F)$.

Definition 3.3.1: Lattice Chain

An $\mathfrak{o}$ -lattice chain in V is a non-empty set $\mathcal{L}$ of $\mathfrak{o}$ -lattices in V such that:

1. $\mathcal{L}$ is linearly ordered under inclusion (if $L, L' \in \mathcal{L}$, then $L \subseteq L'$ or $L' \subseteq L$).
2. $\mathcal{L}$ is stable under multiplication by $F^\times$ (if $L \in \mathcal{L}$ and $x \in F^\times$, then $xL \in \mathcal{L}$).

Given a lattice chain $\mathcal{L}$, one can index its elements as $\mathcal{L} = \{L_i\}_{i \in \mathbb{Z}}$ such that $L_{i+1} \subsetneq L_i$. Stability under $F^\times$ implies the existence of an integer $e = e(\mathcal{L})$, called the *period* of $\mathcal{L}$, such that $\varpi L_i = L_{i+e}$ for all i .

Proposition 3.3.2: Classification of Lattice Chains

Let $\mathcal{L} = \{L_i\}$ be an $\mathfrak{o}$ -lattice chain in V . Then $e(\mathcal{L})$ is either 1 or 2.

1. If $e(\mathcal{L}) = 1$, there exists $g \in G$ such that $gL_i = \mathfrak{p}^i \oplus \mathfrak{p}^i$ for all $i \in \mathbb{Z}$.
2. If $e(\mathcal{L}) = 2$, there exists $g \in G$ such that $gL_{2i} = \mathfrak{p}^i \oplus \mathfrak{p}^i$ and $gL_{2i+1} = \mathfrak{p}^i \oplus \mathfrak{p}^{i+1}$ for all $i \in \mathbb{Z}$.

Proof:

The quotient $L_0/L_e = L_0/\varpi L_0$ is a vector space over the residue field k of dimension 2. The chain $L_0 \supset L_1 \supset \cdots \supset L_e$ defines a flag of subspaces in k^2 . The length of a strictly decreasing flag in dimension 2 is at most 2. Thus $e \in \{1, 2\}$. The basis construction follows from the elementary divisor theorem.

Associated to a lattice chain is a ring of endomorphisms.

Definition 3.3.3: Chain Order

Let $\mathcal{L}$ be a lattice chain. The associated *chain order* (or hereditary order) is:

$$\mathfrak{A}_{\mathcal{L}} = \{x \in A : xL \subseteq L \text{ for all } L \in \mathcal{L}\} = \bigcap_{i \in \mathbb{Z}} \mathrm{End}_{\mathfrak{o}}(L_i).$$

We denote the period $e(\mathcal{L})$ by $e_{\mathfrak{A}}$.

By proposition 3.3.2, there are exactly two conjugacy classes of chain orders in $M_2(F)$:

1. $e_{\mathfrak{A}} = 1$: $\mathfrak{A} \cong M_2(\mathfrak{o})$ (The standard maximal order).
2. $e_{\mathfrak{A}} = 2$: $\mathfrak{A} \cong \mathcal{I} = \begin{pmatrix} \mathfrak{o} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{o} \end{pmatrix}$ (The standard Iwahori order).

Let $\mathfrak{A}$ be a chain order. Its Jacobson radical is denoted by $\mathfrak{P} = \text{rad}(\mathfrak{A})$.

Proposition 3.3.4: Radical of Chain Order

Let $\mathfrak{A}$ be a chain order with period e .

1. $\mathfrak{P} = \{x \in A : xL_i \subseteq L_{i+1} \text{ for all } i\}$.
2. $\mathfrak{P}$ is an invertible two-sided ideal of $\mathfrak{A}$, and $\mathfrak{P}^e = \mathfrak{p}\mathfrak{A}$.
3. There exists $\Pi \in G \cap \mathfrak{A}$ such that $\mathfrak{P} = \Pi\mathfrak{A} = \mathfrak{A}\Pi$. Such an element is called a *prime element* of $\mathfrak{A}$.

For the standard orders:

- If $\mathfrak{A} = M_2(\mathfrak{o})$, $\mathfrak{P} = \varpi M_2(\mathfrak{o})$. A prime element is ϖI .
- If $\mathfrak{A} = \mathcal{I}$, $\mathfrak{P} = \begin{pmatrix} \mathfrak{p} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{p} \end{pmatrix}$. A prime element is $\Pi = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$.

This structure defines a filtration of compact open subgroups. Let $U_{\mathfrak{A}} = \mathfrak{A}^{\times}$ be the group of units. For $n \geq 1$, define:

$$U_{\mathfrak{A}}^n = 1 + \mathfrak{P}^n.$$

These are compact open normal subgroups of $U_{\mathfrak{A}}$. For integers $n > m \geq 1$ such that $2m \geq n$, the map $x \mapsto 1 + x$ induces an isomorphism $\mathfrak{P}^m / \mathfrak{P}^n \xrightarrow{\sim} U_{\mathfrak{A}}^m / U_{\mathfrak{A}}^n$.

We also require the normalizer of the order in G :

$$\mathcal{K}_{\mathfrak{A}} = \{g \in G : g\mathfrak{A}g^{-1} = \mathfrak{A}\}.$$

If $\mathfrak{A} = M_2(\mathfrak{o})$, $\mathcal{K}_{\mathfrak{A}} = F^{\times} \text{GL}_2(\mathfrak{o})$. If $\mathfrak{A} = \mathcal{I}$, $\mathcal{K}_{\mathfrak{A}}$ is generated by $F^{\times} \mathcal{I}$ and the prime element $\Pi = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$. Note that $\mathcal{K}_{\mathfrak{A}}$ is open, contains Z , and is compact modulo Z .

Strata and Characters

We fix an additive character $\psi \in \widehat{F}$ of level one (kernel $\mathfrak{p}$). Define a character on the algebra A by $\psi_A(x) = \psi(\text{tr}(x))$. This induces an isomorphism $A \rightarrow \widehat{A}$ via $a \mapsto (x \mapsto \psi_A(ax))$. Under this duality, if $\mathfrak{A}$ is a chain order, the dual lattice of $\mathfrak{P}^n$ is $\mathfrak{P}^{1-n}$.

Definition 3.3.5: Stratum

A *stratum* in A is a triple $(\mathfrak{A}, n, \alpha)$ consisting of:

- A chain order $\mathfrak{A}$;
- An integer $n \geq 0$;
- An element $\alpha \in \mathfrak{P}^{-n}$.

Two strata $(\mathfrak{A}, n, \alpha_1)$ and $(\mathfrak{A}, n, \alpha_2)$ are *equivalent* if $\alpha_1 \equiv \alpha_2 \pmod{\mathfrak{P}^{1-n}}$.

If $n \geq 1$, a stratum $(\mathfrak{A}, n, \alpha)$ determines a character ψ_α on the group $U_{\mathfrak{A}}^n$:

$$\psi_\alpha(1+x) = \psi_A(\alpha x), \quad x \in \mathfrak{P}^n.$$

Since $\alpha \in \mathfrak{P}^{-n}$ and $x \in \mathfrak{P}^n$, the product is in $\mathfrak{A}$. However, the character is well-defined on the quotient $U_{\mathfrak{A}}^n/U_{\mathfrak{A}}^{n+1}$ because if $y \in \mathfrak{P}^{n+1}$, $\text{tr}(\alpha y) \in \text{tr}(\mathfrak{P}) = \mathfrak{p}$, on which ψ is trivial. The equivalence class of the stratum corresponds bijectively to the character ψ_α .

Proposition 3.3.6: Intertwining of Strata

Let $(\mathfrak{A}, n, \alpha)$ and $(\mathfrak{A}, n, \beta)$ be strata with $n \geq 1$. Let $g \in G$. The following are equivalent:

1. g intertwines the character ψ_α with ψ_β (on $U_{\mathfrak{A}}^n$).
2. The cosets intersect: $g^{-1}(\alpha + \mathfrak{P}^{1-n})g \cap (\beta + \mathfrak{P}^{1-n}) \neq \emptyset$.

Proof:

Let $K_n = U_{\mathfrak{A}}^n$. The intertwining condition is that $\psi_\alpha(gxg^{-1}) = \psi_\beta(x)$ for $x \in K_n \cap g^{-1}K_n g$. In additive terms, $\psi_A(\alpha gxg^{-1}) = \psi_A(g^{-1}\alpha gx) = \psi_A(\beta x)$ for all $x \in \mathfrak{P}^n \cap g^{-1}\mathfrak{P}^n g$. This means $g^{-1}\alpha g - \beta$ lies in the dual of the lattice $\mathfrak{P}^n \cap g^{-1}\mathfrak{P}^n g$. By lattice duality properties, $(\mathfrak{P}^n \cap g^{-1}\mathfrak{P}^n g)^* = \mathfrak{P}^{1-n} + g^{-1}\mathfrak{P}^{1-n}g$. Thus $g^{-1}\alpha g - \beta = y_1 + g^{-1}y_2g$ with $y_i \in \mathfrak{P}^{1-n}$, which rearranges to the intersection condition.

Definition 3.3.7: Fundamental Stratum

A stratum $(\mathfrak{A}, n, \alpha)$ is *fundamental* if the coset $\alpha + \mathfrak{P}^{1-n}$ contains no nilpotent element of A .

Proposition 3.3.8: Criterion for Fundamental Strata

A stratum $(\mathfrak{A}, n, \alpha)$ is fundamental if and only if for every $r \geq 1$, $\alpha^r \notin \mathfrak{P}^{1-rn}$.

This concept is linked to the normalized level of a representation.

Definition 3.3.9: Normalized Level

Let π be an irreducible smooth representation of G . The *normalized level* is

$$\ell(\pi) = \min\{n/e_{\mathfrak{A}} : (\mathfrak{A}, n) \text{ such that } \pi \text{ contains the trivial character of } U_{\mathfrak{A}}^{n+1}\}.$$

Theorem 3.3.10: Strata and Representation Level

Let π be an irreducible smooth representation. Suppose π contains the character ψ_α associated to a stratum $(\mathfrak{A}, n, \alpha)$ with $n \geq 1$. Then $(\mathfrak{A}, n, \alpha)$ is fundamental if and only if $\ell(\pi) = n/e_{\mathfrak{A}}$.

Proof:

If the stratum is not fundamental, one can approximate α by a nilpotent element, or conjugate the order to show that ψ_α is trivial on a subgroup $U_{\mathfrak{A}'}^{n'}$, where $n'/e_{\mathfrak{A}'} < n/e_{\mathfrak{A}}$. If it is fundamental, intertwining arguments show the level cannot be lowered.

3.4 Classification of Fundamental Strata

We classify fundamental strata $(\mathfrak{A}, n, \alpha)$ based on the arithmetic of the element α . Let $\nu_{\mathfrak{A}}(\alpha) = -n$.

Definition 3.4.1: Types of Strata

Let $(\mathfrak{A}, n, \alpha)$ be a fundamental stratum.

1. **Essentially Scalar:** $e_{\mathfrak{A}} = 1$ and the image of α in $\mathfrak{A}/\mathfrak{P} \cong M_2(k)$ has a repeated non-zero eigenvalue.
2. **Split:** $e_{\mathfrak{A}} = 1$ and the image of α has distinct eigenvalues in k .
3. **Unramified Simple:** $e_{\mathfrak{A}} = 1$ and the characteristic polynomial of the image of α is irreducible over k .
4. **Ramified Simple:** $e_{\mathfrak{A}} = 2$ and n is odd.

The last two cases are collectively called **Simple Strata**. They are characterized by the fact that the algebra $E = F[\alpha]$ is a field, and α is “minimal” in the following sense:

Definition 3.4.2: Minimal Element

An element $\alpha \in G \setminus Z$ is *minimal over F* if $E = F[\alpha]$ is a field, and setting $n = -\nu_E(\alpha)$:

- If E/F is ramified, n is odd.
- If E/F is unramified, $\gcd(n, 2) = 1$ (relative to the valuation of E , this corresponds to n being coprime to the ramification index, which is 1 here, but the definition is formalized via the residue characteristic polynomial).

Proposition 3.4.3: Properties of Simple Strata

Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum. Then:

1. α is minimal over F .
2. $E^\times \subset \mathcal{K}_{\mathfrak{A}}$.
3. $e(E|F) = e_{\mathfrak{A}}$.

3.5 The Classification of Cuspidal Representations

We assume π is an irreducible smooth representation of G .

3.5.1 Exhaustion

Proposition 3.5.1: Principal Series Strata

Let χ be a character of T . Let $\Sigma = \text{Ind}_B^G \chi$.

1. If χ is sufficiently ramified, Σ contains a Split fundamental stratum.
2. If $\chi_1 \chi_2^{-1}$ is trivial on units but ramified, Σ contains an Essentially Scalar stratum.
3. If χ is unramified, Σ contains the trivial character of the Iwahori U_I^1 .

Using this, we characterize cuspidals.

Theorem 3.5.2: Exhaustion Theorem

Let π be an irreducible smooth representation of G . Assume $\ell(\pi) \leq \ell(\chi\pi)$ for all characters χ of $F^\times$ (“minimal level”). The following are equivalent:

1. π is cuspidal.
2. Either:
 - (a) $\ell(\pi) = 0$ and π contains a representation of $K = \text{GL}_2(\mathfrak{o})$ inflated from an irreducible *cuspidal* representation of $\text{GL}_2(k)$.
 - (b) $\ell(\pi) > 0$ and π contains a *Simple Stratum*.

Proof:

If π is not cuspidal, it is a subquotient of a principal series. By proposition 3.5.1, it must contain a split or essentially scalar stratum, or be level 0 non-cuspidal. Conversely, the intertwining properties of simple strata (below) prevent them from occurring in induced representations from B .

3.5.2 Cuspidal Types

To generate the representations, we extend the simple stratum character. Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum with $E = F[\alpha]$. We define a filtration of open subgroups in $\mathcal{K}_{\mathfrak{A}}$:

$$H_{\alpha} = U_E^1 U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}, \quad J_{\alpha} = U_E U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}, \quad J = E^{\times} U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}.$$

(Note: if n is odd, $J_{\alpha} = J$). The character ψ_{α} on $U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}$ extends naturally to H_{α} (using $\psi_A(\alpha \cdot)$).

Theorem 3.5.3: Intertwining of Simple Characters

Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum.

1. An element $g \in G$ intertwines the character ψ_{α} on $U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}$ if and only if $g \in J$.
2. For any irreducible representation Λ of J containing ψ_{α} , the induced representation $\text{c-Ind}_J^G \Lambda$ is irreducible and cuspidal.

Definition 3.5.4: Cuspidal Type

A *cuspidal type* is a triple $(\mathfrak{A}, J, \Lambda)$ where:

1. $\mathfrak{A}$ is a chain order.
2. J is an open subgroup of $\mathcal{K}_{\mathfrak{A}}$ as defined above (associated to a simple stratum or level 0).
3. Λ is an irreducible representation of J constructed as follows:
 - **Level 0:** $J = F^{\times} \text{GL}_2(\mathfrak{o})$, Λ is an extension of a cuspidal rep of $\text{GL}_2(k)$.
 - **Positive Level:** $\Lambda|_{U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}}$ is a multiple of ψ_{α} for a simple stratum α .

Theorem 3.5.5: Classification of Cuspidal Representations

The mapping

$$(\mathfrak{A}, J, \Lambda) \mapsto \pi_{\Lambda} = \text{c-Ind}_J^G \Lambda$$

induces a bijection between the set of G -conjugacy classes of cuspidal types and the set of equivalence classes of irreducible cuspidal representations of G .

Proof:

Injectivity: Follows from the Intertwining Theorem. Since g intertwines Λ iff $g \in J$, the Hecke algebra $\mathcal{H}(G, \Lambda)$ is 1-dimensional, implying irreducibility. **Surjectivity:** By the Exhaustion Theorem, π contains a simple stratum (or is level 0). If level > 0 , $\pi|_{U_{\mathfrak{A}}^{\lfloor n/2 \rfloor + 1}}$ contains ψ_{α} . One then ascends the hierarchy $H_{\alpha} \subset J_{\alpha} \subset J$ using Clifford theory (Heisenberg groups appear here for J_{α}/H_{α}). This proves π contains some Λ of the required type.

3.6 Representations with Iwahori-Fixed Vectors

We consider representations π such that $V^I \neq 0$, where I is the Iwahori subgroup.

Proposition 3.6.1: Structure of Iwahori-Fixed Representations

Let (π, V) be irreducible. If $V^I \neq 0$, then π is a subquotient of $\text{Ind}_B^G \chi$ where χ is an unramified character of T . Specifically, π is either:

1. An unramified principal series $\text{Ind}_B^G \chi$ (χ regular).
2. An unramified twist of the Steinberg representation $\chi \otimes \text{St}_G$.
3. An unramified character $\chi \circ \det$.

In particular, such representations are never cuspidal.

Proof:

The category of smooth representations generated by their I -fixed vectors is equivalent to the category of modules over the Hecke algebra $\mathcal{H}(G, I) = e_I * \mathcal{H}(G) * e_I$. The algebra $\mathcal{H}(G, I)$ is generated by two elements T_w (supported on IwI) and T_Π (supported on $IIII$). These satisfy the Affine Hecke Algebra relations (e.g., quadratic relation for T_w). The finite dimensional irreducible modules of this algebra are 1 or 2 dimensional. These correspond precisely to the listed representation types via the functor $M \mapsto \mathcal{H}(G) \otimes_{\mathcal{H}(G, I)} M$.

This analysis is crucial for establishing the square-integrability of the Steinberg representation.

Theorem 3.6.2: Square-Integrability of Steinberg

The Steinberg representation St_G is square-integrable modulo the center. That is, for matrix coefficients f ,

$$\int_{G/Z} |f(g)|^2 d\dot{g} < \infty.$$

Proof:

Let $f(g) = \langle \tilde{v}, \pi(g)v \rangle$ where v is I -fixed. Using the Iwahori decomposition and the specific values of the Hecke operators on the unique I -fixed line in St_G , one calculates that $|f(g)|$ behaves like $q^{-\ell(w)/2}$ where $\ell(w)$ is the length of the element in the affine Weyl group. The number of elements of length k grows linearly/polynomially, while the decay is exponential, ensuring convergence.

Quadratic Tame Extensions

Paramaterizing Tame Cuspidals

We have seen that cuspidal representations are constructed by compact induction, but the construction in the previous chapter was abstract. To establish the Langlands correspondence, we need an explicit parameterization of these representations that can be matched against Galois representations. The goal of this chapter is to provide this explicit parameterization for the “tame” case (when the residual characteristic p is odd). We will show that tame cuspidal representations are in bijection with *Admissible Pairs* $(E/F, \xi)$, consisting of a quadratic extension and a character.

We begin in Section 4.1 by defining admissible pairs and analyzing the arithmetic of tame quadratic extensions. In Section 4.2, we explicitly construct a cuspidal representation π_ξ for each pair $(E/F, \xi)$ using the machinery of fundamental strata. Finally, in Section 4.3, we prove the Parametrization Theorem, showing that this construction yields a bijection between admissible pairs and the set of tame cuspidal representations. This provides the “Group side” of the correspondence that we will later match to the “Galois side.”

4.1 Admissible Pairs

We rely on the arithmetic of tame extensions. Throughout, let $\psi \in \widehat{F}$ be a character of level one. As a reminder:

Definition 4.1.1: Tamely Ramified Extension

A finite field extension E/F is *tamely ramified* if $p \nmid e(E/F)$. Equivalently, $\mathrm{tr}_{E/F}(\mathfrak{o}_E) = \mathfrak{o}_F$.

If $p \neq 2$, every quadratic extension is tamely ramified. If $p = 2$, only the unramified quadratic extension is tame.

Lemma 4.1.2: Properties of Tame Extensions

Let E/F be a tamely ramified extension with ramification index e .

1. For $r \in \mathbb{Z}$, $\text{tr}_{E/F}(\mathfrak{p}_E^{1+r}) = \mathfrak{p}_F^{1+\lfloor r/e \rfloor}$.
2. For $m \geq 1$, the norm $N_{E/F}$ induces an isomorphism

$$U_E^{em}/U_E^{em+1} \xrightarrow{\sim} U_F^m/U_F^{m+1}.$$

3. The map $N_{E/F} : U_E/U_E^1 \rightarrow U_F/U_F^1$ is surjective if E/F is unramified, and has kernel/-cokernel of order $\gcd(e, q-1)$ if totally ramified.

Proof:

Property (1) follows from the fact that the inverse different $\mathfrak{d}_{E/F}^{-1} = \mathfrak{p}_E^{1-e}$. Property (2) follows from the expansion $N(1+x) = 1 + \text{tr}(x) + \dots$. See [6] for more details.

Define $\psi_E = \psi \circ \text{tr}_{E/F}$. By the lemma, ψ_E has level one in E .

Proposition 4.1.3: Norm and Character Levels

Let χ be a character of $F^\times$ of level $m \geq 1$. Let $\chi_E = \chi \circ N_{E/F}$. Then χ_E has level em . If $c \in \mathfrak{p}_F^{-m}$ satisfies $\chi(1+x) = \psi(cx)$ for $x \in \mathfrak{p}_F^{\lfloor m/2 \rfloor + 1}$, then

$$\chi_E(1+y) = \psi_E(cy), \quad y \in \mathfrak{p}_E^{\lfloor em/2 \rfloor + 1}.$$

Definition 4.1.4: Admissible Pair

A pair $(E/F, \chi)$ is called *admissible* if:

1. E/F is a tamely ramified quadratic extension.
2. χ is a character of $E^\times$.
3. χ does not factor through the norm $N_{E/F}$ (i.e., $\chi \neq \varphi \circ N_{E/F}$ for any $\varphi \in \widehat{F^\times}$).
4. If $\chi|_{U_E^1}$ factors through $N_{E/F}$, then E/F is unramified.

Admissible pairs are isomorphic if there is an F -isomorphism $j : E \rightarrow E'$ such that $\chi = \chi' \circ j$. Let $P_2(F)$ denote the set of isomorphism classes.

Definition 4.1.5: Minimality of Pairs

An admissible pair $(E/F, \chi)$ is *minimal* if $\chi|_{U_E^n}$ does not factor through $N_{E/F}$, where n is the level of χ .

Any admissible pair is isomorphic to a twist $(E/F, \chi' \otimes \varphi_E)$ where $(E/F, \chi')$ is minimal.

Proposition 4.1.6: Minimality Criterion

Let $(E/F, \chi)$ be admissible with χ of level $m \geq 1$. Let $\alpha \in \mathfrak{p}_E^{-m}$ satisfy $\chi(1+x) = \psi_E(\alpha x)$ for $x \in \mathfrak{p}_E^m$. Then $(E/F, \chi)$ is minimal if and only if the element α is *minimal over F* .

Proof:

Factoring through the norm on the unit group is equivalent to the additive character condition $\psi_E(\alpha x) = \psi_E(cx)$ for some $c \in F$, which means $\alpha \equiv c \pmod{\mathfrak{p}_E^{1-m}}$. This contradicts minimality of the element α .

4.2 Construction of Representations

We associate an irreducible cuspidal representation π_χ to each admissible pair $(E/F, \chi)$.

4.2.1 Level Zero Construction

Let $(E/F, \chi)$ be an admissible pair where χ has level zero. By definition, E/F must be unramified.

Lemma 4.2.1: Admissibility at Level Zero

Let E/F be unramified quadratic and χ a character of level zero. Let $\sigma \in \text{Gal}(E/F)$ be the non-trivial element. The following are equivalent:

1. $(E/F, \chi)$ is admissible.
2. $\chi \neq \chi^\sigma$.
3. $\chi|_{U_E} \neq \chi^\sigma|_{U_E}$.

Proof:

Since E/F is unramified, $E^\times = F^\times U_E$. The kernel of the norm consists of x/x^σ . χ factors through the norm iff $\chi(x/x^\sigma) = 1$ iff $\chi = \chi^\sigma$.

Embed E into $A = M_2(F)$ such that $E^\times$ normalizes the maximal order $\mathfrak{A} = M_2(\mathfrak{o})$. This induces an embedding of residue fields $k_E \rightarrow M_2(k)$. The character $\chi|_{U_E}$ inflates a character $\tilde{\chi}$ of $k_E^\times$. Since $\tilde{\chi} \neq \tilde{\chi}^q$, this generates a cuspidal representation $\tilde{\pi}$ of $\text{GL}_2(k)$ (see theorem 2.1.14). Let Λ be the extension of the inflation of $\tilde{\pi}$ to $K_{\mathfrak{A}} = F^\times \text{GL}_2(\mathfrak{o})$ such that $\Lambda|_{F^\times}$ is a multiple of $\chi|_{F^\times}$.

Definition 4.2.2: Representation π_χ (Level 0)

For a level zero admissible pair, define $\pi_\chi = \text{c-Ind}_{K_{\mathfrak{A}}}^G \Lambda$.

Proposition 4.2.3: Bijection at Level Zero

The map $(E/F, \chi) \mapsto \pi_\chi$ induces a bijection between $P_2(F)_0$ and $\mathcal{A}_2^0(F)_0$ (cuspidals of level 0).

4.2.2 Positive Level Construction

Let $(E/F, \chi)$ be a minimal admissible pair with χ of level $n \geq 1$. Choose $\alpha \in \mathfrak{p}_E^{-n}$ realizing χ as a character (via ψ_E). Embed E in A such that $E^\times \subset K_{\mathfrak{A}}$ for a chain order $\mathfrak{A}$ (where $e_{\mathfrak{A}} = e(E/F)$). The triple $(\mathfrak{A}, n, \alpha)$ is a simple stratum. Define the groups $J = E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$ and $J^1 = U_E^1 U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$. We must construct a representation $\Lambda \in C(\psi_\alpha, \mathfrak{A})$ of J .

Case 1: n is odd. In this case, $J = E^\times U_{\mathfrak{A}}^{(n+1)/2}$. The character ψ_α is defined on $U_{\mathfrak{A}}^{(n+1)/2}$.

Definition 4.2.4: Representation Λ (Odd)

Define the character Λ of J by:

$$\Lambda|_{U_{\mathfrak{A}}^{(n+1)/2}} = \psi_\alpha, \quad \Lambda|_{E^\times} = \chi.$$

These are consistent since χ corresponds to α .

Case 2: n is even. Here $n = 2m$. E/F is unramified. $J = E^\times U_{\mathfrak{A}}^m$. Let $H^1 = U_E^1 U_{\mathfrak{A}}^{m+1}$ and $J^1 = U_E^1 U_{\mathfrak{A}}^m$. Define a character θ on H^1 by $\theta(ux) = \chi(u)\psi_\alpha(x)$ for $u \in U_E^1, x \in U_{\mathfrak{A}}^{m+1}$.

Lemma 4.2.5: Heisenberg Extension

There is a unique irreducible representation η of J^1 such that $\eta|_{H^1}$ contains θ .

1. $\dim \eta = q$.
2. $\eta|_{H^1}$ is a multiple of θ .
3. There is a unique extension $\tilde{\eta}$ of η to the group $\mu_E J^1$ (where μ_E are roots of unity) such that $\text{tr} \tilde{\eta}(\zeta) = -\chi(\zeta)$ for $\zeta \in \mu_E \setminus \mu_F$.

Proof:

The quotient $J^1/H^1 \cong \mathfrak{p}_F^m/(\mathfrak{p}_E^m + \mathfrak{p}_F^{m+1}) \cong k \oplus k$. The character commutator form is non-degenerate symplectic. This is the unique representation of the finite Heisenberg group. The trace condition fixes the extension to the cyclic group acting on the Heisenberg group.

Corollary 4.2.6: Representation Λ (Even)

There is a unique irreducible representation Λ of J such that:

1. $\Lambda|_{J^1} \cong \eta$.
2. $\Lambda|_{F^\times}$ is a multiple of $\chi|_{F^\times}$.
3. For $\zeta \in \mu_E \setminus \mu_F$, $\text{tr}\Lambda(\zeta) = -\chi(\zeta)$.

Definition 4.2.7: Representation π_χ (Positive Level)

Let Λ be as defined in Case 1 or Case 2. Define $\pi_\chi = \text{c-Ind}_J^G \Lambda$.

Proposition 4.2.8: Independence of Choices

The equivalence class of π_χ depends only on the isomorphism class of $(E/F, \chi)$.

1. It is independent of the choice of α , ψ , and the embedding $E \rightarrow A$.
2. If φ is a character of $F^\times$ such that $(E/F, \chi\varphi_E)$ is minimal, then $\pi_{\chi\varphi_E} = \varphi\pi_\chi$.

Proof:

Any two embeddings are conjugate. Changing α changes ψ_α by a conjugation or a unit, which is absorbed by the uniqueness of the Heisenberg representation.

This allows us to define π_χ for any admissible pair (not necessarily minimal) by twisting: $\pi_\chi = \varphi\pi_{\chi'}$ where $\chi = \chi'\varphi_E$.

4.3 The Parametrization Theorem

We denote $\mathcal{A}_2^0(F)$ the set of irreducible cuspidal representations. $\mathcal{A}_2^{\text{ur}}(F)$ is the subset of unramified representations (twists of level 0 types).

Theorem 4.3.1: Tame Parametrization Theorem

The construction $(E/F, \chi) \mapsto \pi_\chi$ induces a bijection:

$$P_2(F) \xrightarrow{\sim} \begin{cases} \mathcal{A}_2^0(F) & \text{if } p \neq 2 \\ \mathcal{A}_2^{\text{nr}}(F) & \text{if } p = 2 \end{cases}$$

Properties:

1. $\ell(\pi_\chi) = \ell(\chi)/e(E|F)$.
2. $\omega_{\pi_\chi} = \chi|_{F^\times}$.
3. $\pi_{\check{\chi}} = \check{\pi}_\chi$.
4. $\pi_{\chi\varphi_E} = \varphi\pi_\chi$.

Lemma 4.3.2: Unramified Recognition

A cuspidal representation π is unramified (in $\mathcal{A}_2^{\text{nr}}(F)$) if and only if the chain order $\mathfrak{A}$ in its cuspidal type is isomorphic to $M_2(\mathfrak{o})$.

4.4 Tame Intertwining Properties

We prove the injectivity of the map constructed above.

Proposition 4.4.1: Injectivity of Parametrization

The map $(E/F, \chi) \mapsto \pi_\chi$ is injective on isomorphism classes of minimal pairs.

Proof:

Suppose $\pi_{\chi_1} \cong \pi_{\chi_2}$. Central characters imply $\chi_1|_{F^\times} = \chi_2|_{F^\times}$. The levels must match. The representations contain simple strata $(\mathfrak{A}, n, \alpha_i)$. These strata must intertwine, meaning the characters ψ_{α_i} are conjugate.

Lemma 4.4.2: Conjugacy of Minimal Elements

Let α_1, α_2 be minimal elements generating extensions E_1, E_2 . If their characters intertwine, there exists $u \in U_{\mathfrak{A}}$ such that $uE_1u^{-1} = E_2$ and $u\alpha_1u^{-1} \equiv \alpha_2 \pmod{U_{E_2}^1}$.

Using this lemma, we align the fields $E_1 \cong E_2$. The equality of representations implies equality of the inducing types Λ_i . For odd level, $\Lambda = \chi$, so $\chi_1 = \chi_2$. For even level, the trace relation on roots of unity recovers the character on the full group $E^\times$.

Proposition 4.4.3: Minimality Converse

Let $\pi \in \mathcal{A}_2^0(F)$ (or $\mathcal{A}_2^{\text{nr}}(F)$ if $p = 2$). Suppose $\ell(\pi) \leq \ell(\varphi\pi)$ for all φ . Then there exists a minimal pair $(E/F, \chi)$ such that $\pi \cong \pi_\chi$.

Proof:

π contains a cuspidal type $(\mathfrak{A}, J, \Lambda)$. If $\mathfrak{A} = \mathcal{I}$, we are in the ramified case (odd level), and $\chi = \Lambda|_{E^\times}$ works. If $\mathfrak{A} = M_2(\mathfrak{o})$ and level > 0 , we are in the even level case. We reconstruct χ from $\Lambda|_{J^1}$ (which gives the restriction to units) and the character formula on roots of unity (which gives the rest).

4.5 A Certain Group Extension

We fix a prime number p . Let G be a finite p -group with cyclic centre Z , such that $V = G/Z$ is an elementary abelian p -group. Let θ be a faithful character of Z . The alternating pairing

$$h : V \times V \longrightarrow \mathbb{C}^\times, \quad (x, y) \longmapsto \theta([x, y]),$$

is non-degenerate. By the Stone-von Neumann theorem, there is a unique irreducible representation η of G such that $\eta|_Z$ contains θ .

We introduce a group of automorphisms. Let A be a finite cyclic group of automorphisms of G , of order relatively prime to p . Assume A acts trivially on Z . Since η is the unique representation with central character θ , and A fixes θ , the representation η is stable under A . Thus, η admits an extension to an irreducible representation of the semi-direct product $AG = A \ltimes G$.

Lemma 4.5.1: Extensions of Heisenberg Representations

Let η_0 be some irreducible representation of AG such that $\eta_0|_G \cong \eta$. Let χ range over the irreducible characters of $AG/G \cong A$. The representations $\chi \otimes \eta_0$ are distinct and

$$\text{Ind}_G^{AG} \eta \cong \bigoplus_{\chi \in \hat{A}} (\chi \otimes \eta_0).$$

Proof:

The determinant of the representation $\chi \otimes \eta_0$ is $\chi^{\dim \eta} \det \eta_0$. Since $\dim \eta = |V|^{1/2}$ is a power of p and $|A|$ is relatively prime to p , the map $\chi \mapsto \chi^{\dim \eta}$ is a permutation of $\hat{A}$. Thus the determinants are distinct, implying the representations are distinct. By Frobenius Reciprocity, $\text{Ind}_G^{AG} \eta$ contains each $\chi \otimes \eta_0$. Comparing dimensions ($|A| \dim \eta$ vs $\sum 1 \cdot \dim \eta$), we see they exhaust the induced representation.

We now specialize to the case relevant to GL_2 . We impose the following hypotheses:

- (a) $|V| = q^2$, where q is a power of p .
- (b) $|A| = q + 1$.

(c) If $a \in A, a \neq 1$, then a acts on V with only the trivial fixed point (regular action).

Lemma 4.5.2: The Trace -1 Extension

Under hypotheses (a)–(c), there is a unique irreducible representation η_1 of AG such that:

1. $\eta_1|_G \cong \eta$.
2. $\text{tr}\eta_1(a) = -1$ for all $a \in A, a \neq 1$.

Proof:

Let $\eta_i, 1 \leq i \leq q+1$, be the distinct extensions of η to AG . Consider the induced representation

$$\xi = \text{Ind}_{AZ}^{AG}(1_A \otimes \theta),$$

where $1_A \otimes \theta$ is the character $(a, z) \mapsto \theta(z)$ of the abelian group $AZ \cong A \times Z$. We decompose ξ . Since $\xi|_G \cong \text{Ind}_Z^G \theta \cong (\dim \eta)\eta = q\eta$, we must have

$$\xi \cong \bigoplus_{i=1}^{q+1} m_i \eta_i, \quad \text{with } \sum m_i = q.$$

We compute the character of ξ restricted to AZ .

$$\xi|_{AZ} \cong \bigoplus_{g \in AZ \backslash AG / AZ} \text{Ind}_{AZ \cap g^{-1}AZg}^{AZ}(1_A \otimes \theta)^g.$$

The double coset space $AZ \backslash AG / AZ$ is isomorphic to the orbit space V/A . By hypothesis (c), A acts freely on $V \setminus \{0\}$. Thus we have one trivial orbit (size 1) and $q-1$ regular orbits (size $q+1$). For the trivial orbit ($g=1$), we get $1_A \otimes \theta$. For non-trivial orbits, the intersection of groups is just Z , so we induce from Z to AZ , obtaining $\text{Reg}_A \otimes \theta$.

$$\xi|_{AZ} \cong (1_A \otimes \theta) \oplus (q-1)(\text{Reg}_A \otimes \theta).$$

Calculating the inner product $\langle \xi, \xi \rangle_{AG} = \langle \xi|_{AZ}, 1_A \otimes \theta \rangle_{AZ} = 1 + (q-1) \cdot 1 = q$. Since $\sum m_i = q$ and $\sum m_i^2 = q$, it must be that one m_i is 0 and the rest are 1. Renumber so $m_1 = 0$. Then $\xi = (\bigoplus \eta_i) - \eta_1$. Since $\bigoplus \eta_i \cong \text{Ind}_G^{AG} \eta$, its trace on any $a \in A \setminus \{1\}$ is 0. Therefore, $\text{tr}\xi(a) = -\text{tr}\eta_1(a)$. From the explicit formula above, $\text{tr}\xi(a) = 1 + (q-1)\text{tr}(\text{Reg}_A)(a) = 1 + 0 = 1$. Thus $\text{tr}\eta_1(a) = -1$.

To ensure this applies to our context, we analyze the conjugacy classes.

Lemma 4.5.3: Conjugacy in AG

Let $a \in A, a \neq 1$.

1. Let $g \in G$; the element ag is G -conjugate to an element of the form $az, z \in Z$.
2. For $z \in Z$, the G -conjugacy class and the AG -conjugacy class of az are the same, and this class has q^2 elements.
3. For $z_1, z_2 \in Z$, the elements az_1, az_2 are AG -conjugate if and only if $z_1 = z_2$.

Proof:

Consider the map $\varphi_a : V \rightarrow V$ given by $v \mapsto av a^{-1}v^{-1}$. Since A acts without fixed points on V , this map is injective, hence bijective. This implies every ag can be conjugated by G into aZ , proving (1). Since A is abelian, A centralizes az . Thus the AG -conjugacy class is the same as the G -conjugacy class. The size is $|G|/|C_G(az)|$. The centralizer condition $gag^{-1} = a$ modulo Z is equivalent to g being a fixed point of a in V , which implies $g \in Z$. Thus the centralizer is Z , and the class size is $|G|/|Z| = |V| = q^2$. Part (3) follows because the commutator of AG is contained in G , so $AG/G \cong A$ is abelian. Conjugacy preserves the projection to A .

We verify that the construction in section 4.2 satisfies these hypotheses. We have $n = 2m$.

- $G = J_\alpha^1 / \ker \theta$.
- $Z = H_\alpha^1 / \ker \theta$.
- θ is the character defined by ψ_α and χ .

We have $V = G/Z \cong J_\alpha^1/H_\alpha^1 \cong \mathfrak{p}_F^m/(\mathfrak{p}_E^m + \mathfrak{p}_F^{m+1})$. We identify $A/P \cong M_2(k)$. Since E/F is unramified, the image of $\mathfrak{o}_E$ is the field $k_E \subset M_2(k)$. Then $V \cong M_2(k)/k_E$. This has dimension 2 over k (size q^2). The group $A = \mu_E/\mu_F \cong k_E^\times/k^\times$ has order $(q^2 - 1)/(q - 1) = q + 1$. The action is by conjugation. A scalar $x \in k_E^\times$ fixes $y \in M_2(k)$ iff they commute. Since k_E is a maximal subfield, the centralizer is k_E . Thus the action on the quotient $M_2(k)/k_E$ is free. All hypotheses are satisfied.

5

Characterizing Representations using Functional Equations

We have classified representations of G , but to relate them to Galois groups, we need invariant objects that exist on both sides of the fence. In particular, we shall generalize the notion of characters from [3] and the functional theory of [6] to create new invariants. The goal of this chapter is to define these invariants – the Local L -function $L(\pi, s)$ and the Local Constant (or ε -factor) $\varepsilon(\pi, s, \psi)$ – and to prove that they completely characterize the representation.

We start in Section 5.1 with the thesis of John Tate, establishing the functional equation and ε -factors for $\mathrm{GL}_1(F)$ (characters). We then upgrade this to $\mathrm{GL}_2(F)$ in Section 5.2 using the Godement-Jacquet theory of Zeta integrals. We provide explicit formulas for the ε -factors of cuspidal representations using Gauss sums. The key result of this chapter is the *Converse Theorem* (Section 5.5), which asserts that if two representations have matching L and ε factors for all twists, they must be isomorphic. This theorem is the engine that makes the Langlands correspondence unique.

5.1 Functional Equation for $\mathrm{GL}(1)$

Recall that if $a \in F$ and $\psi \in \widehat{F}$, then $a\psi$ denotes the character $x \mapsto \psi(ax)$. If $\psi \neq 1$, the map $a \mapsto a\psi$ gives an isomorphism $F \cong \widehat{F}$

Definition 5.1.1: Fourier Transform

Let $\psi \in \widehat{F}$, $\psi \neq 1$ to define an isomorphism $F \cong \widehat{F}$, and let μ be a Haar measure on F . For $\Phi \in C_c^\infty(F)$, define the *Fourier transform* $\hat{\Phi}$ of Φ (with respect to μ and ψ) to be:

$$\hat{\Phi}(x) = \int_F \Phi(y) \psi(xy) d\mu(y), \quad x \in F.$$

The integrand is locally constant, and hence the integral reduces to a finite sum.

Proposition 5.1.2: Fourier Transform Properties

1. For $\Phi \in C_c^\infty(F)$, the function $\hat{\Phi}$ lies in $C_c^\infty(F)$.
2. There is a positive real number $c = c(\psi, \mu)$ such that

$$\hat{\hat{\Phi}}(x) = c\Phi(-x), \quad \Phi \in C_c^\infty(F), \quad x \in F.$$

3. For given ψ , there is a unique Haar measure μ_ψ for which $c(\psi, \mu_\psi) = 1$. This measure satisfies $\mu_\psi(\mathfrak{o}) = q^{l/2}$, where l is the level of ψ .
4. For $a \in F^\times$, we have $\mu_{a\psi} = \|a\|^{\frac{1}{2}} \mu_\psi$.

Notice how (1) is completely false in the real analysis case: By the uncertainly principle it is impossible for f and $\hat{f}$ to be compactly supported, but it *is* the case in p -adic analysis case.

Proof:

- 1-2. Let l be the level of ψ so that $\psi(\mathfrak{p}^{-l}) = 1$ and $\psi(\mathfrak{p}^{-l-1}) \neq 1$, and let Φ_j be the characteristic function of $\mathfrak{p}^j$. For $a \in F$, the character $a\psi|_{\mathfrak{p}^j}$ is trivial if and only if $a \in \mathfrak{p}^{l-j}$. The support of $\hat{\Phi}_j$ is therefore $\mathfrak{p}^{l-j}$ and

$$\hat{\Phi}_j(x) = \mu(\mathfrak{o})q^{-j}, \quad x \in \mathfrak{p}^{l-j}.$$

Therefore, (1) and (2) hold for all functions Φ_j , with $c = \mu(\mathfrak{o})^2 q^{-l}$. Let $\Phi \in C_c^\infty(F)$, $a \in F$, and let Ψ denote the function $x \mapsto \Phi(x - a)$. We then have

$$\hat{\Psi}(x) = \int_F \Phi(y - a) \psi(xy) d\mu(y) = a\psi(x) \hat{\Phi}(x).$$

The function $a\psi$ is locally constant, so $\hat{\Psi}$ lies in $C_c^\infty(F)$ provided $\hat{\Phi}$ does. Calculating the Fourier transform again, we get

$$\hat{\hat{\Psi}}(x) = \hat{\Phi}(a + x).$$

If parts (1) and (2) of the Lemma hold for Φ , they also therefore hold for Ψ with the same value of c . However, they do hold for the functions Φ_j , and the functions $x \mapsto \Phi_j(x - a)$, $a \in F$, $j \in \mathbb{Z}$, span $C_c^\infty(F)$. The first two parts of the proposition therefore hold for all $\Phi \in C_c^\infty(F)$, with $c = \mu(\mathfrak{o})^2 q^{-l}$, as above.

3. For $b > 0$, we have $c(\psi, b\mu) = b^2 c(\psi, \mu)$; put another way, to achieve $c(\psi, \mu) = 1$, we must have $\mu(\mathfrak{o})^2 q^{-l} = 1$.
4. immediate.

Let us box the important concepts from the above proof:

Definition 5.1.3: Self-dual Haar measure

The measure μ_ψ is called *the self-dual Haar measure on F , relative to ψ* . Using μ_ψ to compute the Fourier transform:

$$\hat{\Phi}(x) = \int_F \Phi(y) \psi(xy) d\mu_\psi(y),$$

Definition 5.1.4: Fourier Inversion Formula

The 2nd point in proposition 5.1.2 is called the *Fourier inversion formula*

$$\hat{\hat{\Phi}}(x) = \Phi(-x), \quad \Phi \in C_c^\infty(F), \quad x \in F. \quad (5.1)$$

We next codify the multiplication information i.e. the Mellin Transform, i.e. the Gelfand Transform with $(\mathbb{R}_{>0}, \cdot)$ [4].

Definition 5.1.5: Multiplicative Haar Measure

Let μ be a Haar measure on the additive group F . We define a Haar measure μ^* on the multiplicative group $F^\times$ by

$$d\mu^*(x) = \frac{d\mu(x)}{\|x\|}.$$

It is easy to show this measure is scaling-invariant. Hence, when integrating over F^* , it is invariant to units. To that end, let χ be a character of $F^\times$, let $\Phi \in C_c^\infty(F)$, and choose a prime element ϖ of F .

Definition 5.1.6: Mellin Transform

Let $\Phi \in C_c^\infty(M)$. Then it's *Mellin transform* (with respect to s) is a new function $\mathcal{M}(\Phi, -, s)$ that inputs multiplicative characters and outputs:

$$\mathcal{M}(\Phi, \chi, s) = \int_{F^\times} \Phi(x) \chi(x) \|x\|^s d\mu^*(x)$$

This can be great simplified. Recall that the structure of the multiplicative group is (topologically) $F^\times \cong \mathbb{Z} \times U_F$. We can decompose the field into “shells” (or annuli) based on valuation. For $m \in \mathbb{Z}$, let e_m denote the characteristic function of the set $\varpi^m U_F = \mathfrak{p}^m \setminus \mathfrak{p}^{m+1}$.

For $\Phi \in C_c^\infty(F)$, the function $\Phi_m = e_m \Phi$ lies in $C_c^\infty(F^\times) \subset C_c^\infty(F)$. It is identically zero for $m \ll 0$. We may therefore set:

$$z_m = z_m(\Phi, \chi) = \int_{\varpi^m U_F} \Phi(x) \chi(x) d\mu^*(x), \quad m \in \mathbb{Z}. \quad (5.2)$$

On the shell $\varpi^m U_F$, the norm $\|x\|$ is exactly q^{-m} . Therefore, $\|x\|^s = (q^{-m})^s = (q^{-s})^m$. Thus the value can be pulled out and we can re-write it as:

$$\mathcal{M}(\Phi, \chi, s) = \sum_{m \in \mathbb{Z}} (q^{-s})^m \underbrace{\left(\int_{\varpi^m U_F} \Phi(x) \chi(x) d\mu^*(x) \right)}_{\text{This is } z_m}$$

The value z_m captures the correlation between the test function Φ and the character χ specifically on the shell where $v_F(x) = m$. We assemble these into a generating function and let the q^{-s} vary (as it can depend on the choice of q) by choosing a formal variable X :

Definition 5.1.7: Local Zeta Integral (Formal Series)

We define a formal Laurent series $Z(\Phi, \chi, X) \in \mathbb{C}((X))$ by

$$\begin{aligned} \mathcal{Z} : C_c^\infty(F) &\rightarrow \mathbb{C}((X)) \\ Z(\Phi, \chi, X) &= \sum_{m \in \mathbb{Z}} z_m X^m. \end{aligned}$$

We denote its image by $\mathcal{Z}(\chi) = \mathcal{Z}(\chi, X) = \{Z(\Phi, \chi, X) : \Phi \in C_c^\infty(F)\}$.

For $a \in F^\times$ and $\Phi \in C_c^\infty(F)$, we denote by $a \cdot \Phi$ the function $x \mapsto \Phi(a^{-1}x)$. A simple calculation shows:

$$Z(a \cdot \Phi, \chi, X) = \chi(a) X^{v_F(a)} Z(\Phi, \chi, X).$$

Thus $\mathcal{Z}(\chi)$ is a module over the ring $\mathbb{C}[X, X^{-1}]$ of Laurent polynomials.

Recall that on the shell $\varpi^m U_F$, the norm is constant: $\|x\| = q^{-m}$. If we formally substitute $X = q^{-s}$ into our series, we recover the standard integral notation:

$$Z(\Phi, \chi, q^{-s}) = \sum_{m \in \mathbb{Z}} \int_{\varpi^m U_F} \Phi(x) \chi(x) (q^{-m})^s d\mu^*(x) = \int_{F^\times} \Phi(x) \chi(x) \|x\|^s d\mu^*(x).$$

We shall use the notation $\zeta(\Phi, \chi, s) = Z(\Phi, \chi, q^{-s})$ interchangeably. The function z_m acts as the m -th Fourier coefficient of Φ with respect to the valuation dimension of the group $F^\times$.

Proposition 5.1.8: Rationality of Zeta Integrals

Let χ be a character of $F^\times$; then

$$\mathcal{Z}(\chi, X) = P_\chi(X)^{-1} \mathbb{C}[X, X^{-1}],$$

where

$$P_\chi(X) = \begin{cases} 1 - \chi(\varpi)X & \text{if } \chi \text{ is unramified,} \\ 1 & \text{otherwise.} \end{cases}$$

Hence, we can further conclude that $\mathcal{Z}(\chi)$ is a PID.

Proof:

Suppose first that $\Phi(0) = 0$. Thus $\Phi|_{F^\times}$ lies in $C_c^\infty(F^\times)$ and $Z(\Phi, \chi, X)$ has only finitely many non-zero coefficients, so:

$$Z(\Phi, \chi, X) \in \mathbb{C}[X, X^{-1}]$$

If Φ is the indicator function of a sufficiently small neighbourhood of 1, then $Z(\Phi, \chi, X)$ is a positive constant, and so $1 \in \mathcal{Z}(\chi)$ and

$$\{Z(\Phi, \chi, X) : \Phi \in C_c^\infty(F^\times)\} = \mathbb{C}[X, X^{-1}].$$

Let Φ_0 be the characteristic function of $\mathfrak{o}$. Then:

$$Z(\Phi_0, \chi, X) = \sum_{m \geq 0} \chi(\varpi^m) X^m \int_{U_F} \chi(x) d\mu^*(x).$$

The inner integral is $\mu^*(U_F)$ if χ is unramified (trivial on U_F), and is zero otherwise (by orthogonality of characters on the compact group U_F). In other words:

$$\mu^*(U_F)^{-1} Z(\Phi_0, \chi, X) = \begin{cases} (1 - \chi(\varpi)X)^{-1} & \text{if } \chi \text{ is unramified,} \\ 0 & \text{otherwise.} \end{cases}$$

Since $C_c^\infty(F)$ is spanned by Φ_0 and $C_c^\infty(F^\times)$, the result follows.

The polynomial $P_\chi(X)^{-1}$ is given a name:

Definition 5.1.9: Local L-function

The (multiplicative) inverse of the polynomial $P_\chi(X)$ is called the *Local L-function*. It is traditionally denoted $L(\chi, s)$ when $X = q^{-s}$ so that:

$$L(\chi, s) = \begin{cases} \frac{1}{1 - \chi(\varpi)q^{-s}} & \text{if } \chi \text{ is unramified,} \\ 1 & \text{otherwise.} \end{cases}$$

Corollary 5.1.10: Poles of Unramified L-functions

Let χ_1, χ_2 be unramified characters of $F^\times$. The following are equivalent:

1. The meromorphic functions $L(\chi_1, s)$ and $L(\chi_2, s)$ have a pole in common;
2. The meromorphic functions $L(\chi_1, s)$ and $L(\chi_2, s)$ have the same sets of poles;
3. $\chi_1 = \chi_2$.

Proof:

The implication (3) $\Rightarrow$ (2) is trivial, and (2) $\Rightarrow$ (1) is immediate since the set of poles is non-empty.

We prove (1) $\Rightarrow$ (3). The poles of $L(\chi_i, s)$ are the zeros of the polynomial $1 - \chi_i(\varpi)q^{-s}$. Thus, s is a pole if and only if $q^s = \chi_i(\varpi)$.

Suppose the functions share a pole s_0 . Then:

$$\chi_1(\varpi) = q^{s_0} = \chi_2(\varpi).$$

By definition, an unramified character is trivial on the units U_F . Since $F^\times \cong \mathbb{Z} \times U_F$ (generated by ϖ and U_F), an unramified character is completely determined by its value on the prime element ϖ . Therefore $\chi_1 = \chi_2$.

We now turn to the main result of the local theory: the functional equation relating the additive and multiplicative structures. We fix a character $\psi \in \widehat{F}$, $\psi \neq 1$. For $\Phi \in C_c^\infty(F)$, let $\hat{\Phi}$ denote the Fourier transform relative to the self-dual measure μ_ψ .

Theorem 5.1.11: Local Functional Equation ($GL(1)$)

Let χ be a character of $F^\times$. There exists a unique rational function $c(\chi, \psi, X) \in \mathbb{C}(X)$ such that

$$Z(\hat{\Phi}, \check{\chi}, q^{-1}X^{-1}) = c(\chi, \psi, X)Z(\Phi, \chi, X), \quad (5.3)$$

for all $\Phi \in C_c^\infty(F)$. Here $\check{\chi} = \chi^{-1}$.

This equation is often called *Tate's (local) functional equation*

Proof:

Consider the space Λ of linear maps $\lambda : C_c^\infty(F) \rightarrow \mathbb{C}(X)$ satisfying the homogeneity condition:

$$\lambda(a \cdot \Phi) = \chi(a)X^{v_F(a)}\lambda(\Phi), \quad \Phi \in C_c^\infty(F), a \in F^\times.$$

Then Λ is a vector space over $\mathbb{C}(X)$ and contains the map $\lambda_0 : \Phi \mapsto Z(\Phi, \chi, X)$ which is non-zero by proposition 5.1.8.

For $\Phi \in C_c^\infty(F)$ and $a \in F^\times$, a simple calculation using the properties of the Fourier transform yields:

$$\widehat{a \cdot \Phi}(y) = \|a\|a^{-1} \cdot \hat{\Phi}(y).$$

It follows that the map

$$\lambda_1 : \Phi \mapsto Z(\hat{\Phi}, \check{\chi}, q^{-1}X^{-1})$$

also lies in Λ . If Λ is one-dimensional, we can pick our desired constant.

Lemma 5.1.12: Uniqueness of Functionals

The space Λ has dimension one over $\mathbb{C}(X)$.

Proof:

(of Lemma) Choose $n \geq 0$ such that $U_F^n \subset \text{Ker } \chi$. Let Φ_k denote the characteristic function of U_F^k , $k \geq 1$. We consider the evaluation map $\Lambda \rightarrow \mathbb{C}(X)$ given by $\lambda \mapsto \lambda(\Phi_n)$. We show this is injective.

Suppose $\lambda(\Phi_n) = 0$. The defining property of Λ implies $\lambda(a \cdot \Phi_k) = \lambda(\Phi_k)$ for any $a \in U_F^n$ and $k \geq n$. Since the volume of the units scales by indices, we have $\lambda(\Phi_k) = q^{n-k} \lambda(\Phi_n) = 0$ for $k \geq n$.

Next, any $\Phi \in C_c^\infty(F^\times)$ is a finite linear combination of $F^\times$ -translates of functions Φ_k for sufficiently large k . Thus, λ vanishes on $C_c^\infty(F^\times)$. For a general $\Phi \in C_c^\infty(F)$, the value $\lambda(\Phi)$ therefore depends only on $\Phi(0)$ (since $\Phi - \Phi(0)\Phi_{\text{small}}$ vanishes at 0). However, $\lambda(a \cdot \Phi) = \chi(a)X^{v(a)}\lambda(\Phi)$ forces $\lambda(\Phi) = 0$ unless λ is identically zero.

Since Λ is one-dimensional and $\lambda_0 \neq 0$, there exists a unique scalar $c(\chi, \psi, X) \in \mathbb{C}(X)$ such that $\lambda_1 = c(\chi, \psi, X)\lambda_0$.

The above is the notation Tate used. The following is the more traditional notation. Let s be a complex variable.

$$\begin{aligned}\zeta(\Phi, \chi, s) &= Z(\Phi, \chi, q^{-s}), \\ L(\chi, s) &= P_\chi(q^{-s})^{-1}, \\ \gamma(\chi, s, \psi) &= c(\chi, \psi, q^{-s}).\end{aligned}$$

The functional equation then reads:

$$\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\chi, s, \psi)\zeta(\Phi, \chi, s).$$

We shall stick with this notation from now on.

Let us next take a closer look at the structure of $\gamma(\chi, s, \psi)$. It is given a name:

Definition 5.1.13: Epsilon Factor

The *epsilon factor* (or local constant) $\varepsilon(\chi, s, \psi)$ defined to be the function $\varepsilon(\chi, s, \psi) \in \mathbb{C}(q^{-s})$ satisfying the relation:

$$\gamma(\chi, s, \psi) = \varepsilon(\chi, s, \psi) \frac{L(\check{\chi}, 1-s)}{L(\chi, s)}.$$

Using theorem 5.1.11, we also get a functional equation for the epsilon factor.

Corollary 5.1.14: Functional Equation of the Local Constant

The function $\varepsilon(\chi, s, \psi)$ satisfies the functional equation

$$\varepsilon(\chi, s, \psi)\varepsilon(\check{\chi}, 1-s, \psi) = \chi(-1).$$

It is of the form

$$\varepsilon(\chi, s, \psi) = q^{(1/2-s)n(\chi, \psi)}\varepsilon(\chi, 1/2, \psi),$$

for some integer $n(\chi, \psi) \in \mathbb{Z}$.

Proof:

The functional equation in theorem 5.1.11 reads

$$\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\chi, s, \psi) \zeta(\Phi, \chi, s). \quad (5.4)$$

Applying this twice, we get

$$\zeta(\hat{\hat{\Phi}}, \chi, s) = \gamma(\check{\chi}, 1-s, \psi) \zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\check{\chi}, 1-s, \psi) \gamma(\chi, s, \psi) \zeta(\Phi, \chi, s).$$

Using the Fourier inversion formula $\hat{\hat{\Phi}}(x) = \Phi(-x)$, we see that

$$\zeta(\hat{\hat{\Phi}}, \chi, s) = \int_{F^\times} \Phi(-x) \chi(x) \|x\|^s d\mu^*(x) = \chi(-1) \zeta(\Phi, \chi, s).$$

Comparing the two expressions, we obtain

$$\gamma(\check{\chi}, 1-s, \psi) \gamma(\chi, s, \psi) = \varepsilon(\check{\chi}, 1-s, \psi) \varepsilon(\chi, s, \psi) = \chi(-1),$$

which proves the first relation.

For the second assertion, we can rewrite (5.4) in the form

$$\frac{\zeta(\hat{\Phi}, \check{\chi}, 1-s)}{L(\check{\chi}, 1-s)} = \varepsilon(\chi, s, \psi) \frac{\zeta(\Phi, \chi, s)}{L(\chi, s)}.$$

The fraction on either side lies in $\mathbb{C}[q^s, q^{-s}]$ (by the definition of the L-function as the generator of the zeta ideal). As in proposition 5.1.8, we may choose Φ such that $\zeta(\Phi, \chi, s) = L(\chi, s)$. We deduce that $\varepsilon(\chi, s, \psi) \in \mathbb{C}[q^s, q^{-s}]$. Likewise, $\varepsilon(\check{\chi}, 1-s, \psi) \in \mathbb{C}[q^s, q^{-s}]$.

The functional equation implies that $\varepsilon(\chi, s, \psi)$ is an invertible element of the ring $\mathbb{C}[q^s, q^{-s}]$, and hence equal to aq^{ms} , for some $a \in \mathbb{C}^\times, m \in \mathbb{Z}$. This can be written in the required form, completing the proof.

5.1.1 Local Constants

L-functions characterize unramified characters. If the character is ramified, we loose all information:

Lemma 5.1.15: L-function of Ramified Character

If the character χ of $F^\times$ is ramified, then $L(\chi, s) = 1$.

Proof:

By definition, $L(\chi, s)$ is the generator of the ideal $\mathcal{Z}(\chi)$ of all zeta integrals. To prove the generator is 1, it suffices to exhibit a test function $\Phi \in C_c^\infty(F)$ such that the integral $Z(\Phi, \chi, s)$ is a non-zero constant.

Since χ is smooth but ramified, there exists an integer $n \geq 1$ such that χ is non-trivial on U_F but trivial on the subgroup $U_F^n = 1 + \mathfrak{p}^n$. Let Φ be the characteristic function of this set U_F^n . The support of Φ is contained in the units, so $v_F(x) = 0$ (and $\|x\|^s = 1$) for all x in the support.

Computing the integral:

$$Z(\Phi, \chi, s) = \int_{F^\times} \Phi(x) \chi(x) \|x\|^s d\mu^*(x) = \int_{U_F^n} 1 \cdot \chi(x) \cdot 1 d\mu^*(x).$$

Since χ is trivial on U_F^n , this reduces to:

$$Z(\Phi, \chi, s) = \int_{U_F^n} d\mu^*(x) = \mu^*(U_F^n).$$

This is a non-zero constant (independent of s). The ideal $\mathcal{Z}(\chi)$ therefore contains a unit of the ring $\mathbb{C}[q^s, q^{-s}]$, which implies that the generator $L(\chi, s)$ is 1.

Thus, we need more information to characterize irreducible representations using functionals. Theorem 5.5.1 will show the ε -factors are exactly the required information. It will be essential to have a table of values for the local constant $\varepsilon(\chi, s, \psi)$. We first observe that the dependence on the additive character ψ is transparent.

Lemma 5.1.16: Dependence on Additive Character

Let $a \in F^\times$; we then have

$$\varepsilon(\chi, s, a\psi) = \chi(a) \|a\|^{s-1/2} \varepsilon(\chi, s, \psi).$$

Proof:

The L -functions $L(\chi, s)$ and $L(\check{\chi}, 1-s)$ depend only on the character χ and not on the additive character ψ . Therefore, the ratio between $\varepsilon(\chi, s, a\psi)$ and $\varepsilon(\chi, s, \psi)$ is the same as the ratio between the factors $\gamma(\chi, s, a\psi)$ and $\gamma(\chi, s, \psi)$. We calculate the latter using the functional equation.

Let $\psi' = a\psi$. From proposition 5.1.2, the self-dual measure scales as $\mu_{\psi'} = \|a\|^{1/2} \mu_\psi$. The Fourier transform relative to ψ' is given by:

$$\begin{aligned} \hat{\Phi}_{\psi'}(x) &= \int_F \Phi(y) \psi'(axy) d\mu_{\psi'}(y) \\ &= \|a\|^{1/2} \int_F \Phi(y) \psi(axy) d\mu_\psi(y) \\ &= \|a\|^{1/2} \hat{\Phi}_\psi(ax). \end{aligned}$$

We substitute this into the zeta integral on the left side of the functional equation:

$$\zeta(\hat{\Phi}_{\psi'}, \check{\chi}, 1-s) = \int_{F^\times} \|a\|^{1/2} \hat{\Phi}_\psi(ax) \check{\chi}(x) \|x\|^{1-s} d\mu^*(x).$$

We perform the change of variables $y = ax$. Since μ^* is a Haar measure on $F^\times$, $d\mu^*(x) = d\mu^*(y)$. We observe that $x = a^{-1}y$, so $\check{\chi}(x) = \check{\chi}(a^{-1})\check{\chi}(y) = \chi(a)\check{\chi}(y)$ and $\|x\|^{1-s} = \|a\|^{s-1}\|y\|^{1-s}$. The integral becomes:

$$\begin{aligned} \zeta(\hat{\Phi}_{\psi'}, \check{\chi}, 1-s) &= \|a\|^{1/2} \chi(a) \|a\|^{s-1} \int_{F^\times} \hat{\Phi}_\psi(y) \check{\chi}(y) \|y\|^{1-s} d\mu^*(y) \\ &= \chi(a) \|a\|^{s-1/2} \zeta(\hat{\Phi}_\psi, \check{\chi}, 1-s). \end{aligned}$$

Applying the functional equation definition $\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$ to both sides, we obtain:

$$\gamma(\chi, s, a\psi)\zeta(\Phi, \chi, s) = \chi(a)\|a\|^{s-1/2}\gamma(\chi, s, \psi)\zeta(\Phi, \chi, s).$$

Since this holds for all test functions Φ , the result follows.

Hence, using the above lemma we can reduce the computation of $\varepsilon(\chi, s, \psi)$ for characters ψ of level one. The self-dual Haar measure for such a ψ satisfies

$$\int_{\mathfrak{o}} dx = q^{1/2}.$$

For the purposes of calculation, it will be convenient to choose the Haar measure d^*x on $F^\times$ such that

$$\int_{U_F} d^*x = 1.$$

Proposition 5.1.17: Epsilon Factor for Unramified Characters

Suppose that χ is unramified and that ψ has level one. If ϖ is a prime element of F , then

$$\varepsilon(\chi, s, \psi) = q^{s-1/2}\chi(\varpi)^{-1}.$$

Proof:

If Φ denotes the characteristic function of $\mathfrak{o}$, we get $\zeta(\Phi, \chi, s) = L(\chi, s)$, as in the proof of proposition 5.1.8. On the other side, since ψ has level 1, the Fourier transform is supported on $\mathfrak{p}$ and given by $\hat{\Phi}(x) = q^{1/2}\Phi(\varpi^{-1}x)$, so

$$\begin{aligned} \zeta(\hat{\Phi}, \chi^{-1}, s) &= q^{1/2} \int_{F^\times} \Phi(\varpi^{-1}x) \chi(x)^{-1} \|x\|^s d^*x \\ &= q^{1/2-s} \chi(\varpi)^{-1} \int_{F^\times} \Phi(x) \chi(x)^{-1} \|x\|^s d^*x \\ &= q^{1/2-s} \chi(\varpi)^{-1} L(\chi^{-1}, s). \end{aligned}$$

It follows that

$$\varepsilon(\chi, s, \psi) = \frac{\zeta(\hat{\Phi}, \chi^{-1}, 1-s)}{L(\chi^{-1}, 1-s)} = q^{s-1/2}\chi(\varpi)^{-1},$$

as required.

Theorem 5.1.18: Epsilon Factor for Ramified Characters

Suppose that χ has level $n \geq 0$ and is *ramified*. Let $\psi \in \hat{F}$ have level one. Then

$$\varepsilon(\chi, s, \psi) = q^{n(1/2-s)} \left(\sum_{x \in U_F/U_F^{n+1}} \chi(\alpha x)^{-1} \psi(\alpha x) \right) q^{-(n+1)/2},$$

for any $\alpha \in F^\times$ such that $v_F(\alpha) = -n$.

Proof:

Let Φ be the characteristic function of U_F^{n+1} , so that

$$\zeta(\Phi, \chi, s) = \mu^*(U_F^{n+1}).$$

The Fourier transform $\hat{\Phi}$ has support $\mathfrak{p}^{-n}$ and

$$\hat{\Phi}(y) = q^{1/2-n-1} \psi(y), \quad y \in \mathfrak{p}^{-n}.$$

Thus

$$\begin{aligned} \zeta(\hat{\Phi}, \chi^{-1}, s) &= q^{1/2-n-1} \int_{\mathfrak{p}^{-n} \setminus \{0\}} \psi(y) \chi(y)^{-1} \|y\|^s d^*y \\ &= q^{1/2-n-1} \sum_{m \geq -n} z_m q^{-ms}, \end{aligned}$$

where

$$z_m = \int_{\mathfrak{p}^m \setminus \mathfrak{p}^{m+1}} \psi(y) \chi(y)^{-1} d^*y.$$

We know from corollary 5.1.14 that $\zeta(\hat{\Phi}, \chi^{-1}, s)$ must reduce to a monomial in q^s , so only one of the coefficients z_m is non-zero. The first one is

$$z_{-n} = \int_{U_F} \psi(\alpha u) \chi^{-1}(\alpha u) d^*u,$$

for any $\alpha \in F$ with $v_F(\alpha) = -n$. The integrand is constant on cosets of U_F^{n+1} and so the integral reduces to

$$\int_{U_F} \psi(\alpha u) \chi^{-1}(\alpha u) d^*u = \mu^*(U_F^{n+1}) \sum_{x \in U_F/U_F^{n+1}} \chi(\alpha x)^{-1} \psi(\alpha x).$$

If the coefficients $m \geq 1 - n$, the proof is complete. To that end, take $\beta \in F$, with $v_F(\beta) = m > -n$, and consider the coefficient

$$z_m = \int_{U_F} \psi(\beta u) \chi(\beta u)^{-1} d^*u. \quad (5.5)$$

Taking $v \in U_F^n$, we have

$$\int_{U_F} \psi(\beta u) \chi(\beta u)^{-1} d^*u = \int_{U_F} \psi(\beta uv) \chi(\beta uv)^{-1} d^*u.$$

Setting $v = 1 + y$, the first factor reduces to $\psi(\beta u) \psi(\beta uy) = \psi(\beta u)$, since $\beta uy \in \mathfrak{p}^{m+n} \subset \mathfrak{p} \subset \text{Ker } \psi$ (as $m > -n \implies m + n \geq 1$). Thus

$$\int_{U_F} \psi(\beta u) \chi(\beta u)^{-1} d^*u = \chi(v)^{-1} \int_{U_F} \psi(\beta u) \chi(\beta u)^{-1} d^*u.$$

We may certainly choose $v \in U_F^n$ so that $\chi(v) \neq 1$ (by the definition of level), so the integral (5.5) vanishes, completing the proof.

Let us look at the sum within the parenthesis, the “trigonometric sum”:

$$\sum_{x \in U_F/U_F^{n+1}} \chi(\alpha x)^{-1} \psi(\alpha x)$$

where the names comes from the classical case the image of a character would lie in the unit circle. This type of sum mixing an additive and multiplicative character is given a name:

Definition 5.1.19: Gauss Sum

Let χ be a ramified character of $F^\times$ of level $n \geq 0$, and let $\psi \in \widehat{F}$ have level one. Let $c \in F$ satisfy $v_F(c) = -n$. The sum

$$\tau(\chi, \psi) = \sum_{x \in U_F/U_F^{n+1}} \check{\chi}(cx) \psi(cx) \quad (5.6)$$

is called the *Gauss sum* of χ relative to ψ .

Another common convention is to denote the Gauss sum as $\tau(\check{\chi}, \psi)$; the above version is slightly more convenient for the present purposes.

The definition is independent of the choice of coset representative x and element c . Re-writing the equation in theorem 5.1.18 gives:

$$\varepsilon(\chi, s, \psi) = q^{n(1/2-s)} \frac{\tau(\chi, \psi)}{q^{(n+1)/2}}. \quad (5.7)$$

The functional equation for the local constant (corollary 5.1.14) gives:

$$\tau(\chi, \psi) \tau(\check{\chi}, \psi) = \chi(-1) q^{n+1}. \quad (5.8)$$

The sums of characters can often be simplified (think of the proof of summing the powers of roots of unity). Indeed:

Proposition 5.1.20: Simplification of Gauss Sum

Suppose that χ has level $n \geq 1$. Let $c \in F$ satisfy

$$\chi(1+x) = \psi(cx), \quad x \in \mathfrak{p}^{[n/2]+1}.$$

Then

$$\tau(\chi, \psi) = q^{[(n+1)/2]} \sum_y \check{\chi}(cy) \psi(cy), \quad (5.9)$$

where y ranges over $U_F^{[(n+1)/2]} / U_F^{[n/2]+1}$.

Proof:

In the defining sum of equation (5.6), we write $x = y(1+z)$, with $y \in U_F/U_F^{[n/2]+1}$ and

$z \in \mathfrak{p}^{[n/2]+1}/\mathfrak{p}^{n+1}$. Then:

$$\check{\chi}(cy(1+z))\psi(cy(1+z)) = \check{\chi}(cy)\psi(cy)\psi(c(y-1)z).$$

Summing:

$$\tau(\chi, \psi) = \sum_y \check{\chi}(cy)\psi(cy) \sum_z \psi(c(y-1)z).$$

Now, the map $z \mapsto \psi(c(y-1)z)$ is a character of $\mathfrak{p}^{[n/2]+1}/\mathfrak{p}^{n+1}$ since $c(y-1) \in \mathfrak{p}^{-n}$. It is the trivial character if and only if $y \equiv 1 \pmod{\mathfrak{p}^{n-[n/2]}}$, i.e. if and only if $y \in U_F^{[(n+1)/2]}$. The inner sum therefore vanishes unless $y \in U_F^{[(n+1)/2]}$ in which case it takes the value $(\mathfrak{p}^{[n/2]+1} : \mathfrak{p}^{n+1}) = q^{[(n+1)/2]}$, as we sought to show.

The formula of in proposition 5.1.20 has an important application for its use in the proof of the Converse Theorem:

Theorem 5.1.21: Stability Theorem

Let θ, χ be characters of $F^\times$, of level $l \geq 0, n \geq 1$ respectively. Suppose that $2l < n$. Let $\psi \in \widehat{F}$, $\psi \neq 1$, and let $c \in F$ satisfy $\chi(1+x) = \psi(cx)$, $x \in \mathfrak{p}^{[n/2]+1}$. Then

$$\varepsilon(\theta\chi, s, \psi) = \theta(c)^{-1}\varepsilon(\chi, s, \psi).$$

Proof:

By lemma 5.1.16, it is enough to treat the case where ψ has level one. The character $\theta\chi$ has level n , and agrees with χ on $U_F^{[(n+1)/2]}$. Applying proposition 5.1.20, we get

$$\tau(\theta\chi, \psi) = q^{[(n+1)/2]} \sum_y \check{\theta}\check{\chi}(cy)\psi(cy),$$

where the sum is taken over $y \in U_F^{[(n+1)/2]}/U_F^{[n/2]+1}$. Since θ is trivial on $U_F^{[(n+1)/2]}$, this reduces to $\theta(c)^{-1}\tau(\chi, \psi)$, as required.

5.2 Functional Equation for $GL(2)$

The transition from the theory for $GL_1(F)$ (Tate's Thesis) to $GL_2(F)$ (Godement-Jacquet theory) is not merely a change in dimension, but a structural upgrade of every component of the functional equation. The fundamental philosophy remains identical: an L -function is constructed by integrating a “test function” against a “representation” over the group. However, to accommodate the non-abelian nature of the group, scalars must be replaced by matrices, and characters by matrix coefficients.

The ambient space for the test functions shifts from the field F to the full matrix algebra $A = M_2(F)$. Just as the multiplicative group $F^\times$ is an open dense subset of F , the general linear group $G = GL_2(F)$ is an open dense subset of the algebra A . Consequently, the test functions Φ are chosen from the space $C_c^\infty(A)$, the space of locally constant, compactly supported functions on the matrix ring. This space is spanned by characteristic functions of lattices, such as $a + \varpi^j M_2(\mathfrak{o})$.

This change in ambient space necessitates a corresponding change in the Fourier transform. Both theories rely on an additive Fourier transform on the underlying linear space. For GL_1 , the duality on F was given by the product xy . For the matrix algebra $M_2(F)$, the natural non-degenerate bilinear form is given by the trace of the product. Let $\psi \in \widehat{F}$ be a non-trivial additive character. The character ψ_A of the additive group $A = M_2(F)$ is defined by $\psi_A(x) = \psi(\text{tr}_A(x)) = \psi(x_{11} + x_{22})$. The Haar measure μ on A is normalized to be self-dual relative to ψ_A .

Definition 5.2.1: Matrix Fourier Transform

For a test function $\Phi \in C_c^\infty(A)$, the Fourier transform $\hat{\Phi}$ is defined by

$$\hat{\Phi}(x) = \int_A \Phi(y) \psi_A(xy) d\mu(y).$$

Just as in the one-dimensional case, the map $\Phi \mapsto \hat{\Phi}$ is an isomorphism of $C_c^\infty(A)$ onto itself, and the Fourier Inversion Formula holds: $\hat{\hat{\Phi}}(x) = \Phi(-x)$.

The most significant conceptual shift concerns the object being integrated. For GL_1 , the representation is a character $\chi : F^\times \rightarrow \mathbb{C}^\times$. Since it is one-dimensional, the character itself is the function integrated against Φ . For GL_2 , an irreducible representation (π, V) is generally infinite-dimensional. The operator $\pi(g)$ cannot be integrated directly to yield a scalar value. Instead, the theory utilizes *matrix coefficients*.

Definition 5.2.2: Matrix Coefficient

Let (π, V) be a smooth representation of G . For vectors $v \in V$ and $\tilde{v} \in \check{V}$ (the smooth dual), the function

$$f(g) = \gamma_{\tilde{v} \otimes v}(g) = \langle \tilde{v}, \pi(g)v \rangle$$

is called a *matrix coefficient* of π . Let $C(\pi)$ denote the vector space spanned by all such coefficients.

The space $C(\pi)$ carries a smooth representation of $G \times G$ via the action $(g_1, g_2) \cdot f(x) = f(g_1^{-1}xg_2)$. In the context of Zeta integrals, these coefficients replace the character $\chi(x)$. The Zeta integrals are then defined by integrating the test function against the matrix coefficient, weighted by a power of the determinant.

Definition 5.2.3: Zeta Integral for GL_2

Let $\Phi \in C_c^\infty(A)$, $f \in C(\pi)$, and $s \in \mathbb{C}$. The *Zeta integral* is defined as:

$$\zeta(\Phi, f, s) = \int_G \Phi(g) f(g) \|\det g\|^s d^\times g,$$

where $d^\times g$ is a Haar measure on G .

Convergence is the first thing to settle, and it is not automatic: G is not compact modulo the centre, so nothing yet prevents the integral from diverging for every s . The situation is better than it looks. Sorting G by the valuation of the determinant turns the zeta integral into a formal Laurent series whose coefficients are integrals over *compact* sets, and whose negative part is finite.

Throughout, write $X = q^{-s}$, so that $\|\det g\|^s = X^{v_F(\det g)}$, and set

$$G_m = \{g \in G : v_F(\det g) = m\} \quad m \in \mathbb{Z}$$

Lemma 5.2.4: Zeta Integrals as Laurent Series

Let (π, V) be a smooth representation of G , let $\Phi \in C_c^\infty(A)$ and let $f \in C(\pi)$. For $m \in \mathbb{Z}$ put

$$c_m(\Phi, f) = \int_{G_m} \Phi(g) f(g) d^\times g$$

1. The set $\text{supp} \Phi \cap G_m$ is a compact subset of G , and $c_m(\Phi, f)$ is an absolutely convergent integral.
2. There is an integer $N \geq 0$, depending only on $\text{supp} \Phi$, such that $c_m(\Phi, f) = 0$ for all $m < -2N$.

Consequently $Z(\Phi, f, X) = \sum_{m \in \mathbb{Z}} c_m(\Phi, f) X^m$ is a well-defined element of $\mathbb{C}((X))$, and $\zeta(\Phi, f, s)$ is its value at $X = q^{-s}$ at any s for which the series converges.

Proof:

1. The set $\varpi^m U_F$ is compact, hence closed, in F , and $\det: A \rightarrow F$ is a polynomial map, hence continuous. Therefore $G_m = \det^{-1}(\varpi^m U_F)$ is closed in A , and $\text{supp} \Phi \cap G_m$ is a closed subset of the compact set $\text{supp} \Phi$. It is therefore compact, and it lies in G , which carries the topology induced from A since G is open in A .
Both Φ and f are locally constant, so Φf takes only finitely many values on this compact set. A Haar measure is finite on compact sets, so the integral converges absolutely.
2. The lattices $\varpi^{-N} M_2(\mathfrak{o})$, $N \geq 0$, form an increasing open cover of A , so compactness gives N with $\text{supp} \Phi \subseteq \varpi^{-N} M_2(\mathfrak{o})$. For g in the support we then have $\varpi^N g \in M_2(\mathfrak{o})$, whence

$$\varpi^{2N} \det g = \det(\varpi^N g) \in \mathfrak{o}$$

and therefore $v_F(\det g) \geq -2N$, completing the proof.

The classification results already proved reduce the analysis of $Z(\Phi, f, X)$ to two cases, and the two are genuinely different in character. A cuspidal representation has coefficients that vanish outside a set compact modulo the centre (theorem 3.1.3), so only the central direction is unbounded, and that direction is governed by Tate's theory applied to the central character. Every other irreducible representation is built from a character of the torus (theorem 2.4.16), and unfolding along the Iwasawa decomposition reduces it to a *pair* of Tate integrals. To pass from the induced representations to their subquotients, the following is needed.

Lemma 5.2.5: Coefficients of a Subquotient

Let (Σ, X) be an admissible smooth representation of G and let π be a subquotient of Σ . Then $C(\pi) \subseteq C(\Sigma)$.

Proof:

It suffices to treat a subrepresentation and a quotient separately, since a subquotient is a quotient of a subrepresentation.

Quotients. Let $\pi = X/Y$ for a G -subspace Y . A smooth functional on X/Y is precisely a smooth functional on X vanishing on Y , so there is a canonical injection $\iota: (X/Y)^\vee \rightarrow \check{X}$. Given $\tilde{u} \in (X/Y)^\vee$ and $\bar{v} \in X/Y$, lift $\bar{v}$ to $v \in X$; then

$$\langle \tilde{u}, \pi(g)\bar{v} \rangle = \langle \tilde{u}, \overline{\Sigma(g)v} \rangle = \langle \iota(\tilde{u}), \Sigma(g)v \rangle$$

which exhibits the coefficient as one of Σ .

Subrepresentations. Let $\pi = (\Sigma|_Y, Y)$ for a G -subspace $Y \subseteq X$, and let $v \in Y$, $\tilde{v} \in \check{Y}$. Choose a compact open subgroup K fixing both, so $v \in Y^K$ and $\tilde{v} \in \check{Y}^K$. By corollary 1.2.18 applied to Y and to X ,

$$Y = Y^K \oplus Y(K), \quad X = X^K \oplus X(K)$$

The first of these intersections is immediate. For the second, $Y(K) \subseteq Y \cap X(K)$ by definition; conversely, if $y \in Y \cap X(K)$, write $y = y_1 + y_2$ with $y_1 \in Y^K$ and $y_2 \in Y(K)$, so that $y_1 = y - y_2$ lies in $X(K)$ as well as in X^K , forcing $y_1 = 0$ and $y = y_2 \in Y(K)$.

The space $Y(K)$ is spanned by the vectors $y - \pi(k)y$, and $\tilde{v}$ is K -fixed, so $\tilde{v}(y - \pi(k)y) = 0$: thus $\tilde{v}$ annihilates $Y(K)$ and is determined by its restriction to Y^K . Since Σ is admissible, X^K is finite dimensional; extend $\tilde{v}|_{Y^K}$ to a linear functional λ on X^K and let $\tilde{u} \in \check{X}$ be λ composed with the projection $X \rightarrow X^K$ along $X(K)$. Then $\tilde{u}$ is K -fixed, hence smooth, and $\tilde{u}|_Y = \tilde{v}$ because the two agree on Y^K and both annihilate $Y(K)$. Therefore

$$\langle \tilde{v}, \pi(g)v \rangle = \langle \tilde{u}, \Sigma(g)v \rangle$$

for every $g \in G$, as we sought to show.

Proposition 5.2.6: Rationality of Zeta Integrals

Let (π, V) be an irreducible smooth representation of G .

1. The integrals $\zeta(\Phi, f, s)$ represent rational functions of q^{-s} .
2. The $\mathbb{C}[q^s, q^{-s}]$ -module generated by these functions is a principal fractional ideal generated by a function of the form $P(q^{-s})^{-1}$, where $P(t) \in \mathbb{C}[t]$ and $P(0) = 1$.

This generator is defined to be the L -function $L(\pi, s)$.

Proof:

Throughout, π is irreducible, hence admissible by corollary 3.1.5 and equipped with a central character ω_π by corollary 1.2.31.

1. By lemma 5.2.4 the expression $Z(\Phi, f, X)$ is a Laurent series with finitely many negative terms, so the content of the claim is that it is the expansion of a rational function. The argument splits according to whether π is cuspidal, and by theorem 3.1.3 together with theorem 2.4.16 these two cases exhaust the irreducible representations of G . In each case a denominator is produced that depends on π alone and not on Φ or f ; this uniformity

is what part (2) will consume.

The cuspidal case. By theorem 3.1.3 the representation π is γ -cuspidal, so every $f \in C(\pi)$ is compactly supported modulo Z . Both G (proposition 2.2.7) and $Z \cong F^\times$ are unimodular, so G/Z carries an invariant measure; fix Haar measures with $d^\times g = d^\times a d\dot{g}$. Since $\det(ag) = a^2 \det g$ and $f(ag) = \omega_\pi(a)f(g)$ for $a \in F^\times$ identified with $aI \in Z$,

$$Z(\Phi, f, X) = \int_{G/Z} \left(\int_{F^\times} \Phi(a\dot{g}) \omega_\pi(a) X^{2v_F(a)} d^\times a \right) f(\dot{g}) X^{v_F(\det \dot{g})} d\dot{g}$$

Fix $\dot{g} \in G$ and set $\Phi_{\dot{g}}(a) = \Phi(a\dot{g})$ for $a \in F$. The map $a \mapsto a\dot{g}$ is an F -linear injection onto the line $F\dot{g}$, which is closed in A , and it is a homeomorphism onto its image; so $\Phi_{\dot{g}}$ is locally constant with compact support, that is $\Phi_{\dot{g}} \in C_c^\infty(F)$. The inner integral is therefore exactly the Tate zeta integral of definition 5.1.7 attached to $\Phi_{\dot{g}}$, the character ω_π and the variable X^2 . By proposition 5.1.8 it converges for $\operatorname{Re}(s)$ large and lies in

$$P_{\omega_\pi}(X^2)^{-1} \mathbb{C}[X^2, X^{-2}]$$

where $P_{\omega_\pi}(t)^{-1} = L(\omega_\pi, s)$ as in definition 5.1.9.

For the outer integral, f vanishes off a compact subset of G/Z . There is a compact open subgroup K' of K under which Φ is invariant under right translation: pick N_0 and M with $\operatorname{supp} \Phi \subseteq \varpi^{-N_0} M_2(\mathfrak{o})$ and Φ constant on cosets of $\varpi^M M_2(\mathfrak{o})$, and take $K' = K \cap (1 + \varpi^{M+N_0} M_2(\mathfrak{o}))$. For $y \in \varpi^{-N_0} M_2(\mathfrak{o})$ and $k \in K'$ one has $yk - y = y(k - 1) \in \varpi^M M_2(\mathfrak{o})$, so $\Phi(yk) = \Phi(y)$; and for y outside that lattice both values vanish, since k is a unit of $M_2(\mathfrak{o})$ and so preserves it. Shrinking K' further so that it also fixes f , the integrand is constant on the cosets of K' meeting the compact support, of which there are finitely many. The outer integral is thus a finite linear combination, and

$$Z(\Phi, f, X) \in P_{\omega_\pi}(X^2)^{-1} \mathbb{C}[X, X^{-1}]$$

a rational function of X with denominator depending only on ω_π .

The non-cuspidal case. By theorem 2.4.16 the representation π is either an irreducible $\iota_B^G \chi$, or $\varphi \circ \det$, or $\varphi \cdot \operatorname{St}_G$; in the latter two cases the exact sequence (2.7) exhibits π as a subquotient of $\operatorname{Ind}_B^G \varphi_T$. In every case π is a subquotient of $\Sigma = \operatorname{Ind}_B^G \chi$ for some character $\chi = \chi_1 \otimes \chi_2$ of T , and Σ is admissible. By lemma 5.2.5 it is enough to prove the claim for $f \in C(\Sigma)$.

Realize Σ on the space X_χ of smooth $\varphi: G \rightarrow \mathbb{C}$ with $\varphi(bg) = \chi(b)\varphi(g)$. Since $B \backslash G$ is compact (corollary 2.2.2), the Duality Theorem (theorem 1.3.12) gives $\check{X}_\chi \cong \operatorname{Ind}_B^G \delta_B^{-1} \check{\chi}$ with the pairing $\langle \tilde{\varphi}, \varphi \rangle = \int_K \tilde{\varphi}(k) \varphi(k) d\mu_K(k)$, so every $f \in C(\Sigma)$ has the form

$$f(g) = \int_K \tilde{\varphi}(k) \varphi(kg) d\mu_K(k)$$

Substituting this and replacing g by $k^{-1}g$, which leaves $d^\times g$ invariant and does not change $v_F(\det g)$ because $\det k \in U_F$, gives

$$Z(\Phi, f, X) = \int_K \tilde{\varphi}(k) \left(\int_G \Phi^k(g) \varphi(g) X^{v_F(\det g)} d^\times g \right) d\mu_K(k), \quad \Phi^k(g) = \Phi(k^{-1}g)$$

and $\Phi^k \in C_c^\infty(A)$. Apply the Iwasawa decomposition $G = BK$ (proposition 2.2.1) and the associated integration formula (corollary 2.2.9). The resulting integration formula (2.4) reads

$$\int_G F(g) d\mu_G(g) = \int_K \int_B F(bk') d\mu_B(b) d\mu_K(k')$$

for a suitable left Haar measure μ_B on B ; the normalizing constant relating μ_G to $d^\times g$ is a positive scalar and affects nothing below, since it does not change the $\mathbb{C}[X, X^{-1}]$ -module generated.

Writing $b = tn$ with $t = \text{diag}(t_1, t_2) \in T$ and $n = n(x) \in N$, a left Haar measure on $B = T \ltimes N$ is $d\mu_B(tn) = d^\times t d\mu_N(n)$: left translation by $t_0 n_0$ sends (t, n) to $(t_0 t, (t^{-1} n_0 t) n)$, which preserves $d^\times t$ in the outer variable and is a translation of N for each fixed t . Since χ is trivial on N and $\det(tn) = t_1 t_2$, the inner integral becomes

$$\int_K \varphi(k') \left(\int_T \int_N \Phi^k(t n(x) k') \chi_1(t_1) \chi_2(t_2) X^{v_F(t_1) + v_F(t_2)} d\mu_N(x) d^\times t \right) d\mu_K(k')$$

Fix k, k' and put $\Phi'(y) = \Phi(k^{-1} y k')$, again in $C_c^\infty(A)$. In the inner double integral substitute $y = t_1 x$, so that $d\mu_N(x) = \|t_1\|^{-1} d\mu_N(y)$ and

$$\int_N \Phi' \begin{pmatrix} t_1 & t_1 x \\ 0 & t_2 \end{pmatrix} d\mu_N(x) = \|t_1\|^{-1} \tilde{\Psi}(t_1, t_2), \quad \tilde{\Psi}(t_1, t_2) = \int_F \Phi' \begin{pmatrix} t_1 & y \\ 0 & t_2 \end{pmatrix} d\mu_N(y)$$

The change of variable is what removes the apparent divergence: the function $\tilde{\Psi}$ lies in $C_c^\infty(F \times F)$. Indeed, choose N_0 with $\text{supp} \Phi' \subseteq \varpi^{-N_0} M_2(\mathfrak{o})$; then the integrand vanishes unless $v_F(t_1), v_F(t_2), v_F(y) \geq -N_0$, so the y -integration runs over the *fixed* compact set $\varpi^{-N_0} \mathfrak{o}$ and $\tilde{\Psi}$ is supported in $\varpi^{-N_0} \mathfrak{o} \times \varpi^{-N_0} \mathfrak{o}$. Local constancy is inherited from Φ' : if Φ' is invariant under translation by $\varpi^M M_2(\mathfrak{o})$ and $t_i \equiv t'_i \pmod{\varpi^M \mathfrak{o}}$, the two matrices differ by an element of that lattice.

The inner double integral is therefore

$$\int_{F^\times} \int_{F^\times} \tilde{\Psi}(t_1, t_2) \chi_1(t_1) \|t_1\|^{-1} \chi_2(t_2) X^{v_F(t_1) + v_F(t_2)} d^\times t_1 d^\times t_2$$

Being locally constant with compact support, $\tilde{\Psi}$ is constant on the cosets of $\varpi^m \mathfrak{o} \times \varpi^m \mathfrak{o}$ for some m and vanishes outside finitely many of them; the argument of proposition 2.2.6, run on $F \times F$, therefore writes it as a finite sum $\sum_j \tilde{\Psi}_j^{(1)} \otimes \tilde{\Psi}_j^{(2)}$ of products of elements of $C_c^\infty(F)$, so the double integral is the finite sum

$$\sum_j Z(\tilde{\Psi}_j^{(1)}, \chi_1 \| - \|^{-1}, X) Z(\tilde{\Psi}_j^{(2)}, \chi_2, X)$$

of products of Tate zeta integrals. By proposition 5.1.8 each summand lies in

$$P_{\chi_1 \| - \|^{-1}}(X)^{-1} P_{\chi_2}(X)^{-1} \mathbb{C}[X, X^{-1}]$$

All the integrands above are bounded by their absolute values in the same way, and the Tate integrals converge absolutely for $\text{Re}(s)$ large, so Fubini's theorem justifies the interchanges on that half plane; both sides are then rational, so the identity persists. Finally the k and k' integrations run over the compact group K with locally constant integrands, hence contribute finite linear combinations, and $Z(\Phi, f, X)$ lies in the displayed fractional ideal, whose denominator depends only on χ .

2. Let $\mathcal{Z}(\pi)$ denote the $\mathbb{C}[X, X^{-1}]$ -submodule of $\mathbb{C}(X)$ generated by the $Z(\Phi, f, X)$, for $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$. It is non-zero: choose $f \in C(\pi)$ with $f(1) \neq 0$ and let Φ be the characteristic function of a small enough compact open neighbourhood of 1 in A on which f is constant and which meets only G_0 , so that $Z(\Phi, f, X)$ is a non-zero constant.

By part (1) there is a single polynomial $D \in \mathbb{C}[X]$, depending only on π , with $\mathcal{Z}(\pi) \subseteq D(X)^{-1}\mathbb{C}[X, X^{-1}]$. The ring $\mathbb{C}[X, X^{-1}]$ is a localization of $\mathbb{C}[X]$, hence a principal ideal domain, and $D(X)^{-1}\mathbb{C}[X, X^{-1}]$ is a free module of rank one over it. A submodule of a free rank-one module over a principal ideal domain is again free of rank at most one, so $\mathcal{Z}(\pi)$ is generated by a single non-zero element $F(X) \in \mathbb{C}(X)$.

It remains to normalize the generator. The units of $\mathbb{C}[X, X^{-1}]$ are exactly the monomials cX^k with $c \in \mathbb{C}^\times$, and multiplying a generator by a unit again gives a generator. Let $c_0 \in \mathbb{C}^\times$ be the constant found above. Since $c_0 \in \mathcal{Z}(\pi) = F\mathbb{C}[X, X^{-1}]$, there is $u \in \mathbb{C}[X, X^{-1}]$ with $c_0 = Fu$, so $F = c_0u^{-1}$. Write $u = X^k\tilde{u}$ with $\tilde{u} \in \mathbb{C}[X]$ and $\tilde{u}(0) \neq 0$; then

$$F = c_0X^{-k}\tilde{u}(X)^{-1}$$

and since c_0X^{-k} is a unit, $\tilde{u}(X)^{-1}$ generates $\mathcal{Z}(\pi)$ as well. Put $P = \tilde{u}/\tilde{u}(0)$, which lies in $\mathbb{C}[X]$ and has $P(0) = 1$; it differs from $\tilde{u}$ by the unit $\tilde{u}(0)$, so $P(X)^{-1}$ is again a generator.

This P is unique. If $P(X)^{-1}$ and $Q(X)^{-1}$ both generate, with $P, Q \in \mathbb{C}[X]$ and $P(0) = Q(0) = 1$, then Q/P is a unit cX^k of $\mathbb{C}[X, X^{-1}]$; both P and Q have non-zero constant term, which forces $k = 0$, and evaluating at $X = 0$ gives $c = 1$. Hence $P = Q$, and $L(\pi, s) = P(q^{-s})^{-1}$ is well defined, completing the proof.

The generator produced by part (2) is unique: two generators of the same $\mathbb{C}[X, X^{-1}]$ -module differ by a unit cX^k , and the normalizations $A = 1$ and $P(0) = 1$ remove exactly that ambiguity. The proof also records two facts that the next sections use directly. For π cuspidal the denominator can be taken to involve only the central character, and for π a subquotient of $\text{Ind}_B^G(\chi_1 \otimes \chi_2)$ it can be taken to be $P_{\chi_1\|\cdot\|^{-1}}P_{\chi_2}$; the sharper statements, that $L(\pi, s) = 1$ in the first case and that the L -factor is exactly a product of two GL_1 factors in the second, are proved in section 5.3 and section 5.4 respectively.

The functional equation for GL_2 involves a subtle normalization shift compared to GL_1 . In Tate's theory, the equation relates s to $1 - s$. For GL_2 , the center of symmetry is shifted. Let $\check{\pi}$ denote the contragredient representation. If $f(g) = \langle \check{v}, \pi(g)v \rangle$ is a coefficient of π , then $\check{f}(g) = f(g^{-1}) = \langle v, \check{\pi}(g)\check{v} \rangle$ is a coefficient of $\check{\pi}$.

Theorem 5.2.7: Functional Equation (Godement-Jacquet)

Let (π, V) be an irreducible smooth representation. There exists a rational function $\gamma(\pi, s, \psi)$ such that:

$$\zeta\left(\hat{\Phi}, \check{f}, \frac{3}{2} - s\right) = \gamma(\pi, s, \psi)\zeta\left(\Phi, f, s + \frac{1}{2}\right) \quad (5.10)$$

for all $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$.

The shift of $+1/2$ on the right and the evaluation at $3/2 - s$ on the left (which is $2 - (s + 1/2)$) ensures that the resulting ε -factor satisfies the standard symmetry $s \leftrightarrow 1 - s$.

The proof runs on the same principle as Tate's (theorem 5.1.11): exhibit both sides of the equation

as elements of one space of equivariant functionals, and show that space is a line. What plays the role of $F^\times$ acting on F is the group $G \times G$ acting on the matrix algebra A by $(g, h): x \mapsto g^{-1}xh$, and the relevant functionals take values not in $\mathbb{C}$ but in the field $\mathbb{C}(X)$, carrying the action

$$(g, h) \cdot P(X) = X^{v_F(\det(gh^{-1}))} P(X)$$

Write $\mathbb{C}(X)_\Theta$ for $\mathbb{C}(X)$ with this action, and set

$$\mathcal{B}(\pi) = \text{Hom}_{G \times G}(C_c^\infty(A) \otimes C(\pi), \mathbb{C}(X)_\Theta)$$

where $G \times G$ acts on $C_c^\infty(A)$ as above and on $C(\pi)$ by $(g, h)f: x \mapsto f(g^{-1}xh)$.

Lemma 5.2.8: Uniqueness of Invariant Bilinear Forms

Let (π, V) be an irreducible smooth representation of G . Then $\dim_{\mathbb{C}(X)} \mathcal{B}(\pi) \leq 1$.

Proof Key ideas:

The argument is the one place in this chapter that reaches beyond the machinery developed here, so the structure is given and the technical input is cited. Stratify A by matrix rank. The rank-two locus is G itself, open in A , and the complement $A \setminus G$ is closed and consists of the rank-one matrices together with 0; extension by zero and restriction give a short exact sequence of smooth $G \times G$ -representations

$$0 \rightarrow C_c^\infty(G) \rightarrow C_c^\infty(A) \rightarrow C_c^\infty(A \setminus G) \rightarrow 0$$

Applying $\text{Hom}_{G \times G}(- \otimes C(\pi), \mathbb{C}(X)_\Theta)$ turns this into a left exact sequence, so the claim reduces to two statements: that the open stratum contributes at most a line, and that the closed stratum contributes nothing.

For the open stratum, $C_c^\infty(G) \otimes C(\pi)$ has a one-dimensional space of such functionals because $C(\pi) \cong \tilde{V} \otimes V$ is irreducible over $G \times G$ for irreducible admissible π , exactly as established for cuspidal π in the proof of proposition 3.1.2, and an equivariant functional on $C_c^\infty(G) \otimes C(\pi)$ is determined by integration against a single matrix coefficient.

For the closed stratum, the rank-one matrices form a single $G \times G$ -orbit, since over a field every rank-one 2×2 matrix is $g e_{11} h$ for suitable $g, h \in G$. Hence $C_c^\infty(A \setminus G)$ is built from a representation compactly induced from the stabilizer of e_{11} , and Frobenius reciprocity for compact induction converts the Hom space into one over that stabilizer. The resulting condition forces X to satisfy a non-trivial polynomial equation, which cannot hold identically in $\mathbb{C}(X)$; so the contribution vanishes.

The step this book does not supply is the last one: Frobenius reciprocity for *compact* induction is deferred in section 1.2 and is what converts the orbit description into a computable Hom space. See [1, §24-§25] for the argument in full.

Granting the lemma, the functional equation is a computation with the two equivariences.

Proof:

Step 1: the zeta integral is equivariant. Let $\Phi \in C_c^\infty(A)$, $f \in C(\pi)$ and $(g, h) \in G \times G$. Substituting $y = gzh^{-1}$, which preserves $d^\times z$ because G is unimodular (proposition 2.2.7), and

using $v_F(\det y) = v_F(\det z) + v_F(\det(gh^{-1}))$,

$$\begin{aligned} Z((g, h)\Phi, (g, h)f, X) &= \int_G \Phi(g^{-1}yh) f(g^{-1}yh) X^{v_F(\det y)} d^\times y \\ &= X^{v_F(\det(gh^{-1}))} \int_G \Phi(z) f(z) X^{v_F(\det z)} d^\times z \end{aligned}$$

so $Z(-, -, X)$ lies in $\mathcal{B}(\pi)$. It is non-zero by the argument in part (2) of proposition 5.2.6.

Step 2: the transformed integral has the same equivariance. The map $z \mapsto gzh^{-1}$ is F -linear on the four-dimensional space A with determinant $\det(g)^2 \det(h)^{-2}$, so $d\mu(y) = \|\det(gh^{-1})\|^2 d\mu(z)$. Together with $\text{tr}(xgzh^{-1}) = \text{tr}((h^{-1}xg)z)$ this gives

$$\widehat{(g, h)\Phi} = \|\det(gh^{-1})\|^2 (h, g)\hat{\Phi}$$

and directly from $\check{f}(x) = f(x^{-1})$,

$$((g, h)f)^\vee(x) = f((h^{-1}xg)^{-1}) = ((h, g)\check{f})(x)$$

Note that both operations *swap* the two factors. Define $W(\Phi, f, X) = Z(\hat{\Phi}, \check{f}, q^{-2}X^{-1})$ and put $m = v_F(\det(gh^{-1}))$. Applying Step 1 to the pair (h, g) , whose determinant valuation is $-m$,

$$W((g, h)\Phi, (g, h)f, X) = q^{-2m} (q^{-2}X^{-1})^{-m} W(\Phi, f, X) = q^{-2m} q^{2m} X^m W(\Phi, f, X)$$

The two powers of q cancel exactly, leaving the factor X^m . Thus W lies in $\mathcal{B}(\pi)$ as well: the swap introduced by the Fourier transform is undone by the reflection $X \mapsto q^{-2}X^{-1}$, and the normalization of $\hat{\Phi}$ is what makes the cancellation exact.

Step 3: proportionality. By lemma 5.2.8 the space $\mathcal{B}(\pi)$ has dimension at most one over $\mathbb{C}(X)$, and Step 1 shows it is non-zero. Hence $\mathcal{B}(\pi)$ is a line and W is a $\mathbb{C}(X)$ -multiple of Z : there is $\gamma \in \mathbb{C}(X)$ with

$$Z(\hat{\Phi}, \check{f}, q^{-2}X^{-1}) = \gamma(X) Z(\Phi, f, X)$$

for all Φ and f . Writing $X = q^{-s}$ turns $q^{-2}X^{-1}$ into $q^{-(2-s)}$, so this reads $\zeta(\hat{\Phi}, \check{f}, 2-s) = \gamma \zeta(\Phi, f, s)$. Replacing s by $s + 1/2$, so that $2-s$ becomes $3/2-s$, gives (5.10) with $\gamma(\pi, s, \psi) = \gamma(q^{-s-1/2})$, completing the proof.

Corollary 5.2.9: Epsilon Factor for GL_2

Define the local constant $\varepsilon(\pi, s, \psi)$ by:

$$\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi) \frac{L(\check{\pi}, 1-s)}{L(\pi, s)}$$

Then $\varepsilon(\pi, s, \psi)$ is a monomial of the form cq^{-fs} with $c \in \mathbb{C}^\times$ and $f \in \mathbb{Z}$, and satisfies

$$\varepsilon(\pi, s, \psi) \varepsilon(\check{\pi}, 1-s, \psi) = \omega_\pi(-1)$$

Proof:

The argument is the GL_2 form of corollary 5.1.14. Apply the functional equation twice. Fourier inversion gives $\hat{\Phi}(x) = \Phi(-x)$, and $\check{f}^\vee = f$, so

$$\zeta(\hat{\Phi}, f, s) = \int_G \Phi(-g) f(g) \|\det g\|^s d^\times g = \omega_\pi(-1) \zeta(\Phi, f, s)$$

where the substitution $g \mapsto -g$ leaves $\|\det g\|^s$ unchanged because $\det(-g) = (-1)^2 \det g$ in rank two, and $f(-g) = f((-I)g) = \omega_\pi(-1)f(g)$. Comparing with two applications of (5.10) gives

$$\gamma(\pi, s, \psi) \gamma(\check{\pi}, 1 - s, \psi) = \omega_\pi(-1)$$

and substituting the defining relation for ε on each factor, the two L -ratios cancel, leaving the stated identity for ε .

For the shape of ε , rewrite the functional equation as

$$\frac{\zeta(\hat{\Phi}, \check{f}, \frac{3}{2} - s)}{L(\check{\pi}, 1 - s)} = \varepsilon(\pi, s, \psi) \frac{\zeta(\Phi, f, s + \frac{1}{2})}{L(\pi, s)}$$

By proposition 5.2.6 the L -function is a generator of the module of zeta integrals, so both fractions lie in $\mathbb{C}[q^s, q^{-s}]$, and there are choices of Φ and f making the right-hand numerator equal to $L(\pi, s)$. Hence $\varepsilon(\pi, s, \psi) \in \mathbb{C}[q^s, q^{-s}]$, and the same argument applied to $\check{\pi}$ puts $\varepsilon(\check{\pi}, 1 - s, \psi)$ there too. Their product is the non-zero constant $\omega_\pi(-1)$, so $\varepsilon(\pi, s, \psi)$ is a unit of $\mathbb{C}[q^s, q^{-s}]$. The units of that ring are exactly the monomials, giving $\varepsilon(\pi, s, \psi) = cq^{-fs}$ with $c \in \mathbb{C}^\times$ and $f \in \mathbb{Z}$, as we sought to show.

Summary

Concept	Tate ($\mathrm{GL}(1)$)	Godement-Jacquet ($\mathrm{GL}(2)$)
Ambient Space	F	$M_2(F)$
Representation	$\chi(x)$	$f(g) = \langle \tilde{v}, \pi(g)v \rangle$
Dual Rep	χ^{-1}	$\check{f}(g) = f(g^{-1})$
Duality Pairing	xy	$\mathrm{tr}(xy)$
Norm Factor	$\ x\ ^s$	$\ \det g\ ^s$
Symmetry	$s \longleftrightarrow 1 - s$	$s + \frac{1}{2} \longleftrightarrow \frac{3}{2} - s$

5.3 Epsilon Factors for Cuspidal

here

5.4 Epsilon Factors for non-Cuspidal

here

5.5 Converse Theorems

(super important! Representations are *characterized* by L-function and ε -factors)

Theorem 5.5.1: Converse Theorem ($\mathrm{gl}(1)$)

here

Proof:

here

6

Representation of Galois Theory: Weil Groups

We now cross the bridge to the other side of the correspondence: the Galois side. The absolute Galois group $\text{Gal}(\overline{F}/F)$ is the key object of study, but its profinite topology is too coarse to see the complex representations that match the smooth representations of $\text{GL}_2(F)$. The goal of this chapter is to refine the Galois group into the *Weil Group* $\mathcal{W}_F$ and to develop its representation theory to a point where it matches the complexity of the $\text{GL}_2(F)$ side.

We begin by defining the Weil group and its topology. We then review Local Class Field Theory (Section 6.2), which essentially constitutes the Langlands correspondence for $n = 1$. We define L -functions and ε -factors for representations of $\mathcal{W}_F$, parallel to those defined for G in the previous chapter. Finally, we introduce *Deligne Representations* (Section 6.4), which incorporate a monodromy operator and allows us to generalize from complex representations to choosing any algebraically closed field of characteristic different from the residual characteristic of F . This modification is necessary to account for the difference between the semisimple Galois representations and the non-semisimple (Special) representations of G , and to align the complex theory with ℓ -adic geometry.

6.1 Weil Group and Smooth Representations

We start by recalling some features of the Galois theory of F . Throughout, let p means the characteristic of the residue class field $\mathbf{k} = \mathfrak{o}/\mathfrak{p}$. Choose a separable algebraic closure $\overline{F}$ of F . We set $\Omega_F = \text{Gal}(\overline{F}/F)$, and view it as a profinite group with its natural (Krull) topology. Thus

$$\Omega_F = \varprojlim \text{Gal}(E/F),$$

where E/F ranges over finite Galois extensions with $E \subset \overline{F}$. If K/F is a finite field extension, with $K \subset \overline{F}$, denote by Ω_K the open subgroup $\text{Gal}(\overline{F}/K)$ of Ω_F . If $\tilde{F}/F$ is some other separable algebraic closure of F , an F -isomorphism $\tilde{F} \cong \overline{F}$ induces an isomorphism $\text{Gal}(\tilde{F}/F) \cong \Omega_F$. This does depend on the choice of isomorphism $\tilde{F} \cong \overline{F}$. Fortunately, it is only up to inner automorphism, so in particular, it induces a *canonical* bijection between the sets of equivalence classes of irreducible smooth representations of the two groups $\text{Gal}(\tilde{F}/F)$, Ω_F .

Fixing an $\overline{F}/F$, consider extensions E/F where $E \subset \overline{F}$. Then the field F admits a unique unramified extension F_m/F of degree m , for each $m \geq 1$. Let F_∞ denote the composite of all these fields. Then F_∞/F is the unique maximal unramified extension of F . The extension F_m/F is Galois and $\text{Gal}(F_m/F)$ is cyclic. An F -automorphism of F_m is determined by its action on the residue field $\mathbf{k}_{F_m} \cong \mathbb{F}_{q^m}$ (see [6, section 1.6]). In particular, there is a unique element φ_m of $\text{Gal}(F_m/F)$ which acts on $\mathbf{k}_{F_m}$ as $x \mapsto x^q$. Set Φ_m to be the map for $\text{Gal}(F_m/F)$ corresponding to the Frobenius map φ_m for the extension of finite fields. The map $\Phi_m \mapsto 1$ gives a canonical isomorphism $\text{Gal}(F_m/F) \rightarrow \mathbb{Z}/m\mathbb{Z}$. Taking the limit over m , we get a canonical isomorphism of topological groups

$$\text{Gal}(F_\infty/F) \cong \varprojlim \mathbb{Z}/m\mathbb{Z} = \hat{\mathbb{Z}}$$

and a unique element $\Phi_F \in \text{Gal}(F_\infty/F)$ which acts on F_m as Φ_m , for all m . Note that $\langle \Phi_F \rangle \cong \mathbb{Z}$ and hence cannot generate the entire $\text{Gal}(F_\infty/F)$, however it is *dense* (on every finite group the restriction generates the entire group).

Definition 6.1.1: Geometric Frobenius

The element Φ_F is the *geometric Frobenius substitution* on F_∞ . The map φ_F for the limit of galois groups of finite fields is the *arithmetic Frobenius substitution*.

An element of Ω_F is called a *geometric Frobenius element* (over F) if its image in $\text{Gal}(F_\infty/F)$ is Φ_F . The Chinese Remainder Theorem gives a canonical isomorphism of topological groups

$$\hat{\mathbb{Z}} \cong \prod_{\ell} \mathbb{Z}_{\ell},$$

where ℓ ranges over all prime numbers and $\mathbb{Z}_{\ell}$ is the (additive) group of ℓ -adic integers.

Let $\mathcal{I}_F = \text{Gal}(\overline{F}/F_\infty)$ be the *inertia group* of F . This fits in the exact sequence of topological groups

$$1 \rightarrow \mathcal{I}_F \rightarrow \Omega_F \rightarrow \text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \cong \hat{\mathbb{Z}} \rightarrow 0.$$

Let us now extend to the maximal tamely ramified extensions. For each integer $n \geq 1$, $p \nmid n$, the field F_∞ has a unique extension E_n/F_∞ of degree n . If ϖ is a prime element of F , the extension E_n is generated over F_∞ by an n -th root of ϖ . The extension E_n/F_∞ is cyclic, and the Galois group $\text{Gal}(E_n/F_\infty)$ admits a canonical description using Kummer theory (presented in [9, section 18.4]). Choose $\alpha \in E_n$ such that $\alpha^n = \varpi$. The map

$$\begin{aligned} \text{Gal}(E_n/F_\infty) &\longrightarrow \mu_n, \\ \sigma &\longmapsto \sigma(\alpha)/\alpha, \end{aligned} \tag{6.1}$$

gives a canonical isomorphism of $\text{Gal}(E_n/F_\infty)$ with the group μ_n of n -th roots of unity in F_∞ . The composite E_∞ of all extensions E_n/F_∞ is the maximal tamely ramified extension of F . Taking the

limit of the isomorphisms (28.3.1), we get topological isomorphisms

$$\mathrm{Gal}(E_\infty/F_\infty) \cong \varprojlim \mu_n \cong \prod_{\ell \neq p} \mathbb{Z}_\ell.$$

In the limit, n ranges over all positive integers not divisible by p ; in the product, ℓ ranges over all prime numbers other than p .

Finally, let $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty)$ be the *wild inertia group*, completing the tower

$$\overline{F}/E_\infty/F_\infty/F$$

Any finite extension of E_∞ has degree a power of p , so the profinite group $\mathcal{P}_F$ is a pro p -group. It is the unique pro p -Sylow subgroup of $\mathcal{I}_F$, and hence it is normal, giving the isomorphism $\mathcal{I}_F/\mathcal{P}_F \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$ which is uniquely determined up to multiplication by an element of $\prod \mathbb{Z}_\ell^\times$.

The group $\mathrm{Gal}(F_\infty/F)$ acts on $\mathrm{Gal}(E_\infty/F_\infty)$ by conjugation. An isomorphism transfers this action to one on the group $\prod_{\ell \neq p} \mathbb{Z}_\ell$. For future reference, we box this result:

Proposition 6.1.2: Tame Inertia Action

If $t_0 : \mathcal{I}_F/\mathcal{P}_F \rightarrow \prod_{\ell \neq p} \mathbb{Z}_\ell$ is a topological isomorphism, then

$$t_0(\Phi_F \sigma \Phi_F^{-1}) = q^{-1} t_0(\sigma), \quad \sigma \in \mathcal{I}_F/\mathcal{P}_F.$$

Let ${}_a\mathcal{W}_F$ be defined by:

$${}_a\mathcal{W}_F = \{\sigma \in \Omega_F : \sigma|_{F_\infty} \in \langle \Phi_F \rangle\} = \mathrm{res}^{-1}(\langle \Phi_F \rangle) \subseteq \Omega_F,$$

where $\mathrm{res} : \Omega_F \rightarrow \mathrm{Gal}(F_\infty/F)$ is the restriction map. As any $\varphi \in \mathcal{I}_F$ restricts to the identity, $\mathcal{I}_F \subseteq {}_a\mathcal{W}_F$. For If $G_E = \mathrm{Gal}(E/F)$ for a finite F , Then the restriction to E is surjective, and hence ${}_a\mathcal{W}_F$ is *dense* in $\mathrm{Gal}(\overline{F}/F)$.

Next, the map:

$$\mathrm{res} : {}_a\mathcal{W}_F \rightarrow \mathrm{Gal}(F_\infty/F) \quad \sigma \mapsto \sigma|_{F_\infty}$$

is by construction surjective onto $\langle \Phi_F \rangle$ with kernel $\mathcal{I}_F$ forming a short exact sequence:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathcal{I}_F & \longrightarrow & G_F & \longrightarrow & \hat{Z} \cong \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \longrightarrow \\ & & = \downarrow & & \uparrow & & \uparrow \\ 1 & \longrightarrow & \mathcal{I}_F & \longrightarrow & {}_a\mathcal{W}_F & \longrightarrow & \mathbb{Z} \longrightarrow \end{array}$$

Notice the difference between these two quotients:

$${}_a\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z} \quad (\text{Generated by Frobenius}) \quad \text{and} \quad \mathrm{Gal}(F_\infty/F) \cong \widehat{\mathbb{Z}} \quad (\text{Profinite completion})$$

Inertia		Total Group		Unramified
$\mathcal{I}_F$	$\subset$	Ω_F (Galois)	$\twoheadrightarrow$	$\widehat{\mathbb{Z}}$
$\downarrow$		$\cup$		$\cup$
$\mathcal{I}_F$	$\subset$	${}_a\mathcal{W}_F$ (Weil)	$\twoheadrightarrow$	$\mathbb{Z}$

For a concrete element in Ω_F that is not in ${}_a\mathcal{W}_F$, assume $\text{char} F \neq 2$ so that 2 is invertible in F (or else pick 3). Let α be the element that is $1/2 \bmod p^n$ for each n and $p \neq 2$. Let $\sigma \in \Omega_F$ associate to this element and notice that $\sigma^2|_{F_\infty} = \Phi_F$, showing that it cannot be in ${}_a\mathcal{W}_F$.

Now, ${}_a\mathcal{W}_F$ is a profinite group, hence not “enough” representations to correspond to irreducible representations mentioned earlier. We need a better topology to hold the arithmetic information of ${}_a\mathcal{W}_F$, that is a locally profinite topology:

Definition 6.1.3: Weil Group

The Weil group $\mathcal{W}_F$ of F (relative to $\bar{F}/F$) is the topological group, with underlying abstract group ${}_a\mathcal{W}_F$, so that

1. $\mathcal{I}_F$ is an open subgroup of $\mathcal{W}_F$, and
2. the topology on $\mathcal{I}_F$, as subspace of $\mathcal{W}_F$, coincides with its natural topology as $\text{Gal}(\bar{F}/F_\infty) \subset \Omega_F$.

As $\mathcal{I}_F$ is open, $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ must be discrete, and as it is infinite it must be that $\mathcal{W}_F$ is no longer compact, and hence it is *locally profinite* (recall in a profinite group a set is open if and only if it is finite index). The definition of $\mathcal{W}_F$ does depend on the choice of $\bar{F}/F$ up to inner automorphism of Ω_F .

Lemma 6.1.4: Injection is Continuous

The identity map $\iota_F : \mathcal{W}_F \rightarrow {}_a\mathcal{W}_F \subset \Omega_F$ is a continuous injection.

Proof:

The topology of $\mathcal{W}_F$ is strictly finer.

Let $\nu_F : \mathcal{W}_F \rightarrow \mathbb{Z}$ be the canonical map taking a geometric Frobenius element to 1 and define:

$$\|x\| = q^{-\nu_F(x)}, \quad x \in \mathcal{W}_F.$$

Proposition 6.1.5: Weil subgroup and Finite Extensions

1. Let E/F be a finite extension, $E \subset \overline{F}$.
 - (a) The group $\mathcal{W}_F$ has a unique subgroup $\mathcal{W}_F^E$ such that

$$\iota_F(\mathcal{W}_F^E) = {}_a\mathcal{W}_F \cap \Omega_E.$$
 - (b) The subgroup $\mathcal{W}_F^E$ is open and of finite index in $\mathcal{W}_F$; it is normal in $\mathcal{W}_F$ if and only if E/F is Galois.
 - (c) The canonical map $\mathcal{W}_F^E \backslash \mathcal{W}_F \rightarrow \Omega_E \backslash \Omega_F$ is a bijection.
 - (d) The canonical map $\iota_E : \mathcal{W}_E \rightarrow \Omega_E$ induces a topological isomorphism $\mathcal{W}_E \cong \mathcal{W}_F^E$.
2. The map $E/F \mapsto \mathcal{W}_F^E$ is a bijection between the set of finite extensions E of F inside $\overline{F}$ and the set of open subgroups of $\mathcal{W}_F$ of finite index.

Proof:
exercise.

We henceforward, identify $\mathcal{W}_E$ with the subgroup $\mathcal{W}_F^E$ of $\mathcal{W}_F$.

(Complex) Smooth Representation of Weil Group

We now look at smooth representations of $\mathcal{W}_F$. Note that as Ω_F is profinite, smooth representations must be semi-simple, while $\mathcal{W}_F$ being locally profinite (as it admits a discrete quotient $\mathbb{Z}$) must admit non-semisimple representations. As we will see, the irreducible representations of $\mathcal{W}_F$ are closely related to the irreducible representations of Ω_F , but the reducible ones exhibit a greater variety, namely the principal series shall be associated to them, and these are “hidden” if we only look at Ω_F . Just like in the previous chapters, we focus on *complex* representations. P-adic representations are left for [ref:HERE](#).

Lemma 6.1.6: Irreducible Representations of Weil Group is Finite

Let (ρ, V) be an irreducible representation of $\mathcal{W}_F$. Then V is finite dimensional.

Proof:

let $v \in V$. As ρ is smooth it is fixed by a compact open subgroup $K \subseteq \mathcal{W}_F$. Let $\mathcal{J} = K \cap \mathcal{I}_F$ so that v is fixed by an open subgroup of $\mathcal{I}_F$. As $\mathcal{I}_F$ has the same topology as the subspace topology of $\mathcal{I}_F \subseteq \Omega_F$ and $\mathcal{J}$ is open, it corresponds to a finite extension E/F . By taking the galois closure of E/F , we get a smaller group that is still in the stabilizer of v , hence we may assume E/F is galois hence $\mathcal{J}$ is normal in $\mathcal{W}_F$. It is an exercise in classical representation theory to show that if ρ is irreducible and $N \trianglelefteq G$ is normal, if N fixes a point v , N acts trivially on all of V . Hence, $\mathcal{J} \subseteq \ker \rho$.

Now, as $\mathcal{J} \subseteq \mathcal{I}_K$ is an open subgroup, $\mathcal{I}_K/\mathcal{J}$ is finite. Let $\Phi \in \mathcal{W}_F$ be a Frobenius element (an element of the coset $1 + \mathcal{I}_F$ in the map $\mathcal{W}_F \xrightarrow{\text{res}} \langle \Phi_F \rangle \cong \mathbb{Z}$). As both $\mathcal{I}_K$ and $\mathcal{J}$ are normal in

$\mathcal{W}_F$ (they are both associated to a galois group of an extension of F), Φ acts by conjugation on $\mathcal{I}_K/\mathcal{J}$, and as this group is finite some Φ^d acts trivially on this group. Thus:

$$\Phi^d \cdot x \cdot \Phi^{-d} = xj$$

for appropriate $j \in \mathcal{J}$. Applying ρ :

$$\rho(\Phi^d)\rho(x)\rho(\Phi^{-d}) = \rho(x)\rho(j)$$

As $\mathcal{J} \subseteq \ker \rho$:

$$\rho(\Phi^d)\rho(x)\rho(\Phi^{-d}) = \rho(x) \cdot I = \rho(x)$$

Moving terms around:

$$\rho(\Phi^d)\rho(x) = \rho(x)\rho(\Phi^d)$$

Thus the map $\rho(\Phi^d)$ is in the center of $\rho(\mathcal{W}_F) \subseteq \text{Aut}(V)$, and as ρ is irreducible this means $\rho(\Phi^d)$ acts like scalar multiplication.

Using this, I claim that for the v that was fixed at the very beginning of the proof, the space

$$U = \{\rho(g) \cdot v : g \in \mathcal{W}_F\}$$

is finite dimensional. By definition of $\mathcal{W}_F$, $g = \Phi^k t$ for $t \in \mathcal{I}_F$. By taking the finite $\mathcal{I}_F/\mathcal{J}$ we have coset representatives $r_1, \dots, r_n$ so that:

$$\rho(g) \cdot v = \rho(\Phi^k)\rho(r_\alpha)\rho(j) \cdot v$$

where $j \in \mathcal{J}$ is any element. As $\mathcal{J} \subseteq \ker \rho$ we simplify to

$$\rho(g) \cdot v = \rho(\Phi^k)\rho(r_\alpha) \cdot v$$

Next, write $\Phi^k = \Phi^{ad+c}$ where $0 \leq c < d$ so that we can use the fact that $\rho(\Phi^d)$ is scalar multiplication to get

$$\rho(\Phi^d) = \lambda^a \rho(\Phi^c)$$

But now, we get that there finitely many possible vectors that are used to span U

$$\rho(\Phi^c)\rho(r_\alpha) \cdot v \quad 0 \leq c < d, \text{ and } 0 < \alpha < n$$

showing it is finite. Certainly this space is $\mathcal{W}_F$ -stable, and as V is irreducible it must be that $U = V$, as we sought to show.

Hence the representations of $\mathcal{W}_F$ of interest will be finite! Let ρ be a finite representation of $\mathcal{W}_F$. Then the restriction to $\mathcal{I}_F$ (a profinite group) will be semi-simple by an easy generalization of Maschke's theorem (proposition 1.2.9), and $\rho(\mathcal{I}_F)$ is finite. Therefore, all the complexity comes from the Frobenius elements of $\mathcal{W}_F$.

Proposition 6.1.7: Irreducible in Galois, then in Weil

Let $\mathcal{W}_F$ be the Weil group, $\iota_F : \mathcal{W}_F \rightarrow {}_a\mathcal{W}_F$ the identity map (a continuous but not homeomorphic bijection), (ρ, V) a smooth representation of Ω_F , the absolute galois group of F so that $\rho \circ \iota_F$ can be thought of as the restriction to $\mathcal{W}_F$. Then:

1. if ρ is irreducible (on Ω_F), then $\rho \circ \iota_F$ is irreducible (on $\mathcal{W}_F$)
2. If ρ_1, ρ_2 are irreducible on Ω_F , then $\rho_1 \cong \rho_2$ if and only if $\rho_1 \circ \iota_F \cong \rho_2 \circ \iota_F$.

Proof:

1. Take $\ker \rho \subseteq \Omega_F$. This is an open subset. Let $K = \ker \rho$. Then gK is open for all $g \in \Omega_F$. As $\mathcal{W}_F$ is dense, $gK \cap \mathcal{W}_F \neq \emptyset$. Then $gk = w$ for some $k \in K$ and $w \in \mathcal{W}_F$, in particular:

$$g = wk^{-1} \quad w \in \mathcal{W}_F, \quad k^{-1} \in K$$

As this is true for all $g \in \Omega_F$:

$$\Omega_F = \ker(\rho) \cdot \mathcal{W}_F$$

Taking ρ of both sides we get:

$$\rho(\Omega_F) = \rho(\mathcal{W}_F)$$

Thus, if there are no nontrivial $\rho(\Omega_F)$ -invariant subspaces, there are no nontrivial $\rho(\mathcal{W}_F)$ -invariant subspaces.

2. If $\rho_1 \rightarrow \rho_2$ is an Ω_F -isomorphism, restriction certainly makes it a $\mathcal{W}_F$ -isomorphism, and the equality $\rho(\Omega_F) = \rho(\mathcal{W}_F)$ proven above shows the other way is true too, completing the proof.

The converse is true *up to tensoring* with an unramified character:

Definition 6.1.8: Unramified Character

Let χ be a (complex) character on Ω_F or $\mathcal{W}_F$. Then χ is *unramified* if it is trivial on $\mathcal{I}_F$ ($\chi|_{\mathcal{I}_F}$ is trivial).

If ρ is a smooth, finite-dimensional representation of Ω_F or $\mathcal{W}_F$, the map $x \mapsto \det(\rho(x))$ is a character, $\det \rho$.

Proposition 6.1.9: Irreducible and Finite in Weil, then in Galois

Let τ be an irreducible smooth representation of $\mathcal{W}_F$. The following are equivalent:

1. the group $\tau(\mathcal{W}_F)$ is finite;
2. $\tau \cong \rho \circ \iota_F$, for some irreducible smooth representation ρ of Ω_F ;
3. the character $\det \tau$ has finite order.

For any irreducible smooth representation τ of $\mathcal{W}_F$, there is an unramified character χ of $\mathcal{W}_F$ such that $\chi \otimes \tau$ satisfies (1)–(3).

Proof:

- (1) $\Rightarrow$ (2) We need to find the appropriate representation ρ of Ω_F . Let $\Phi \in \mathcal{W}_F$ be a Frobenius element. Then $\tau(\Phi)$ has finite order, say d . Since $\hat{\mathbb{Z}}/d\hat{\mathbb{Z}} \cong \mathbb{Z}/d\mathbb{Z}$, for $a \in \hat{\mathbb{Z}}$, there exists an integer $\bar{a}$ such that $\bar{a} \equiv a \pmod{d\hat{\mathbb{Z}}}$ up to $d\mathbb{Z}$. Now, as shown earlier any $w \in \Omega_F$ can be written as

$$w = \Phi^a \sigma \quad a \in \hat{\mathbb{Z}}, \sigma \in \mathcal{I}_F$$

Define ρ of Ω_F as :

$$\rho(w) = \rho(\Phi^a \sigma) = \tau(\Phi)^{\bar{a}} \tau(\sigma)$$

Certainly $\tau = \rho \circ \iota_F$ and ρ is irreducible

- (2) $\Rightarrow$ (3) Let τ be an irreducible representation that extends to ρ so that $\det \tau = (\det \rho)|_{\mathcal{W}_F}$. Let $\chi = \det \rho$. Then it is a *character*:

$$\chi : \Omega_F \rightarrow \mathbb{C}^*$$

As χ is smooth, its kernel K is open and as Ω_F is compact Ω_F/K is finite. Thus, χ factors through Ω_F/K which has finite order, hence $\chi^n = \text{id}$ for appropriate n , hence $\tau^n = 1$.

- (3) $\Rightarrow$ (1) Assume $\det \tau$ has finite order. First, as $\mathcal{I}_F$ is compact, $\tau(\mathcal{I}_F)$ is finite, hence some order of the conjugation by $\tau(\varphi)$ acts trivially, say $\tau(\varphi)^k$. Trivially, it commutes with $\tau(\Phi)$, and hence with all of $\mathcal{W}_F$, and so by Schur's lemma $\tau(\Phi)^k = \lambda I$.

Now, letting $n = \dim \tau$ (which exists as τ is irreducible by lemma 6.1.6), $\det(\tau(\Phi)^k) = \det(\lambda I) = \lambda^n I$. As $\det \tau$ has finite order (say m):

$$\det(\tau(\Phi)^k)^m = \lambda^{nm} = 1$$

Hence λ is a root of unity. Hence $\tau(\Phi)^k = \lambda I$ has finite order. As $\mathcal{W}_F = \langle \Phi \rangle \cdot \mathcal{I}_F$ and the τ of both groups on the right hand side has finite order, $\tau(\mathcal{W}_F)$ has finite order.

For the last part, notice the proof that $\tau(\Phi)^k$ did not rely on any assumption, but on the compactness of $\mathcal{I}_F$. Say $\tau(\Phi)^k = cI$. Using this c , pick a $z_0 \in \mathbb{C}$ such that $(z_0^n = c$. The define:

$$\chi : \mathcal{W}_F \rightarrow \mathbb{C}^* \quad \chi(\Phi^n \mathcal{I}_F) = z_0^n$$

This is certainly a smooth character and unramified by construction. Then letting $\tau' = \chi^{-1} \otimes \tau$ we get:

$$\tau'(\Phi)^k = (\chi^{-1}(\Phi)\tau(\Phi))^k = \chi(\Phi)^{-k}\tau(\Phi)^k = c^{-1} \cdot (c \cdot I) = I$$

Hence τ' has finite order, and like in the proof of (3) $\Rightarrow$ (1) as $\mathcal{W}_F = \langle \Phi \rangle \cdot \mathcal{I}_F$ and τ' is finite on both pieces, $\tau'(\mathcal{W}_F)$ is finite, satisfying the first condition and completing the proof.

This result is as good as it can get, that is there exist irreducible representations τ of $\mathcal{W}_F$ where $\tau(\mathcal{W}_F)$ has infinite order:

Example 6.1.10: Irreducible in Weil But Not Defined in Galois Group

Let $\chi : \mathcal{W}_F \rightarrow \mathbb{C}^*$, $\omega(w) = |w| = q^{-\nu_F(w)}$. This is certainly a smooth one-dimensional irreducible representation, and $\chi(\mathcal{W}_F) = \mathbb{Z}$. This map simply does not exist over Ω_F (the image must be compact).

Next, let us relate representations of $\mathcal{W}_F$ and $\mathcal{W}_E$ where E/F is a finite-separable/galois extension where $E \subseteq \bar{F}$.

Definition 6.1.11: Induction On Weil Representation

Let E/F be a finite separable extension where $E \subseteq \bar{F}$. Then write:

$$\text{Ind}_{E/F} := \text{Ind}_{\mathcal{W}_E}^{\mathcal{W}_F}$$

for the functor of smooth induction from $\mathcal{W}_E$ to $\mathcal{W}_F$, and:

$$\text{Res}_{E/F} := \text{Res}_{\mathcal{W}_E}^{\mathcal{W}_F}$$

From now on, if ρ is a smooth representation of $\mathcal{W}_F$, $\rho|_E = \text{Res}_{E/F}\rho = \rho|_{\mathcal{W}_E}$. Recall lemma 1.2.37, for ease of reference the following translates this lemma with our groups of interest:

Lemma 6.1.12: Weil Representation of Finite Extension

Let E/F be a finite separable field extension, $E \subset \bar{F}$.

1. Let ρ be a smooth representation of $\mathcal{W}_F$; then ρ is semisimple if and only if $\rho|_E$ is semisimple.
2. Let τ be a smooth representation of $\mathcal{W}_E$; then τ is semisimple if and only if $\text{Ind}_{E/F}\tau$ is semisimple.

Proof:

Directly apply lemma 1.2.37.

Definition 6.1.13: Class of Semisimple Weil Representations

Let F be a local field. For each integer $n \geq 1$, let

$$\mathcal{G}_n^{\text{ss}}(F)$$

denote the set of isomorphism classes of semisimple smooth representations of $\mathcal{W}_F$ of dimension n . Let

$$\mathcal{G}_n^0(F)$$

denote the set of isomorphism classes of *irreducible* smooth representations of W_F of dimension n .

If E/F is a finite extension where $E \subseteq \bar{F}$. Then lemma 6.1.12 gives the induction and restriction maps

$$\text{Ind}_{E/F} : \mathcal{G}_n^{\text{ss}}(E) \rightarrow \mathcal{G}_{nd}^{\text{ss}}(F),$$

$$\text{Res}_{E/F} : \mathcal{G}_n^{\text{ss}}(F) \rightarrow \mathcal{G}_n^{\text{ss}}(E),$$

where $d = [E : F]$.

It is important to take a moment to recognise that we made choices in the definition of the Weil group (the field closure, the Frobenius element), but W_F is, colloquially speaking, well-defined up to inner automorphism of Ω_F . The sets $\mathcal{G}_n^0(F)$, $\mathcal{G}_n^{\text{ss}}(F)$ are thus well-defined and independent of all choices (and so are the above two maps). In particular, given two separable field extensions E/F , E'/F' of finite degree d and an isomorphism $\alpha : E \rightarrow E'$ such that $\alpha(F) = F'$. We get a well-defined bijection $\mathcal{G}_n^{\text{ss}}(F) \rightarrow \mathcal{G}_n^{\text{ss}}(F')$ and the diagram commutes

$$\begin{array}{ccc} \mathcal{G}_n^{\text{ss}}(E) & \longrightarrow & \mathcal{G}_n^{\text{ss}}(E') \\ \downarrow \text{Ind}_{E/F} & & \downarrow \text{Ind}_{E'/F'} \\ \mathcal{G}_{nd}^{\text{ss}}(F) & \longrightarrow & \mathcal{G}_{nd}^{\text{ss}}(F') \end{array}$$

Similarly for restriction. The induction and restriction map are defined for any finite separable extension E/F , independent of any choice of F -embedding of E in $\bar{F}$.

Proposition 6.1.14: Semisimple Weil Representation

Let (ρ, V) be a smooth representation of $\mathcal{W}_F$ of finite dimension, and let $\Phi \in \mathcal{W}_F$ be a Frobenius element. The following are equivalent:

- (1) the representation ρ is semisimple;
- (2) the automorphism $\rho(\Phi) \in \text{Aut}_{\mathbb{C}}(V)$ is semisimple;
- (3) the automorphism $\rho(\Psi) \in \text{Aut}_{\mathbb{C}}(V)$ is semisimple, for every element Ψ of $\mathcal{W}_F$.

Proof:

(3) $\Rightarrow$ (2) is obvious, so let's prove (2) $\Rightarrow$ (1) and then (1) $\Rightarrow$ (3)

(2) $\Rightarrow$ (1) Recall a semisimple operator means it is diagonalizable. As $\rho(\Phi)$ is semisimple, so is $\rho(\Phi)^d$ for any d . Pick d so that $\rho(\Phi)^d$ commutes with $\rho(\mathcal{I}_F)$ (as it is finite,

such a d exists). Then as Φ^d and $\mathcal{I}_F$ commutes, $\rho|_H$ where $H = \langle \Phi^d, \mathcal{I}_F \rangle$ is semisimple. Now, G/H is finite and $\rho|_H$ is semisimple, hence by lemma 6.1.12 ρ is semisimple

- (1) $\Rightarrow$ (3) Let ρ be semisimple and pick any $\Psi \in \mathcal{W}_F$. If $\Psi \in \mathcal{I}_F$, it $\rho(\Psi)$ semisimple as $\mathcal{I}_F$ is profinite, so assume $\Psi \notin \mathcal{I}_F$. Now, let $n = \nu_F(\Psi)$ which is nonzero as $\Psi \notin \mathcal{I}_F$. Choose an unramified degree n extension E/F (such an extension exists, see [6]) and consider $\mathcal{W}_E$ which by proposition 6.1.5 is an open subgroup of $\mathcal{W}_F$ of finite index and consists of elements whose valuation is divisible by n . Certainly Ψ belongs in this group as:

$$\nu_E(\Psi) = \frac{1}{n} \nu_F(\Psi) = \frac{1}{n} \cdot n = 1$$

and $\mathcal{I}_E = \mathcal{I}_F$ (certainly the bijective as E/F is unramified). For the generation notice that:

$$1 \rightarrow \mathcal{I}_E \rightarrow \mathcal{W}_E \xrightarrow{\nu_F} \langle \Psi \rangle \rightarrow 1$$

And hence $\mathcal{W}_E = \langle \Psi, \mathcal{I}_E \rangle$. As ρ is assumed to be semisimple and E/F is finite, by lemma 6.1.6 $\rho|_E$ is semisimple. As we want to show $\rho(\Psi)$ is semisimple, we may focus on E , or equivalently we may as well assume $E = F$. As we want to prove it is semisimple, we need to prove each block-diagonal is diagonalizable, so we may further reduce to assuming ρ is irreducible. But now, by proposition 6.1.9, any irreducible representation can be twisted by an unramified character χ to get a finite image. By untwisting ($c = \chi(\Psi)^{-1}$), we get:

$$\rho(\Psi) = c\varphi$$

where φ has finite order. But then by regular Maschke's theorem $\rho(\Psi)$ is semisimple, as we sought to show.

6.2 Weil Group and Local Class Field Theory

Let $\mathcal{W}_F^{\text{der}}$ denote the closure of the commutator subgroup of $\mathcal{W}_F$ and let

$$\mathcal{W}_F^{\text{ab}} = \mathcal{W}_F / \mathcal{W}_F^{\text{der}}$$

The group $\mathcal{W}_F^{\text{ab}}$ is a locally profinite abelian group (defined independently of the choices made in the definition of $\mathcal{W}_F$). The following summarizes the core result of CFT:

Theorem 6.2.1: Local Class Field Theory

There is a canonical continuous group homomorphism

$$\mathbf{a}_F: \mathcal{W}_F \longrightarrow F^\times$$

Then:

1. The map $\mathbf{a}_F$ induces a topological isomorphism $\mathcal{W}_F^{\text{ab}} \cong F^\times$.
2. An element $x \in \mathcal{W}_F$ is a geometric Frobenius if and only if $\mathbf{a}_F(x)$ is a prime element of F .
3. We have $\mathbf{a}_F(\mathcal{I}_F) = U_F$ and $\mathbf{a}_F(\mathcal{P}_F) = U_F^1$.
4. If E/F is a finite separable extension, the diagram

$$\begin{array}{ccc} \mathcal{W}_E & \xrightarrow{\mathbf{a}_E} & E^\times \\ \downarrow & & \downarrow N_{E/F} \\ \mathcal{W}_F & \xrightarrow{\mathbf{a}_F} & F^\times \end{array}$$

commutes.

5. Let $\alpha: F \rightarrow F'$ be an isomorphism of fields. The map α induces an isomorphism $\alpha: \mathcal{W}_F^{\text{ab}} \rightarrow \mathcal{W}_{F'}^{\text{ab}}$, and the diagram

$$\begin{array}{ccc} \mathcal{W}_F^{\text{ab}} & \xrightarrow{\alpha} & \mathcal{W}_{F'}^{\text{ab}} \\ \mathbf{a}_F \downarrow & & \downarrow \mathbf{a}_{F'} \\ F^\times & \xrightarrow{\alpha} & F'^\times \end{array}$$

commutes.

The map $\mathbf{a}_F$ is the Artin Reciprocity Map.

Proof:

This is the ultimate result of *EYNTKA Class Field Theory* (see [6])

Theorem 6.2.2: Local Class Theory: Norm

1. The map $E/F \mapsto N_{E/F}(E^\times)$ gives a bijection between the set of finite abelian extensions E/F , $E \subset \bar{F}$, and the set of open subgroups of $F^\times$ of finite index.
2. The map $\mathbf{a}_F$ induces an isomorphism $\text{Gal}(E/F) \cong F^\times / N_{E/F}(E^\times)$, for any finite abelian extension E/F .
3. Let E/F be a finite separable extension. If E_1/F is the maximal abelian sub-extension of E/F , then $N_{E/F}(E^\times) = N_{E_1/F}(E_1^\times)$.

Proof:

This is the ultimate result of *EYNTKA Class Field Theory* (see [6])

There is also the need for “lifting” a representation given E/F . To that end, let G be some group and H a subgroup of G of finite index. Let $D(G)$ be the commutator subgroup of G (denoted $[G, G]$ in [6] and [3] but abbreviated to $D(G)$ since it is all “derivation”). Choose a section $t: G/H \rightarrow G$ of the quotient map $G \rightarrow G/H$. For $x \in G/H$ and $g \in G$, define an element $h_{x,g} \in H$ by $gt(x) = t(gx)h_{x,g}$. Then we can define the map:

Definition 6.2.3: Verlagerung Map

$$\begin{aligned} \text{ver}_{G/H}: G/D(G) &\longrightarrow H/D(H), \\ gD(G) &\longmapsto \prod_{x \in G/H} h_{x,g} D(H). \end{aligned}$$

The map $\text{ver}_{G/H}$, called the *transfer* or *Verlagerung*, and is a homomorphism of abelian groups.

If G is locally compact, we write G^{der} for the closure of $D(G)$ in G and $G^{\text{ab}} = G/G^{\text{der}}$. If H is a closed subgroup of G of finite index, the transfer then induces a continuous homomorphism $G^{\text{ab}} \rightarrow H^{\text{ab}}$, which we continue to denote $\text{ver}_{G/H}$. In particular, if E/F is a finite separable field extension, the transfer gives a map $\text{ver}_{E/F}: \mathcal{W}_F^{\text{ab}} \rightarrow \mathcal{W}_E^{\text{ab}}$.

Theorem 6.2.4: Transfer Theorem

If E/F is a finite separable field extension, the diagram

$$\begin{array}{ccc} \mathcal{W}_E^{\text{ab}} & \xrightarrow{\mathbf{a}_E} & E^\times \\ \text{ver}_{E/F} \uparrow & & \uparrow \\ \mathcal{W}_F^{\text{ab}} & \xrightarrow{\mathbf{a}_F} & F^\times \end{array}$$

commutes, where the map $F^\times \rightarrow E^\times$ is inclusion.

Let us now link this to representations. As $\mathcal{W}^{\text{ab}}$ is abelian, all representations are characters. By theorem 6.2.1, there is an isomorphism:

$$\text{Hom}_{\mathbf{Top}}(F^*, \mathbb{C}) \xrightarrow{\sim} \text{Hom}_{\mathbf{Top}}(\mathcal{W}_F^{\text{ab}}, \mathbb{C}) \quad \chi \mapsto \chi \circ \mathbf{a}_F$$

The isomorphism preserves unramified characters, and maps tamely ramified characters of F^* to characters of $\mathcal{W}_F$ trivial on $\mathcal{P}_F$. Thus, without loss of generality we may simply refer to characters of $\mathcal{W}_F$ without composing with the Artin map.

An important character to consider is $\det \rho$ where $\rho \in \mathcal{G}_d^{ss}(F)$ viewed as a character on F^* . It shall often come up as it natural reduces an n -representation to a character.

Let K/F is a separable extension of finite degree d and 1_K denotes the trivial character of $\mathcal{W}_K$.

Then its regular representation is:

$$R_{K/F} = \text{Ind}_{K/F} 1_K \in \mathcal{G}_d^{\text{ss}}(F),$$

and its determinant

$$\varkappa_{K/F} = \det R_{K/F}.$$

Thus, $\varkappa_{K/F}$ is a character of $F^\times$ satisfying $\varkappa_{K/F}^2 = 1$ (as $\varkappa_{K/F} \in \{1, -1\}$). It arises in the following context:

Proposition 6.2.5: Determinant of Inflation

Let K/F be a separable extension of degree d , let $\rho \in \mathcal{G}_m^{\text{ss}}(K)$ and put $\sigma = \text{Ind}_{K/F} \rho \in \mathcal{G}_{md}^{\text{ss}}(F)$. Then

$$\det \sigma = \varkappa_{K/F}^m \otimes \det \rho|_{F^\times}$$

Proof:

This follows from the Transfer Theorem and this lemma that is left as an exercise:

Lemma 6.2.6

Let G be a group, let H be a subgroup of G of finite index, and ρ a finite dimensional representation of H . Set $\sigma = \text{Ind}_H^G \rho$. Then

$$\det \sigma = \det(R_{G/H})^{\dim \rho} \otimes (\det \rho \circ \text{ver}_{G/H}),$$

where $R_{G/H} = \text{Ind}_H^G 1_H$.

6.3 *L*-functions and Epsilon-Factors

We now turn to the construction of the fundamental invariants – *L*-functions and ε -factors – attached to finite-dimensional, semisimple, smooth representations of the Weil group $\mathcal{W}_F$. This theory generalizes the Tate theory for characters of $F^\times$ (which corresponds to the 1-dimensional case via Class Field Theory), but requires significantly more machinery to establish existence in higher dimensions.

Take the Artin reciprocity map $\mathbf{a}_F : \mathcal{W}_F \rightarrow F^\times$ and identify characters of $F^\times$ with 1-dimensional representations of $\mathcal{W}_F$. For such a character χ , we have already defined the local *L*-function $L(\chi, s)$ and the local constant $\varepsilon(\chi, s, \psi)$ in section 5.1. We translate these to $\mathcal{W}_F$ by setting:

$$L(\chi \circ \mathbf{a}_F, s) = L(\chi, s), \quad \varepsilon(\chi \circ \mathbf{a}_F, s, \psi) = \varepsilon(\chi, s, \psi).$$

The objective is to extend these definitions to $\sigma \in \mathcal{G}_n^{\text{ss}}(F)$.

The extension of the *L*-function is straightforward. For an irreducible representation $\sigma \in \mathcal{G}_n^0(F)$, the definition divides based on dimension. If $n = 1$, it is given by the formula above. If $n \geq 2$:

$$L(\sigma, s) = 1.$$

We then extend this to semisimple representations $\sigma = \bigoplus_i \sigma_i$ by additivity (multiplicativity of the function):

$$L(\sigma, s) = \prod_i L(\sigma_i, s).$$

6.3.1 Artin L-functions There is a more uniform description. Let (σ, V) be a representation of $\mathcal{W}_F$. The space $V^{\mathcal{I}_F}$ of inertia-fixed vectors carries a representation of $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ (generated by a geometric Frobenius Φ). The Artin L-function is defined as:

$$L(\sigma, s) = \det(1 - \sigma|_{V^{\mathcal{I}_F}}(\Phi)q^{-s})^{-1}.$$

This matches the definition given above: if $\dim(\sigma) \geq 2$ and σ is irreducible, the inertia-fixed space $V^{\mathcal{I}_F}$ is usually zero (unless σ is unramified, which implies dimension 1).

The definition of the local constant is far more involved. It is specified axiomatically by its behavior under induction.

Theorem 6.3.2: Existence of the Langlands-Deligne Local Constant

Let $\psi \in \widehat{F}, \psi \neq 1$. Let E/F range over finite extensions inside $\bar{F}$, and let $\psi_E = \psi \circ \text{tr}_{E/F}$. There exists a unique family of functions

$$\mathcal{G}^{\text{ss}}(E) \rightarrow \mathbb{C}[q^s, q^{-s}]^\times, \quad \rho \mapsto \varepsilon(\rho, s, \psi_E),$$

satisfying the following properties:

1. **Dimension 1:** If χ is a character of $E^\times$, then $\varepsilon(\chi \circ \mathbf{a}_E, s, \psi_E) = \varepsilon(\chi, s, \psi_E)$.
2. **Additivity:** If $\rho_1, \rho_2 \in \mathcal{G}^{\text{ss}}(E)$, then

$$\varepsilon(\rho_1 \oplus \rho_2, s, \psi_E) = \varepsilon(\rho_1, s, \psi_E) \varepsilon(\rho_2, s, \psi_E).$$

3. **Inductivity:** If $\rho \in \mathcal{G}_n^{\text{ss}}(E)$ and $E \supset K \supset F$, then

$$\varepsilon(\text{Ind}_{E/K} \rho, s, \psi_K) = \varepsilon(\rho, s, \psi_E) \left(\frac{\varepsilon(R_{E/K}, s, \psi_K)}{\varepsilon(1_E, s, \psi_E)} \right)^n, \quad (6.2)$$

where $R_{E/K} = \text{Ind}_{E/K} 1_E$ is the regular representation.

The goal for the rest of this section is to prove this theorem. Before proving it, we will prove these essential properties.

Proposition 6.3.3: Properties of the Local Constant

Let $\psi \in \widehat{F}$, $\psi \neq 1$, and let $\rho \in \mathcal{G}^{\text{ss}}(F)$.

1. There is an integer $n(\rho, \psi)$ such that

$$\varepsilon(\rho, s, \psi) = q^{n(\rho, \psi)(1/2-s)} \varepsilon(\rho, 1/2, \psi).$$

2. Let $a \in F^\times$. Then:

$$\varepsilon(\rho, s, a\psi) = \det \rho(a) \|a\|^{\dim(\rho)(s-1/2)} \varepsilon(\rho, s, \psi).$$

Consequently, $n(\rho, a\psi) = n(\rho, \psi) + v_F(a) \dim \rho$.

3. The local constants satisfy the functional equation

$$\varepsilon(\rho, s, \psi) \varepsilon(\check{\rho}, 1-s, \psi) = \det \rho(-1).$$

4. There is an integer n_ρ such that, if χ is a character of $F^\times$ of level $k \geq n_\rho$, then

$$\varepsilon(\chi \otimes \rho, s, \psi) = \det \rho(c(\chi))^{-1} \varepsilon(\chi, s, \psi)^{\dim \rho},$$

for any $c(\chi) \in F^\times$ such that $\chi(1+x) = \psi(c(\chi)x)$ for $x \in \mathfrak{p}^{[k/2]+1}$.

The proof of the existence theorem and the proposition requires more induction machinery.

Induction Constants

Let G be a profinite group. Let K_0G be the Grothendieck group of G , the free abelian group generated by isomorphism classes of irreducible smooth representations. Define $\mathcal{K}_0G = \bigoplus_H K_0H$, where H ranges over open subgroups of G . Let $\Gamma(G) = \bigoplus_H \text{Hom}(H, \mathbb{C}^\times)$ be the set of pairs (H, χ) of open subgroups and characters.

Definition 6.3.4: Induction Constant and Division

Let A be an abelian group.

1. An *induction constant* on G is a function $\mathcal{F} : \mathcal{K}_0G \rightarrow A$ such that:

- (a) For each open H , $\mathcal{F}|_{K_0H}$ is a homomorphism.
- (b) If $H \subset J$ are open and ρ is a representation of H of dimension 0 (virtual), then $\mathcal{F}(J, \text{Ind}_H^J \rho) = \mathcal{F}(H, \rho)$.

2. A *division* on G is a function $D : \Gamma(G) \rightarrow A$.

An induction constant defines a division $\partial\mathcal{F}$ by restriction; $\partial\mathcal{F}$ is called the *boundary* of $\mathcal{F}$. Conversely, a division D is *pre-inductive* if $D = \partial\mathcal{F}$ for some induction constant.

Lemma 6.3.5: Uniqueness of Induction Constants

Let $\mathcal{F}$ be an induction constant on G . The division $\partial\mathcal{F}$ determines $\mathcal{F}$ uniquely.

Proof:

It suffices to treat the case where G is finite. Let ρ be an irreducible representation of $H \subset G$ of dimension $m \geq 1$. By the Brauer Induction Theorem, there exist characters χ_i of subgroups H_i and integers n_i such that

$$[\rho] - m[1_H] = \sum_i n_i (\text{Ind}_{H_i}^H [\chi_i] - [1_{H_i}]). \quad (6.3)$$

Writing A multiplicatively, axiom (b) of the definition implies:

$$\mathcal{F}(H, \rho) = \mathcal{F}(H, 1_H)^m \prod_i \left(\frac{\mathcal{F}(H_i, \chi_i)}{\mathcal{F}(H_i, 1_{H_i})} \right)^{n_i}.$$

All terms on the RHS are values of the division $\partial\mathcal{F}$.

Furthermore, one can reduce to finite groups.

Lemma 6.3.6: Reduction to Finite Quotients

If $\mathcal{H}$ is a family of open normal subgroups such that $G \cong \lim G/H$, and a division D restricts to a pre-inductive division on every quotient G/H , then D is pre-inductive on G .

We apply this to the Galois group of a finite Galois extension L/F . Let $G = \text{Gal}(L/F)$. Via Class Field Theory, $\Gamma(G)$ corresponds to pairs (E, χ) where $F \subset E \subset L$ and χ is a character of $E^\times$ trivial on $N_{L/E}(L^\times)$.

Theorem 6.3.7: Existence of Pre-Inductive Division

Let L/F be a finite Galois extension and $G = \text{Gal}(L/F)$. There exists $\psi \in \widehat{F}, \psi \neq 1$, such that the division

$$D_{L/F}^\psi : (E, \chi) \mapsto \varepsilon(\chi, s, \psi_E)$$

is pre-inductive on G .

Before proving this (which requires global methods), we establish a technical local lemma regarding levels.

Lemma 6.3.8: Level Separation

Let L/F be a finite Galois extension. There exists an integer $n_{L/F} \geq 1$ such that if α is a character of $F^\times$ of level $n_{L/F}$ and $(E, \chi) \in \Gamma(G)$, the level of $\alpha \circ N_{E/F}$ is greater than twice the level of χ .

Proof:

For each intermediate field E , there is a level $\ell(E)$ such that $U_E^{\ell(E)} \subset N_{L/E}(L^\times)$. If $(E, \chi) \in \Gamma(G)$, the level of χ is bounded by $\ell(E)$. Set $m(E) = 2\ell(E) + 1$. Choose $n(E)$ such that $U_F^{n(E)} \subset N_{E/F}(U_E^{m(E)})$. The maximum of these $n(E)$ suffices.

Proof of Properties and Extension to Weil Groups

Assuming theorem 6.3.7, we can prove the main results. First, we remove the restriction on ψ . By lemma 6.3.5, the division $\varepsilon(\chi, s, \psi_E)$ defines an induction constant $\varepsilon(\rho, s, \psi_E)$ on Ω_F . For $a \in F^\times$, consider the function

$$(E, \rho) \mapsto \det \rho(a) \|a\|^{(s-1/2) \dim \rho_E}.$$

This is clearly an induction constant. Its boundary is the division $(E, \chi) \mapsto \chi(a) \|a\|_E^{s-1/2}$. Recall from lemma 5.1.16 that $\varepsilon(\chi, s, a\psi_E) = \chi(a) \|a\|_E^{s-1/2} \varepsilon(\chi, s, \psi_E)$. Since the product of pre-inductive divisions is pre-inductive, the division $(E, \chi) \mapsto \varepsilon(\chi, s, a\psi_E)$ is pre-inductive. Thus the theorem holds for all ψ .

This establishes properties (1) and (2) of proposition 6.3.3 for Galois representations. To extend to the Weil group, we use the unramified twist.

Proposition 6.3.9: Unramified Twist of Epsilon Factor

Let $(\Omega_E, \rho) \in \mathcal{K}_0 \Omega_F$, and let φ be an unramified character of $F^\times$ of finite order. Let $\varphi(\varpi) = q^{-s(\varphi)}$. Then $\varepsilon(\varphi_E \otimes \rho, s, \psi_E) = \varepsilon(\rho, s + s(\varphi), \psi_E)$.

Proof:

Both sides define induction constants (as functions of ρ) with the same boundary on 1-dimensional characters. By uniqueness, they are equal.

Next, define the following:

Definition 6.3.10: Langlands Constant

Let E/F be a finite extension, let 1_E be the trivial character of $\mathcal{W}_E$ and set $R_{E/F} = \text{Ind}_{E/F} 1_E$. We define the *Langlands Constant*:

$$\lambda_{E/F}(s, \psi) = \frac{\varepsilon(R_{E/F}, s, \psi)}{\varepsilon(1_E, s, \psi_E)}.$$

Corollary 6.3.11: Langlands Constant Property

The function $\lambda_{E/F}(s, \psi)$ is constant in s .

Proof:

Using the proposition above, $\lambda_{E/F}(s + s(\varphi), \psi) = \lambda_{E/F}(s, \psi)$. Since φ can be any character of finite order, this function is periodic with dense periods, and thus must be constant in s . We write $\lambda_{E/F}(\psi)$.

Now, let ρ be an irreducible representation of $\mathcal{W}_E$. There exists an unramified character φ such that $\varphi_E \otimes \rho$ factors through a representation ρ_0 of the Galois group Ω_E (see proposition 6.1.9). We define:

$$\varepsilon(\rho, s, \psi_E) = \varepsilon(\rho_0, s - s(\varphi), \psi_E).$$

This definition is well-defined and satisfies the inductive properties.

We now prove the remaining properties of proposition 6.3.3.

- **Functional Equation (3):** The maps $\tau \mapsto \varepsilon(\tau, s, \psi)\varepsilon(\check{\tau}, 1 - s, \psi)$ and $\tau \mapsto \det \tau(-1)$ are both induction constants. Their boundaries are equal by the functional equation for $\mathrm{GL}(1)$ (corollary 5.1.14). Thus the maps are equal.
- **Stability (4):** It suffices to consider ρ irreducible. Use the Brauer relation $[\rho] - m[1_F] = \sum (\mathrm{Ind} \varphi_i - 1)$. We have

$$\frac{\varepsilon(\chi \otimes \rho, s, \psi)}{\varepsilon(\chi, s, \psi)^m} = \prod \frac{\varepsilon(\chi_{E_i} \varphi_i, s, \psi_{E_i})}{\varepsilon(\chi_{E_i}, s, \psi_{E_i})}.$$

If χ has sufficiently large level (using lemma 6.3.8), the Stability Theorem for $\mathrm{GL}(1)$ (theorem 5.1.21) applies to each term in the product. The RHS becomes $\prod \varphi_i(c(\chi))^{-1}$. Taking the determinant of the Brauer relation yields $\det \rho = \prod \varphi_i \circ \mathrm{ver}$, which implies the result.

The Global Existence Argument

Finally, we sketch the proof of theorem 6.3.7. This requires global methods (see [6] for more details). Let K be a global field and $\mathbb{A}_K$ its adele ring. For a representation ρ of Ω_K , the global Artin L-function is defined as the product of local L-functions $L(\rho, s) = \prod_v L(\rho_v, s)$. This satisfies a global functional equation $L(\rho, s) = \varepsilon(\rho, s)L(\check{\rho}, 1 - s)$, where the global epsilon factor is the product of local factors $\varepsilon(\rho, s) = \prod_v \varepsilon(\rho_v, s, \psi_v)$. Critically, $\varepsilon(\mathrm{Ind}_{L/K} \tau, s) = \varepsilon(\tau, s)$.

Lemma 6.3.12: Properties of Global L-functions

1. Let L/K be a finite extension, $L \subset \bar{K}$, and let τ be a smooth representation of $\Omega_L = \text{Gal}(\bar{K}/L)$ of finite dimension. Then

$$L(\text{Ind}_{L/K}\tau, s) = L(\tau, s).$$

2. Let ρ be a finite-dimensional smooth representation of Ω_K . The product defining the global L-function converges in some half-plane and admits analytic continuation to a meromorphic function on the whole s-plane. It satisfies a functional equation

$$L(\rho, s) = \varepsilon(\rho, s)L(\check{\rho}, 1 - s), \quad (6.4)$$

for a uniquely determined exponential function $\varepsilon(\rho, s)$ having neither zeros nor poles.

3. In the situation of (1) we have

$$\varepsilon(\text{Ind}_{L/K}\tau, s) = \varepsilon(\tau, s). \quad (6.5)$$

Proof:

Part (1) is given by an elementary argument, based on the definitions of the local factors (which align with the definition of Artin L-functions). Part (2) then follows from the Brauer Induction Theorem (as in lemma 6.3.5) and the known case for 1-dimensional characters (Hecke L-functions). Part (3) is an immediate consequence of (1) and the functional equation (2).

Lemma 6.3.13: Global Realization

There exists a finite Galois extension $\mathcal{L}/\mathcal{F}$ of global fields and a place v_0 of $\mathcal{F}$ such that:

1. The completion $\mathcal{F}_{v_0} \cong F$.
2. There is a unique place u_0 of $\mathcal{L}$ above v_0 , and $\mathcal{L}_{u_0} \cong L$ (the local field).
3. $\mathcal{F}$ has no real embeddings.

Proof:

This follows from the Weak Approximation Theorem. If $\text{char} F = 0$, we approximate local polynomials to ensure the local behavior at v_0 and full decomposition at real places.

We define local integers n_v for places $v \neq v_0$ based on the ramification of $\mathcal{L}/\mathcal{F}$.

Lemma 6.3.14: Existence of Global Character

By the Grunwald-Wang Theorem, there exists a global character α of the idele class group $C_{\mathcal{F}}$ such that:

1. $\alpha_{v_0} = 1$, and
2. the level l_v of α_v satisfies $l_v \geq n_v$, for every non-Archimedean $v \neq v_0$.

Finally:

Proof:

(of theorem 6.3.7 and theorem 6.3.2)

Let Ψ be a global additive character. Set $\psi = \Psi_{v_0}$. For any (E, χ) in the local Galois group $\text{Gal}(L/F)$, we lift it to the global setting (using lemma 6.3.13). We analyze the product formula for the global ε -factor:

$$\varepsilon(\chi\alpha_E, s) = \prod_w \varepsilon((\chi\alpha_E)_w, s, \Psi_w).$$

We analyze the factors at the different places w :

- At w_0 (the unique place above v_0): We constructed α such that $\alpha_{v_0} = 1$. Thus, this factor is simply $\varepsilon(\chi, s, \psi_E)$, which is the local term we wish to define.
- At $w \neq w_0$: The level of α is sufficiently high (by construction in lemma 6.3.14). We apply the 1-dimensional Stability Theorem (theorem 5.1.21) to substitute these factors with terms depending only on α_E , effectively removing χ .

We now rearrange the global functional equation. By lemma 6.3.12(3), the global epsilon factor is inductive: $\varepsilon(\text{Ind}(\chi\alpha)_E) = \varepsilon((\chi\alpha)_E)$. We can therefore express the unknown local term $\varepsilon(\chi, s, \psi_E)$ as a quotient of terms that are known to be inductive (global epsilon factors, character values, and local constants of α):

$$\varepsilon(\chi, s, \psi_E) = \frac{\varepsilon(\text{Ind}(\chi\alpha_E))}{\prod_{w \neq w_0} \varepsilon((\chi\alpha_E)_w, s, \Psi_w)}.$$

Since the right-hand side is built from inductive terms, the local division $D_{L/F}^{\psi}$ is pre-inductive. By lemma 6.3.5 and lemma 6.3.6, this uniquely defines the induction constant for the local field, proving the existence theorem.

6.4 Deligne Representations

We now encounter an interesting friction in the theory due to our choice to work with *complex* representations. Our goal is to relate representations of $G = \text{GL}_2(F)$ to representations of the Weil group $\mathcal{W}_F$. On the group side, we have encountered representations like the Steinberg representation St_G , which is an irreducible quotient of a reducible principal series. On the Galois side, however, representations of $\mathcal{W}_F$ are often constructed from geometry (e.g., étale cohomology), naturally appearing as representations on $\overline{\mathbb{Q}}_{\ell}$ -vector spaces rather than $\mathbb{C}$.

These ℓ -adic representations exhibit behaviors—specifically regarding continuity and image size—that do not perfectly align with the smooth complex representations we have studied so far. To resolve this, and to ensure our theory is as independent of the coefficient field as possible, we introduce *Deligne Representations*. These objects decouple the “semisimple part” of the action from the “monodromy” (nilpotent) part, allowing us to classify representations over $\mathbb{C}$ that correspond to continuous representations over $\overline{\mathbb{Q}}_\ell$.

6.4.1 Definition and Algebraic Structure

A Deligne representation enriches a smooth representation with a nilpotent operator that tracks how the norm scales under conjugation.

Definition 6.4.1: Deligne Representation

A *Deligne representation* of $\mathcal{W}_F$ is a triple $(\rho, V, \mathfrak{n})$, where:

1. (ρ, V) is a finite-dimensional, smooth representation of $\mathcal{W}_F$ on a complex vector space V .
2. $\mathfrak{n} \in \text{End}_{\mathbb{C}}(V)$ is a nilpotent endomorphism.
3. The operators satisfy the compatibility relation:

$$\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}, \quad \forall w \in \mathcal{W}_F. \quad (6.6)$$

A Deligne representation is called *semisimple* if the underlying smooth representation ρ is semisimple. We denote by $\mathcal{G}_n(F)$ the set of equivalence classes of n -dimensional, semisimple Deligne representations. Note that we can identify the “ordinary” smooth representations $\mathcal{G}_n^{\text{ss}}(F)$ with the subset of triples where $\mathfrak{n} = 0$.

Example 6.4.2: The Special Representation $Sp(n)$

Let $V = \mathbb{C}^n$ with basis $v_0, \dots, v_{n-1}$. Let $\mathfrak{n}$ be the standard principal nilpotent element (Jordan block): $\mathfrak{n}v_{n-1} = 0$ and $\mathfrak{n}v_i = v_{i+1}$. Define a smooth representation $\rho_0(w)v_i = \|w\|^i v_i$. We define $\rho(w) = \|w\|^{(1-n)/2} \rho_0(w)$. The triple $(\rho, \mathbb{C}^n, \mathfrak{n})$ is a semisimple Deligne representation denoted $Sp(n)$ (not to be confused with the spin group!)

The standard operations on representations must be adapted to handle the operator $\mathfrak{n}$. These definitions are chosen specifically to mirror the behavior of ℓ -adic representations (as we shall see in section 6.4.2).

Definition 6.4.3: Operations on Deligne Representations

Let $\sigma_1 = (\rho_1, V_1, \mathfrak{n}_1)$ and $\sigma_2 = (\rho_2, V_2, \mathfrak{n}_2)$ be Deligne representations.

1. **Direct Sum:** $\sigma_1 \oplus \sigma_2 = (\rho_1 \oplus \rho_2, V_1 \oplus V_2, \mathfrak{n}_1 \oplus \mathfrak{n}_2)$.
2. **Tensor Product:** $\sigma_1 \otimes \sigma_2 = (\rho_1 \otimes \rho_2, V_1 \otimes V_2, \mathfrak{n}_1 \otimes 1 + 1 \otimes \mathfrak{n}_2)$.
3. **Dual:** $\check{\sigma}_1 = (\check{\rho}_1, \check{V}_1, -\check{\mathfrak{n}}_1)$.

Exercise 6.4.1

Show that the indecomposable, semisimple Deligne representations of $\mathcal{W}_F$ are those of the form $\rho \otimes Sp(n)$, for $n \geq 1$ and $\rho \in \mathcal{G}_m^0(F)$.

6.4.2 Relation with ℓ -adic Representations

We now motivate the definition above. Let ℓ be a prime with $\ell \neq p$. In nature (arithmetic geometry), representations of $\mathcal{W}_F$ act on vector spaces over $\overline{\mathbb{Q}}_\ell$. These are continuous in the ℓ -adic topology, but almost never smooth (the image of inertia is usually infinite).

Fix a continuous surjection $t_\ell : \mathcal{I}_F \rightarrow \mathbb{Z}_\ell$ (this is unique up to a unit in $\mathbb{Z}_\ell^\times$). It satisfies the conjugation relation $t_\ell(gxg^{-1}) = \|g\|t_\ell(x)$.

Theorem 6.4.4: Grothendieck's Monodromy Theorem

Let (σ, V) be a finite-dimensional, continuous representation of $\mathcal{W}_F$ over $\overline{\mathbb{Q}}_\ell$. There exists a unique nilpotent endomorphism $\mathbf{n}_\sigma \in \text{End}(V)$ such that, for x in some open subgroup of $\mathcal{I}_F$:

$$\sigma(x) = \exp(t_\ell(x)\mathbf{n}_\sigma).$$

Proof:

Uniqueness is straightforward. For existence, we view σ as a homomorphism into $\text{GL}_d(\overline{\mathbb{Q}}_\ell)$. The kernel of t_ℓ has pro-order prime to ℓ , so its image must be finite (and eventually trivial in the pro- ℓ subgroups of GL_d). Thus σ factors through t_ℓ on an open subgroup. The image of inertia must satisfy $\sigma(\Phi h \Phi^{-1})^q = \sigma(h)$. This forces the eigenvalues of $\sigma(h)$ to be roots of unity that are permuted by the q -power map. Since the image is in a pro- ℓ group, we use the following lemma:

Lemma 6.4.5: Integrality of Roots

If $\alpha \in \overline{\mathbb{Q}}_\ell$ is a root of unity such that $(\alpha - 1)/\ell^2$ is integral, then $\alpha = 1$.

Proof:

If α has order ℓ^r , the valuation of $\alpha - 1$ is $1/(\ell^r - \ell^{r-1})$, which is too small. If the order is prime to ℓ , $\alpha - 1$ is a unit.

Using the matrix logarithm (which converges on the pro- ℓ group), we define $\mathbf{n} = t_\ell(h_0)^{-1} \log \sigma(h_0)$ for a suitable h_0 .

This theorem allows us to strip away the non-smooth part of the representation. Define a new representation σ_Φ by:

$$\sigma_\Phi(\Phi^a x) = \sigma(\Phi^a x) \exp(-t_\ell(x)\mathbf{n}_\sigma), \quad a \in \mathbb{Z}, x \in \mathcal{I}_F.$$

This representation σ_Φ is trivial on an open subgroup of inertia, hence *smooth*. The triple $(\sigma_\Phi, V, \mathbf{n}_\sigma)$ is a Deligne representation over $\overline{\mathbb{Q}}_\ell$.

Theorem 6.4.6: Equivalence of Categories (ℓ -adic to Deligne)

Let $\text{Rep}_{\overline{\mathbb{Q}}_\ell}^f(\mathcal{W}_F)$ be the category of finite-dimensional continuous ℓ -adic representations. The functor

$$(\sigma, V) \longmapsto (\sigma_\Phi, V, \mathbf{n}_\sigma)$$

is an equivalence of categories $\text{Rep}_{\overline{\mathbb{Q}}_\ell}^f(\mathcal{W}_F) \xrightarrow{\sim} \text{D-Rep}_{\overline{\mathbb{Q}}_\ell}(\mathcal{W}_F)$.

Proof:

We sketched the construction of the object. The inverse functor is given by $\tau(g) = \tau_\Phi(g) \exp(t_\ell(x)\mathbf{n})$. The check that this is well-defined and functorial relies on the conjugation relation $\rho(w)\mathbf{n}\rho(w)^{-1} = \|w\|\mathbf{n}$, which corresponds exactly to the relation satisfied by the Monodromy operator.

Finally, we clarify the notion of semisimplicity.

Proposition 6.4.7: Semisimplicity of ℓ -adic Representations

Let (σ, V) be a continuous ℓ -adic representation. The following are equivalent:

1. The associated Deligne representation is semisimple (i.e., σ_Φ is semisimple).
2. There is a Frobenius element Ψ such that $\sigma(\Psi)$ is semisimple.
3. $\sigma(g)$ is semisimple for every $g \in \mathcal{W}_F \setminus \mathcal{I}_F$.

We call such representations Φ -semisimple.

Corollary 6.4.8: Bijection of Classes

The isomorphism classes of n -dimensional Φ -semisimple representations over $\overline{\mathbb{Q}}_\ell$ are in canonical bijection with isomorphism classes of n -dimensional semisimple Deligne representations over $\mathbb{C}$ (once an isomorphism $\mathbb{C} \cong \overline{\mathbb{Q}}_\ell$ is chosen).

6.4.3 L-functions and Epsilon-factors

We extend the definitions from section 6.3 to Deligne representations. The presence of the operator $\mathbf{n}$ modifies the L-function by restricting to the kernel of the monodromy.

Let $\sigma = (\rho, V, \mathbf{n})$ be a semisimple Deligne representation. Let $V_{\mathbf{n}} = \ker(\mathbf{n})$. This subspace is invariant under ρ (due to the relation (6.6)). We define:

$$L(\sigma, s) = L(\rho|_{V_{\mathbf{n}}}, s). \quad (6.7)$$

For the epsilon factor, we incorporate the full structure and the dual:

$$\varepsilon(\sigma, s, \psi) = \varepsilon(\rho, s, \psi) \frac{L(\check{\rho}, 1-s)}{L(\rho, s)} \frac{L(\rho, s)}{L(\check{\rho}|_{V_{\mathbf{n}}}, 1-s)}. \quad (6.8)$$

This simplifies to $\varepsilon(\sigma, s, \psi) = \varepsilon(\rho, s, \psi) \frac{L(\check{\rho}, 1-s)}{L(\check{\sigma}, 1-s)}$.

Example 6.4.9: L-function of $Sp(2)$

Consider $\sigma = Sp(2)$. Here $\rho = \|\cdot\|^{-1/2} \oplus \|\cdot\|^{1/2}$. The kernel of $\mathfrak{n}$ corresponds to the subrepresentation $\|\cdot\|^{1/2}$. Let $\zeta_F(s) = (1 - q^{-s})^{-1}$. Then:

$$L(Sp(2), s) = L(\|\cdot\|^{1/2}, s) = \zeta_F(s + 1/2).$$

In contrast, $L(\rho, s) = \zeta_F(s - 1/2)\zeta_F(s + 1/2)$. The Monodromy operator “kills” the pole at $s = 1/2$. Calculating the epsilon factor (assuming ψ has level 1):

$$\varepsilon(Sp(2), s, \psi) = -q^{s-1/2} \frac{\zeta_F(s + 1/2)}{\zeta_F(3/2 - s)}.$$

This calculation perfectly matches the local constant for the Steinberg representation of $GL_2(F)$.

The Langlands Correspondence

We have now assembled all the necessary components to state and prove the local Langlands correspondence for $\mathrm{GL}_2(F)$. This correspondence establishes a canonical bijection between two-dimensional representations of the Weil group $\mathcal{W}_F$ and irreducible smooth representations of $G = \mathrm{GL}_2(F)$. It is determined by the preservation of arithmetic invariants—specifically, L -functions and ε -factors—which we explored in Chapter 5.

In this chapter, we first state the general correspondence. We then handle the “reducible” cases (where the Galois representation is a sum of characters) which correspond to the Principal Series representations. Then, we look at the irreducible cases. For odd residual characteristic ($p \neq 2$), we establish the correspondence explicitly using the method of Admissible Pairs from Chapter 4. The dyadic case ($p = 2$) is significantly more involved and is reserved for future chapters, though the existence theorem stated here covers it.

7.1 Statement of the Correspondence

Recall the notations for our categories of representations.

Definition 7.1.1: Representation Sets

Let F be a non-Archimedean local field.

1. Let $\mathcal{G}_2(F)$ denote the set of equivalence classes of 2-dimensional, semisimple, Deligne representations of the Weil group $\mathcal{W}_F$.
2. Let $\mathcal{A}_2(F)$ denote the set of equivalence classes of irreducible smooth representations of $G = \mathrm{GL}_2(F)$.

As usual, if χ is a character of $F^\times$, we identify it via Local Class Field Theory with a character of $\mathcal{W}_F$, denoted $\chi \circ \mathbf{a}_F$ or simply χ .

Theorem 7.1.2: The Local Langlands Correspondence

Let $\psi \in \widehat{F}$ be a non-trivial additive character ($\psi \neq 1$). There exists a unique surjective map:

$$\pi_F : \mathcal{G}_2(F) \rightarrow \mathcal{A}_2(F)$$

such that for all $\rho \in \mathcal{G}_2(F)$ and all characters χ of $F^\times$:

$$L(\chi \otimes \pi_F(\rho), s) = L(\chi \otimes \rho, s), \quad (7.1)$$

$$\varepsilon(\chi \otimes \pi_F(\rho), s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi). \quad (7.2)$$

Furthermore:

1. The map π_F is a bijection.
2. The defining relations hold for *all* $\psi \neq 1$.
3. The central character matches the determinant: $\omega_{\pi_F(\rho)} = \det \rho$.
4. The correspondence commutes with twisting: $\pi_F(\chi \otimes \rho) = \chi \pi_F(\rho)$.
5. The correspondence commutes with contragredients: $\pi_F(\check{\rho}) = \pi_F(\rho)^\vee$.

Proof:

The uniqueness of such a map follows immediately from the Converse Theorem (Theorem 5.5.1), which states that representations in $\mathcal{A}_2(F)$ are determined by their twisted ε -factors (for cuspidals) or L -functions (for non-cuspidals).

The existence and bijectivity are established by decomposing the sets $\mathcal{G}_2(F)$ and $\mathcal{A}_2(F)$ based on reducibility and matching them piece by piece. We detail this decomposition below.

The correspondence respects the natural partition of representations into “reducible” and “irreducible” types.

- **Galois Side:** $\mathcal{G}_2(F) = \mathcal{G}_2^1(F) \cup \mathcal{G}_2^0(F)$.
 - $\mathcal{G}_2^1(F)$: The set of pairs $(\rho, V, \mathbf{n})$ where the underlying smooth representation ρ is *reducible* (sum of 1-dimensional characters).
 - $\mathcal{G}_2^0(F)$: The set of pairs where ρ is *irreducible* (implies $\mathbf{n} = 0$).
- **GL(2) Side:** $\mathcal{A}_2(F) = \mathcal{A}_2^1(F) \cup \mathcal{A}_2^0(F)$.
 - $\mathcal{A}_2^1(F)$: The non-cuspidal representations (Principal Series, Special/Steinberg, 1-dimensional).
 - $\mathcal{A}_2^0(F)$: The supercuspidal representations.

As shown in chapter 5 $\pi \in \mathcal{A}_2^0(F)$ if and only if $L(\chi\pi, s) = 1$ for all χ . Similarly $\rho \in \mathcal{G}_2^0(F)$ if and only if $L(\chi \otimes \rho, s) = 1$ for all χ . Thus, any map satisfying the Main Theorem must map $\mathcal{G}_2^1$ to $\mathcal{A}_2^1$ and $\mathcal{G}_2^0$ to $\mathcal{A}_2^0$.

7.2 The Reducible Correspondence

We first handle the set $\mathcal{G}_2^1(F)$. Since ρ is semisimple and reducible, it must be a direct sum of characters:

$$\rho = \chi_1 \oplus \chi_2.$$

This corresponds to the principal series representations discussed in Chapter 2.

Theorem 7.2.1: Correspondence for $\mathcal{G}_2^1(F)$

There is a unique bijection $\pi_1 : \mathcal{G}_2^1(F) \rightarrow \mathcal{A}_2^1(F)$ preserving L and ε factors. It is defined as follows for $\rho = \chi_1 \oplus \chi_2$:

1. **Principal Series:** If $\chi_1\chi_2^{-1} \neq \|\cdot\|^{\pm 1}$, then $\pi_1(\rho) = \pi(\chi_1, \chi_2)$, the irreducible principal series representation (normalized induction $\iota_B^G(\chi_1 \otimes \chi_2)$).
2. **Special/Steinberg:** If $\chi_1\chi_2^{-1} = \|\cdot\|$, so $\rho = \chi \oplus \chi\|\cdot\|$, then $\pi_1(\rho) = \chi\text{St}_G$, the twisted Steinberg representation.

The case $\chi_1\chi_2^{-1} = \|\cdot\|^{-1}$ is symmetric to the Steinberg case (swapping χ_1, χ_2), and the case where $\mathfrak{n} \neq 0$ in the Deligne representation corresponds to the Special representations. The 1-dimensional representations of G do *not* appear in the image of semisimple Galois representations; they correspond to non-semisimple Deligne representations which we have excluded from $\mathcal{G}_2(F)$.

Proof:

This is a direct verification using the formulas for L -functions and ε -factors derived in Chapter 5. For example, for the principal series $\pi(\chi_1, \chi_2)$, we proved that $L(\pi, s) = L(\chi_1, s)L(\chi_2, s)$ and $\varepsilon(\pi, s) = \varepsilon(\chi_1, s)\varepsilon(\chi_2, s)$, which exactly matches $\rho = \chi_1 \oplus \chi_2$.

7.3 The Tame Correspondence ($p \neq 2$)

We now turn to the heart of the matter: matching irreducible representations of $\mathcal{W}_F$ ($\mathcal{G}_2^0(F)$) with supercuspidal representations of G ($\mathcal{A}_2^0(F)$).

When the residual characteristic p is odd, we can use the classification via *Admissible Pairs* from Chapter 4. Recall that $P_2(F)$ is the set of isomorphism classes of admissible pairs $(E/F, \xi)$, where E/F is a quadratic extension and ξ is a character of $E^\times$ not factoring through the norm.

We have two “naive” maps from $P_2(F)$:

1. **To Galois:** $(E/F, \xi) \mapsto \rho_\xi = \text{Ind}_{E/F} \xi \in \mathcal{G}_2^0(F)$.
2. **To $\text{GL}(2)$:** $(E/F, \xi) \mapsto \pi_\xi$, the cuspidal representation constructed via compact induction in Chapter 4.

If $p \neq 2$, both maps are bijections. However, the map $\rho_\xi \mapsto \pi_\xi$ is *not* the Langlands correspondence. There is a subtle mismatch in the central characters.

$$\det(\rho_\xi) = \kappa_{E/F} \cdot (\xi|_{F^\times}) \quad \text{but} \quad \omega_{\pi_\xi} = \xi|_{F^\times}.$$

Here $\kappa_{E/F}$ is the quadratic character of $F^\times$ associated to E/F by local class field theory. To fix this, we must twist by a “rectifier” character.

Definition 7.3.1: Rectifier Character Δ_ξ

For an admissible pair $(E/F, \xi)$, we define a character Δ_ξ of $E^\times$ as follows:

1. If E/F is unramified, Δ_ξ is the unique unramified character of $E^\times$ of order 2.
2. If E/F is totally ramified, let $\lambda_{E/F}(\psi)$ be the Langlands constant defined in 6.3.10. Let α be an element relating ξ to the additive character ψ_E . Then Δ_ξ is the unique character such that:

$$\Delta_\xi|_{U_E^1} = 1, \quad \Delta_\xi|_{F^\times} = \kappa_{E/F}, \quad \Delta_\xi(\varpi_E) = \kappa_{E/F}(\zeta)\lambda_{E/F}(\psi)^n$$

where n is the level of ξ and ζ is a specific root of unity derived from α .

Crucially, Δ_ξ depends only on the isomorphism class of the pair.

Theorem 7.3.2: The Tame Correspondence

Let $p \neq 2$. The Langlands correspondence $\pi_F : \mathcal{G}_2^0(F) \rightarrow \mathcal{A}_2^0(F)$ is given by:

$$\pi_F(\rho_\xi) = \pi_{\xi'} \quad \text{where } \xi' = \xi \cdot \Delta_\xi.$$

Proof:

(Sketch) We verify the conditions of Theorem 7.1.2.

1. **Determinant:** $\det(\rho_\xi) = \kappa_{E/F} \cdot \xi|_F$. For the $\mathrm{GL}(2)$ side, $\omega_{\pi_{\xi'}} = \xi'|_F = (\xi\Delta_\xi)|_F = \xi|_F \cdot \kappa_{E/F}$. They match.
2. **Epsilon Factors:** The calculation of $\varepsilon(\pi_{\xi\Delta_\xi}, s, \psi)$ involves the Gauss sums of the inducing types. The specific definition of Δ_ξ is engineered precisely to cancel the factor $\lambda_{E/F}(\psi)$ that appears in the inductive formula for the Galois ε -factor.

Specifically, for $\rho = \mathrm{Ind}_{E/F}\xi$, the inductivity of ε gives:

$$\varepsilon(\rho, s, \psi) = \lambda_{E/F}(\psi)\varepsilon(\xi, s, \psi_E).$$

On the group side, the construction of $\pi_{\xi'}$ yields an epsilon factor related to $\varepsilon(\xi', s, \psi_E)$. The character Δ_ξ bridges the gap between $\lambda_{E/F}$ and the difference between ξ and ξ' .

Since the map is a composition of bijections $(P_2(F) \rightarrow \mathcal{G}_2^0(F) \text{ and } P_2(F) \rightarrow \mathcal{A}_2^0(F))$, it is a bijection.

7.4 The ℓ -adic Correspondence

So far, we have worked with representations over $\mathbb{C}$. However, in arithmetic geometry (e.g., the cohomology of modular curves), representations naturally arise over $\overline{\mathbb{Q}_\ell}$, where $\ell \neq p$.

There always exists a (choice of) isomorphism $\iota : \mathbb{C} \cong \overline{\mathbb{Q}}_\ell$, but using it directly is problematic because the correspondence depends on $\sqrt{q}$. The automorphism group $\text{Aut}(\mathbb{C})$ acts on the sets $\mathcal{G}_2(F)$ and $\mathcal{A}_2(F)$, but the correspondence π_F is *not* $\text{Aut}(\mathbb{C})$ -equivariant. Specifically, if $\sigma \in \text{Aut}(\mathbb{C})$, generally $\pi_F(\sigma\rho) \neq \sigma\pi_F(\rho)$.

To fix this, we introduce a normalization.

Definition 7.4.1: Normalized Galois Representation

For $\rho \in \mathcal{G}_2(F)$, define the twist:

$$\tilde{\rho} = \rho \otimes \|\cdot\|^{-1/2}.$$

(Note: This is a twist by a character of the Weil group).

Theorem 7.4.2: The ℓ -adic Correspondence

Define a modified correspondence $\Pi_F : \mathcal{G}_2(F) \rightarrow \mathcal{A}_2(F)$ by:

$$\Pi_F(\rho) = \pi_F(\tilde{\rho}) = \pi_F(\rho \otimes \|\cdot\|^{-1/2}).$$

This map Π_F commutes with the action of $\text{Aut}(\mathbb{C})$. Consequently, for any $\ell \neq p$, it defines a unique, canonical bijection between ℓ -adic representations of W_F and ℓ -adic representations of $\text{GL}_2(F)$.

Proof:

The non-equivariance of the original π_F comes from the factor $q^{1/2}$ in the functional equation. The twist by $\|\cdot\|^{-1/2}$ introduces a factor that exactly compensates for the action of automorphisms on the square root of q .

Explicitly, if $\sigma \in \text{Aut}(\mathbb{C})$, one checks that the defining relations for Π_F are preserved under σ . This allows us to descend the correspondence from $\mathbb{C}$ to $\overline{\mathbb{Q}}_\ell$ (or even fields of definition) without ambiguity.

7.5 Summary

We have established the Local Langlands Correspondence for $\text{GL}_2(F)$:

7.5.1 The Correspondence Maps

- **Principal Series** $\pi(\chi_1, \chi_2) \longleftrightarrow \chi_1 \oplus \chi_2$ (Reducible, decomposable).
- **Special Representations** $\chi\text{St}_G \longleftrightarrow \chi \otimes \text{Sp}(2)$ (Reducible, indecomposable Deligne rep).
- **Supercuspidal Representations** $\pi \longleftrightarrow \rho$ (Irreducible).

For the supercuspidal case with $p \neq 2$:

1. Take an admissible pair $(E/F, \xi)$.

2. Construct the Weil representation $\rho = \text{Ind}_{E/F}\xi$.
3. Construct the rectifying character Δ_ξ .
4. The matching $\text{GL}(2)$ representation is the compactly induced representation $\pi_{\xi\Delta_\xi}$ constructed in Chapter 4.

This satisfies the preservation of L -functions (which are 1 for cuspids) and ε -factors (which encode the arithmetic information, including Gauss sums).

A

The Projectivity Theorem

Throughout the text, we have focused on the construction and classification of representations. In this appendix, we establish a categorical property of the cuspidal representations. The goal of this appendix is to prove that irreducible cuspidal representations are *projective objects* in the category of smooth representations with a fixed central character. This result highlights the “isolated” nature of cuspidal representations—they cannot be deformed or connected to other representations and provides a rigorous underpinning for some of the homological algebra used implicitly in the main text. We prove this using the Schur Orthogonality Relations for square-integrable representations.

The main result is:

Theorem A.0.1: Projectivity Theorem

Let G be a unimodular locally profinite group satisfying the countability hypothesis, with centre Z . Assume that any character $\chi : Z \rightarrow \mathbb{R}_+^\times$ extends to a character $G \rightarrow \mathbb{R}_+^\times$.

Let (π, V) be an irreducible γ -cuspidal representation of G . Let (τ, U) be a smooth representation of G admitting ω_π as central character. Let $f : U \rightarrow V$ be a surjective G -homomorphism. Then there exists a G -homomorphism $\varphi : V \rightarrow U$ such that $f \circ \varphi = 1_V$ (i.e., π is projective in this category).

The proof requires the orthogonality relations for square-integrable representations. The group G/Z is locally profinite and unimodular. Let $d\dot{g}$ be a Haar measure on G/Z .

Theorem A.0.2: Schur's Orthogonality Relation

Let (π, V) be an irreducible γ -cuspidal representation. For $v_1, v_2 \in V$ and $\tilde{v}_1, \tilde{v}_2 \in \check{V}$,

$$\int_{G/Z} \langle \check{\pi}(g)\tilde{v}_1, v_1 \rangle \langle \tilde{v}_2, \pi(g)v_2 \rangle d\dot{g} = d(\pi)^{-1} \langle \tilde{v}_1, v_2 \rangle \langle \tilde{v}_2, v_1 \rangle,$$

for a constant $d(\pi) > 0$ depending only on π and the measure $d\dot{g}$.

Proof:

Since π is γ -cuspidal, the integrand has compact support in G/Z , so the integral converges. Fixing $\tilde{v}_1$ and v_2 , the integral defines a G -invariant pairing $\check{V} \times V \rightarrow \mathbb{C}$, given by a homomorphism $\Theta : \check{V} \rightarrow \check{V}$. Since π is admissible and irreducible, Schur's Lemma implies Θ is a scalar. Thus the integral is equal to $c(\tilde{v}_1, v_2) \langle \tilde{v}_2, v_1 \rangle$. The bilinear form $c(-, -)$ is again G -invariant, hence proportional to the evaluation pairing. This gives the formula. To show $d(\pi) > 0$, one twists to make ω_π unitary, then defines a G -invariant Hermitian form using the integral. Schur's Lemma implies this form is unique up to a scalar, which must be positive definite.

The constant $d(\pi)$ is given a name as it generalizes the notion of dimension:

Definition A.0.3: Formal Degree

The constant $d(\pi)$ is called the *formal degree* of π .

Proof:

of theorem A.0.1

Let χ be a character of $F^\times$. Let $\mathcal{H}_\chi(G)$ be the space of locally constant functions $f : G \rightarrow \mathbb{C}$, compactly supported modulo Z , such that $f(zg) = \chi(z)^{-1}f(g)$. This is an algebra under convolution. Let $\omega = \omega_\pi$. Since π is γ -cuspidal, matrix coefficients of π lie in $\mathcal{H}_\omega(G)$. Let $u \in U$ such that $f(u) \neq 0$. The map $H_\omega(G) \rightarrow V$ given by $\varphi \mapsto f(\tau(\varphi)u)$ is surjective (using the irreducibility of π). It suffices to show this map splits. We choose $\tilde{v}_0 \in \check{V}$ such that $d(\pi)^{-1} \langle \tilde{v}_0, f(u) \rangle = 1$. The function $\varphi_0(g) = \langle \check{\pi}(g)\tilde{v}_0, f(u) \rangle$ lies in $\mathcal{H}_\omega(G)$. Define $\Phi : V \rightarrow H_\omega(G)$ by $v \mapsto \varphi_v$, where $\varphi_v(g) = \langle \check{\pi}(g)\tilde{v}_0, v \rangle$. The composite map $V \rightarrow H_\omega(G) \rightarrow V$ sends $w \in V$ to

$$\pi(\varphi_w)f(u) = \int_{G/Z} \langle \check{\pi}(g)\tilde{v}_0, w \rangle \pi(g)f(u) d\dot{g}.$$

By Schur's orthogonality relations, this equals $\langle \tilde{v}_0, f(u) \rangle d(\pi)^{-1}w = w$. Thus the map splits.

Index

- Admissible Representation, 18
- Central Character, 29
- Class of Semisimple Weil Representations, 129
- Compact Induction, 27
- Continuous Character, 16
- Cuspidal Representation, 47
- Epsilon Factor, 103
- Fourier Inversion Formula, 99
- Fourier Transform, 97
- Frobenius Reciprocity, 26
- Geometric Frobenius, 122
- Haar Integral, 32
- Haar Measure, 34
- Haar Module of G , 36
- Hecke Algebra, 40
- Induced Representation, 25
- Induction On Weil Representation, 129
- Induction Space of Smooth Functions, 37
- Inflation, 45
- Isotypic Component, 23
- Jacquet Module, 62
- Local L-function, 101
- Locally Profinite Group, 15
- Matrix Subgroups, 41
- Mellin Transform, 99
- Mirabolic Subgroup, 56
- Principal Series Representation, 62
- Principle Series Representations, 47
- Profinite Completion, 12
- Profinite Group, 12
- Representation Category, 22
- Schur's Lemma For Smooth Representations, 29
- Self-dual Haar measure, 99
- Semi-invariant Measure, 38
- Semisimple Representation, 23
- Smooth Dual (Contragradient), 30
- Smooth Map For Locally Profinite Groups, 17
- Smooth Representation, 18
- Smooth Representation Admitting Central Character, 29
- Topology on p-adic field, 10
- Twist On Representation, 65
- Unimodular Group, 34
- unitary character, 16
- Unramified Character, 127
- Verlagerung, 133
- Weil Group, 124

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